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Bacterial toxins and uses thereof as Ras specific proteases for treating cell proliferation diseases and disorders

Abstract

Disclosed are bacterial toxins and uses thereof as specific proteases for Ras sarcoma oncoproteins (Ras proteins). The bacterial toxins may be modified for use as pharmaceutical agents for treating Ras-dependent diseases and disorders including cell proliferation diseases and disorders such as cancer.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/093,628 filed Nov. 9, 2020, which is a continuation of U.S. application Ser. No. 15/957,396 filed Apr. 19, 2018, granted as U.S. Pat. No. 10,829,752 on Nov. 10, 2020, which claims the benefit of priority to U.S. Provisional Application

62/487,217, filed Apr. 19, 2017. U.S. application Ser. No. 15/957,396 is also a continuation-in-part of U.S. application Ser. No. 14/816,724, filed Aug. 3, 2015, which claims the benefit of priority to U.S. Provisional Applications 62/032,330, filed Aug. 1, 2014, and 62/172,432, filed Jun. 8, 2015. This application also claims the benefit of priority to U.S. Provisional Application 63/190,779, filed May 19, 2021. The contents of each application are incorporated herein by reference in their entireties.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

(1) This application is being filed electronically via EFS-Web and includes an electronically submitted Sequence Listing in .txt format. The contents of the electronic sequence listing (70258102152 ST25.txt; Size 99,201 bytes and Date of Creation: Apr. 30, 2025) is herein incorporated by reference in its entirety.

BACKGROUND

(2) The field of the invention relates to bacterial toxins. In particular, the field of the invention relates to bacterial toxins that are specific proteases for Ras sarcoma oncoproteins (Ras) and uses therefor for treating Ras-dependent diseases and disorders.

(3) Rat sarcoma (Ras) oncoproteins (e.g., KRas, HRas, and NRas) regulate cell growth, differentiation, and survival by mediating specific signal transduction within cells. Mutational activation of Ras genes is associated with 33% of human cancers, making it one of the most frequent oncogenic mutations. Cancer research has focused on developing several strategies to block mutant Ras and to inhibit the over-activation of the downstream signaling. However, three decades after the discovery of Ras, no drugs or therapeutics that target Ras proteins directly or act on Ras-driven human cancers have been developed successfully.

(4) Here, we discovered a novel protease that cleaves Ras. This protein, known as domain of unknown function in the fifth position (DUF5) otherwise known as Rasaap1-specific endopeptidase (RRSP), is an effector domain of the Multifunctional-Autoprocessing Repeats-in-Toxins (MARTX) family of bacterial toxins. The domain is present in the toxin secreted by some strains of the bacterial pathogen *Vibrio vulnificus*. This domain is also found in the MARTX toxin of other bacterial species and as a toxic domain unlinked to a MARTX toxin in other bacteria, revealing that cleavage of Ras is a conserved toxic function among various bacterial species.

(5) When RRSP (DUF5) from *V. vulnificus* (DUF5.sub.Vv) is released into the cytosol of eukaryotic cells by natural toxin delivery from the bacterium, by transient expression following DNA transfection, or by the anthrax lethal factor N-terminal domain-protective antigen (LF.sub.N-PA) delivery system, it is able to block the Ras pathway, resulting in loss of cell proliferation. Here, we demonstrate, both in vitro and in vivo, that this block occurs because RRSP (DUF5.sub.Vv) is an endopeptidase that cleaves Ras within Switch I, an essential loop for exchange of guanosine nucleotide diphosphate (GDP) with guanosine nucleotide triphosphate (GTP) to activate Ras and for the interaction with several Ras-binding partners. The binding of guanosine nucleosides and binding partners then regulate the Ras downstream pathways that control cell growth, motility, differentiation and response to cell stress.

(6) Because in many cancers Ras is constitutively activated by specific mutations, developing treatments against tumors harboring Ras mutations remains one of the most challenging goals in modern medicine. The use of protein toxin-based therapeutic approaches is a consolidated and alternative way of treating cancer disease compared to conventional radiation or chemical therapy. Because DUF5.sub.Vv specifically cleaves Ras, including Ras mutant isoforms found in cancer cells, resulting in loss of proliferation, we have found that Ras/Rap1-specific endopeptidase (RSP, DUF5.sub.Vv) and proteins similar to DUF5.sub.Vv can be used as the toxic component to create new conjugated toxin-based therapies for cancer treatment. In addition, DUF5.sub.Vv can be used as a cell biological reagent to rapidly eliminate Ras from cells for research or industrial purposes.

SUMMARY

(7) Disclosed are bacterial toxins and uses thereof as specific proteases for Ras sarcoma oncoproteins (Ras proteins). The bacterial toxins may be modified for use as therapeutic polypeptides pharmaceutical agents for treating Ras-dependent diseases and disorders including cell proliferation diseases and disorders such as cancer. Such modification may include, for example, fusion proteins that are capable of translocating into a cell.
(8) The disclosed bacterial toxins include Ras/Rapt-specific endopeptidase (RRSP, also known as DUF5 proteases) and portions thereof comprising active subdomains thereof such as C2A and/or C2B that exhibit protease activity for Ras proteins, and preferably which exhibit specific protease activity for Ras proteins. Active subdomains of RRSP (DUF5) that exhibit protease activity for Ras proteins may include the C2A subdomain and/or the C2B subdomain.
(9) In some embodiments, the RRSP or portions thereof are fused to a bacteria toxin or bacterial toxin element that facilitates the transport of RRSP into a cell.
(10) The disclosed bacterial toxins may be utilized in methods for treating a cell proliferative disease or disorder in a subject. Contemplated treatment methods may include administering a therapeutic polypeptide comprising a RRSP (DUF5) protease or an active portion thereof comprising the C2A subdomain and the C2B subdomain to the subject. Typically, the cell proliferative disease or disorder is associated with an activating mutation in a Ras protein and is a Ras-dependent cell proliferative disease or disorder such as a Ras-dependent cancer.
(11) The disclosed bacterial toxins include the RRSP (DUF5), a homolog thereof, or an active portion thereof comprising the C2A subdomain and/or the C2B subdomain from a number of microorganisms. These include, but are not limited to *Vibrio vulnificus, Vibrio ordalii, Vibrio cholerae, Vibrio splendidus, Moritella dasanensis, Aeromonas salmonicida, Aeromonas hydrophila, Photorhabdus temperata, Xenorhabdus nematophila, Photorhabdus luminescens, Photorhabdus asymbiotica, Yersinia kristensenii,* and *Pasteurella multocida.*
(12) The disclosed bacterial toxins may be formulated as therapeutic polypeptides for delivery to the cytosol of proliferating cells. In some embodiments of the therapeutic polypeptides, the RRSP (DUF5), a homolog thereof, or a portion thereof comprising the C2A subdomain and/or the C2B subdomain may be fused or complexed with a carrier that facilitates transport of the RRSP (DUF5), the homolog thereof, or the C2A subdomain and/or the C2B sub domain thereof into the cytosol of proliferating cells.
(13) Pharmaceutical compositions and kits comprising the disclosed bacterial toxins for treating cell proliferative diseases or disorders also are contemplated herein. The compositions may include a therapeutic polypeptide comprising a DUF5 protease, a homolog thereof, or a portion thereof comprising the C2A subdomain and/or the C2B subdomain, and a carrier that facilitates transport of the DUF5 protease, the homolog thereof, or the portion thereof comprising the C2A subdomain and/or the C2B subdomain into the cytosol of proliferating cells. In the therapeutic polypeptides of the compositions and kits, the DUF5 protease, the homolog thereof, or the portion thereof comprising the C2A subdomain and/or the C2B subdomain may be fused or conjugated to the carrier or complexed with the carrier. Specifically contemplated are fusion proteins comprising the amino acid sequence of the disclosed bacterial toxins fused to the amino acid sequence of a carrier polypeptide that facilitates transport of the bacterial toxins into proliferating cells. Suitable protein carriers include bacterial toxin elements, including bacterial toxins and region or element that allow for translocation of the fusion polypeptide into a cell, for example, anthrax toxin lethal factor (LF), described more herein.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1**A and FIG. **1**B illustrate the *Vibrio vulnificus* CMCP6 MARTX toxin CMCP6. FIG. **1**A.

Linear schematic shows the overall domain structure of the toxin with the pore forming conserved regions and the autoprocessing cysteine protease (CPD). The cytotoxic and cytopathic "effector domains" are DUF1, RID, ABH, MCF, and DUF5 are described in text. FIG. **1**B. The current model for toxin assembly on the eukaryotic cell membrane to form a pore for translocation of the central domains. After being translocated, the CPD binds inositol hexakisphosphate (InsP6) to initiate autoprocessing between effector domains for release to the cytosol. The five domains are then free to access targets in the cell.

(2) FIG. **2**. Left: DUF5.sub.Vv crystals. Right: Domain structure of DUF5.sub.Vv based on crystal structure. C1/MLD: membrane localization domain; C2A: Smallest active domain; C2B: Putative stabilization or specificity domain.

(3) FIG. **3**A, FIG. **3**B, and FIG. **3**C illustrate that cells intoxicated with DUF5.sub.Vv or C1/C2A show loss of all cellular Ras. FIG. **3**A. Western blot to detect total ERK1/2 (upper panels) or phosphor-ERK1/2 (lower panels). Cellular actin in whole cell lysate was used as the loading control. Prior to collection, HeLa cell were incubated for 24 hr with protective antigen (PA), the N-terminus of Lethal factor (LFn), DUF5.sub.Vv fused to anthrax toxin lethal factor (LF.sub.NDUF5), or mixture of proteins as shown. FIG. **3**B. GLISA activation assay (Cytoskeleton Inc) to quantify total active Ras in the GTP bound form (% active Ras) from Hela cell lysates intoxicated as in panel A. FIG. **3**C. Detection of total Ras in cell lysates by western blot using Ras10 monoclonal antibody (upper panels). Detection with anti-actin antibody was used as the loading control.

(4) FIG. **4**. HeLa cells intoxicated with DUF5.sub.Vv (LF.sub.NDUF5.sub.Vv+PA) show loss of all cellular Ras. Detection of total Ras in cell lysates by western blot using Ras10 monoclonal antibody (upper panels). Detection with anti-actin antibody was used as the loading control.

(5) FIG. **5**. Cells intoxicated with DUF5.sub.Ah show rapid loss of detectable Ras. Western blot detection of total Ras using Ras10 antibody (Upper panel). Prior to collection, HeLa cell were incubated for time shown with DUF5.sub.Ah fused to anthrax toxin lethal factor (LF.sub.NDUF5) in the absence (first lane) or presence of PA.

(6) FIG. **6**A, FIG. **6**B, and FIG. **6**C illustrate that intoxication of cells with DUF5.sub.Vv results in truncation of H-Ras. FIG. **6**A. HeLa cells were transfected to overexpress HA-tagged HRas and then intoxicated with LF.sub.NDUF5.sub.Vv/PA for 24 hr. HA-HRas was immunoprecipitated with anti-HA peptide agarose beads and bound protein was eluted from the bead, separated by SDS-PAGE and visualized using Coomassie Brilliant blue. FIG. **6**B. The 18 kDa band was excised and subjected to peptide mapping by mass spectrometry. Peptides matched to H-Ras region shown in underline. FIG. **6**C Western blot analysis on IP elution fractions using both anti-HA antibody to detect full length expression product and an antibody specific to C-terminus of H-Ras to verify this protein is H-Ras from which the N-terminus comprised of the HA and Ras10 epitopes is absent.

(7) FIG. **7**A, FIG. **7**B and FIG. **7**C illustrate that intoxication of cells with DUF5.sub.Vv (LF.sub.NDUF5.sub.Vv+PA) results in truncation of all Ras isoforms. HeLa cells were transfected to overexpress HA-tagged Ras isoforms as indicated and then intoxicated with LF.sub.NDUF5.sub.Vv/PA for 24 hr. Western blot analysis on HeLa whole cell lysates transfected with HA-KRas (FIG. **7**A), HA-NRas (FIG. **7**B) and HA-HRas (FIG. **7**C). Cells were either untreated (−) or intoxicated with LF.sub.NDUF5.sub.Vv in combination with PA (+).

(8) FIG. **8**. DUF5.sub.Vv directly cleaves Ras isoforms in vitro. Reactions of rDUF5.sub.Vv recombinant Ras isoforms as indicated (1:1 molar ratio) was performed in 50 mM TRIS, 10 mM MgCL.sub.2, 500 mM NaCl pH 7.5 at 37° C. Nucleotides were added as shown. After 10 minutes of incubation, each sample reaction was stopped by addition of 6×SDS-PAGE Loading buffer and boiling for 5 min. Samples were separated on 15% SDS-PAGE gel and bands were visualized with Coomassie brilliant blue.

(9) FIG. **9**. DUF5.sub.Vv directly cleaves K-Ras in vitro. A reaction of rDUF5.sub.Vv with KRas performed in FIG. **8** in the presence of guanosine nucleotides as indicated show no dependence on

nucleotide for proper conformation of rKRas in this reaction.
(10) FIG. **10**. DUF5.sub.Vv cleaves Ras isoforms between Y32 and D33. Bands in FIG. **8** were excised and N-terminal sequence determined by Edman degradation. All isoforms cleaved the same site shown by arrows.
(11) FIG. **11**. rKRas is cleaved by DUF5 from *A. hydrophila* (DUF5.sub.Ah) and by *P. asymbiotica* hypothetical protein PAT3833 (DUF5.sub.Pa). A reaction of rDUF5.sub.Vv with KRas performed in FIG. **8** show that other proteins with homology to DUF5.sub.Vv can also cleave rKRas in vitro.
(12) FIG. **12**. Other small GTPase proteins are not cleaved by DUF5.sub.Vv A reaction performed as in FIG. **8** with rDUF5.sub.Vv with small GTPases proteins purified as fusions to glutathione-S-transfer as indicated. No other small GTPases were cleaved by DUF5.sub.Vv.
(13) FIG. **13**A and FIG. **13**B illustrate that LF.sub.NDUF5.sub.Vv is toxic to colorectal (HCT116, FIG. **13**A) and breast cancer cell lines (MDA-MB-231, FIG. **13**B). Cells were treated with LFNDUF5.sub.Vv in the presence of PA and cytotoxicity was observed.
(14) FIG. **14**. rKRas G12V is cleaved by DUF5.sub.Vv A reaction of rDUF5.sub.Vv with rKRas bearing the common G12V mutation was performed as in FIG. **8**. These data show that DUF5.sub.Vv can also mutant forms of rKRas that are common in cancer.
(15) FIG. **15**A, and FIG. **15**B-H illustrate that DUF5 from *V. vulnificus* MARTX toxin is cytotoxic to HeLa cells. FIG. **15**A. Scale drawing of *V. vulnificus* MARTX and *P. multocida* PMT protein toxins with enlarged region showing C1, C2A, and C2B domains that are shared between the two toxins. FIG. **15**B, FIG. **15**C, FIG. **15**D, FIG. **15**E, FIG. **15**F illustrate epifluorescent and DIC images (200×) of HeLa epithelial cells transfected with pEGFP-N3 plasmid clones expressing EGFP (FIG. **15**B), DUF5.sub.Vv-EGFP (FIG. **15**C, FIG. **15**D, and FIG. **15**E), or C1C2Pm-EGFP (FIG. **15**F). Panels in FIG. **15**D and FIG. **15**E are enlarged 200% to show detail of localization of DUF5.sub.Vv-EGFP and cell blebbing, respectively. Arrows in FIG. **15**E indicate EGFP-positive cells in DIC only image. Percent of rounded cells in each cell type is quantified from three independent experiments (FIG. **15**G) and expression of protein in transfected cells is shown by western blot detection using an anti-GFP antibody (FIG. **15**H).
(16) FIG. **16**A, FIG. **16**B, FIG. **16**C, FIG. **16**D, FIG. **16**E and FIG. **16**F illustrate that the C1 MLD of DUF5.sub.Vv is necessary only for efficient cell rounding. (FIG. **16**A) Structural model of DUF5.sub.Vv generated in HHPRED and Modeller based on published structure of PMT. C1, C2A, and C2B subdomains are indicated. FIG. **16**B, FIG. **16**C, and FIG. **16**D illustrate epifluorescent and DIC images (200×) of HeLa epithelial cells transfected with pEGFP-N3 plasmid clones expressing EGFP (FIG. **16**B), DUF5.sub.Vv-EGFP (FIG. **16**C), or C2-EGFP (FIG. **16**D). FIG. **16**E is a Western blot confirming expression of EGFP (FIG. **16**B), DUF5.sub.Vv-EGFP (FIG. **16**C), or C2-EGFP (FIG. **16**D). FIG. **16**F illustrates the percentage (%) of EGFP-positive cells exhibiting rounding.
(17) FIG. **17**A, FIG. **17**B, FIG. **17**C, FIG. **17**D, FIG. **17**E, FIG. **17**F, FIG. **17**G, FIG. **17**H, FIG. **17**I, FIG. **17**J and FIG. **17**K illustrate that C2A is the cytotoxic subdomain of DUF5.sub.Vv. (FIG. **17**A). Schematic of proteins expressed in the panel. FIG. **17**B, FIG. **17**C, FIG. **17**D, FIG. **17**E, FIG. **17**F, FIG. **17**G, and FIG. **17**H illustrate epifluorescent and DIC images (200×) of HeLa epithelial cells transfected with pEGFP-N3 plasmid clones expressing EGFP (B), DUF5.sub.Vv-EGFP (FIG. **17**C), C2-EGFP (FIG. **17**D), C2A-EGFP (FIG. **17**E) and C2B-EGFP (FIG. **17**F). Average of percent rounded cells in each cell type is quantified from three independent experiments (FIG. **17**G) and expression of protein in transfected cells is shown by western blot detection using and anti-GFP antibody (FIG. **17**H). Note that C2A-EGFP could not be detected due to consistent poor sample recovery from plates due to toxicity of this domain. FIG. **17**I, FIG. **17**J, and FIG. **17**K illustrate DIC images of HeLa cells intoxicated with 7 nM PA in combination with 3 nM purified unmodified LFN (I), LFN fused to DUF5.sub.Vv (FIG. **17**J) and LFN fused to only the C1-C2A subdomains of DUF5.sub.Vv (FIG. **17**K).
(18) FIG. **18**A, FIG. **18**B, FIG. **18**C, FIG. **18**D, FIG. **18**E, FIG. **18**F and FIG. **18**G illustrate that

DUF5 from *A. hydrophila* is cytotoxic and causes cell rounding. LFN-DUF5.sub.Ah caused cell rounding when delivered to HeLa cells (FIG. **18**A compared to FIG. **18**B). Protein purity was assessed with SDS-PAGE (FIG. **18**C). Rounding efficiency was comparable to DUF5.sub.Vv, at all concentrations tested (FIG. **18**D and repeated in FIG. **18**E). Finally, release of LDH from intoxicated cells was measured (FIG. **18**F and repeated in FIG. **18**G), showing that there is no appreciable lysis when cells are intoxicated with either DUF5 protein.

(19) FIG. **19**. Amino acid alignment generated in MacVector 12.6.0 of only the C2A and C2B subdomains. Grey shading indicates 100% identical residues. Triangles indicate that sections of sequence were removed during alignment calculations. Asterisk indicated residues changed to alanine via site directed mutagenesis and boxes indicate two aa identified as important for growth inhibition in yeast. Larger asterisks indicate residues G3948 and V3906 which were mutated to stop codons, while the last large asterisk indicates S3986, which was not targeted in the initial mutagenesis but was later found by structural modeling to potentially interact with R3841.

(20) FIG. **20**A, FIG. **20**B and FIG. **20**C illustrate that DUF5.sub.Vv and the C2 domain cause growth inhibition when expressed in yeast. *S. cerevisiae* strain InvSc1 was transformed with pYC2 NT/A plasmid expressing proteins indicated at left or with empty vector (EV), actin crosslinking domain from *V. cholerae* MARTX (ACD), or the C2 domain with stop codons introduced at V3906 or G3048. Panels show 5 μl of 10-fold serial dilutions spotted to SC agar without uracil supplemented with either glucose (non-inducing) or galactose and raffinose (inducing). Panels at right show 12 h growth curves of cultures in SC broth with galactose. (FIG. **20**A (EV, ACD, DUF5.sub.VV, C2, C1/GFP), FIG. **20**B (C2, C2A, and C2B), and FIG. **20**C (C2, V3906*, and G3948*)).

(21) FIG. **21**A, FIG. **21**B, FIG. **21**C, FIG. **21**D, FIG. **21**E and FIG. **21**F illustrate that D3721 and R3841 are important residues for growth inhibition of yeast. Growth inhibition in yeast for C2-D3721A and C2-D3721E (FIG. **21**A) and DUF5.sub.Vv-D3721A, DUF5.sub.Vv-R3841A, DUF5 with both residues mutated to alanine (DARA) and DUF5 with swapped residues (DRRD) in panel B. See FIG. **17**A-**17**K for details. FIG. **21**C and FIG. **21**D illustrate a structural model of DUF5.sub.Vv showing polar contacts of D3721 and R3841 and potential cross association of R3841 with S3986 in C2B. Panel E shows the purified 6×His-tagged DUF5 and DUF5 D3721A proteins that were used in FTS experiments to determine melting temperature in panel F.

(22) FIG. **22**A, FIG. **22**B, FIG. **22**C, FIG. **22**D, FIG. **22**E, FIG. **22**F and FIG. **22**G illustrate that D3721 and R3841 are important residues for intoxication of HeLa cells. FIG. **22**A, FIG. **22**B, FIG. **22**C, FIG. **22**D, and FIG. **22**E illustrate DIC images of HeLa cells intoxicated for 24 h (upper) or 48 h (lower) with 7 nM PA in combination with 3 nM purified unmodified LFN (FIG. **22**A), LFN fused to DUF5.sub.Vv (FIG. **22**B) and LFN fused to only the C1-C2A subdomains of DUF5.sub.Vv (FIG. **22**C), or LFN fused to only the C1-C2A subdomains of DUF5.sub.Vv with D3721 (FIG. **22**D) or R3841 (FIG. **22**E) point mutations. Protein purity was assessed by SDS-PAGE in FIG. **22**F. Three independent experiments were performed and cells were manually counted (FIG. **22**G).

(23) FIG. **23**. MARTX toxin undergoes autoprocessing upon entry into the host cell. Autoprocessing by the inositol hexakisphosphate bound cysteine protease domain releases other effector domains and allows them to perform their functions. DUF5 has been shown to be a stable protein when all the subdomains are present, and is able to efficiently round cells when the C2 domain is intact. When C2B subdomain in removed from the protein, leaving only C1-C2A, cell rounding is less efficient, presumably due to protein turnover. Therefore, C2B is hypothesized to be involved in stabilizing the interaction between DUF5 and the cellular target. C2A and C2B are required for a stable interaction with the target protein, but C2A alone is sufficient for cytotoxic activity.

(24) FIG. **24**A, FIG. **24**B, FIG. **24**C, FIG. **24**D and FIG. **24**E illustrate DUF5.sub.Vv-dependent disruption of Ras-ERK-dependent cell proliferation. FIG. **24**A illustrates major categories of yeast

mutants enabling growth in the presence of DUF5.sub.Vv-C2. FIG. **24**B and FIG. **24**D provide representative immunoblots (n=3) of lysates prepared from cells treated for 24 h (FIG. **24**B) or time indicated (FIG. **24**D) with LFNDUF5.sub.Vv in the absence (−) or the presence (+) of PA. Trimmed ERK1/2 blots are shown unedited in FIG. **29**. FIG. **24**C and FIG. **24**E illustrate clonogenic colony-formation assay (n=2) of cells treated for 24 (FIG. **24**C) or 1 h (FIG. **24**E). Error bars represent the range of the data.

(25) FIG. **25**A, FIG. **25**B, FIG. **25**C, FIG. **25**D, FIG. **25**E and FIG. **25**F illustrate that DUF5.sub.Vv is a Ras site-specific endopeptidase. FIG. **25**A illustrates a coomassie-stained 18% SDS-polyacrylamide gel of anti-HA immunoprecipitated proteins from cells expressing HA-HRas treated for 24 h as indicated. Lower band (HRas*) was excised for peptide sequencing with HRas peptide coverage highlighted in yellow. FIG. **25**B illustrates the same fractions probed by immunoblotting to detect the N terminus (anti-HA) and C terminus (isotype-specific antibody). FIG. **25**C illustrates lysates from cells expressing HA-tagged KRas, NRas or HRas probed by immunoblotting as indicated. FIG. **25**D illustrates in-vitro cleavage of 10 mM rKRas to KRas* with 10 mM rDUF5.sub.Vv (inset) or concentration indicated. Error bars indicate mean±s.d. (n=3). FIG. **25**E illustrates in-vitro cleavage of 10 mM rKRas, rHRas and rNRas with 10 mM rDUF5.sub.Vv. Identical results of Edman degradation were obtained for all three proteins. In FIG. **25**F, black arrow indicates the cleavage site in the Switch I region of HRas69.

(26) FIG. **26**A, FIG. **26**B, FIG. **26**C, FIG. **26**D, FIG. **26**E and FIG. **26**F illustrate DUF5 homologues and other GTPase substrates. FIG. **26**A illustrates a schematic diagram of DUF5 within the mosaic architecture of effector domains in MARTX toxins from *V. vulnificus* (w), *A. hydrophila* (Ah), *Vibrio splendidus* (Vs), *Xenorhabdus nematophila* (.sub.Xn) and *Yersinia kristensii* (.sub.Yk) or as stand-alone proteins in *Photorhabdus luminescens* (.sub.Pl) and *P. asymbiotica* (.sub.Pa) as previously described.sup.17,20. FIG. **26**B illustrates in-vitro cleavage of 10 mM KRas with 10 mM rDUF5 from various species. FIG. **26**C illustrates LFNDUF5.sub.Ah tested for in-vivo loss of all Ras isoforms after 24 h under the same conditions as in b. FIG. **26**D illustrates amino acid identity in Switch I regions of representative GTPases (left) from five major Ras families (right). (FIG. **26**E illustrates a bar graph of percent GFP-fusion protein cleaved after delivery of LFNDUF5.sub.Vv+PA, quantified from immunoblots (FIG. **34**). Error bars indicate mean±s.d. (n=3). FIG. **26**F illustrates a representative in-vitro cleavage (n=3) of GST-fusion proteins to release GST*. Negative cleavage reactions for nine other substrates are shown in FIG. **35**.

(27) FIG. **27**A, FIG. **27**B, FIG. **27**C and FIG. **27**D illustrate DUF5.sub.Vv during bacterial infection and as a potential treatment of malignancies. FIG. **27**A illustrates MARTX toxin effector domain configuration in *V. vulnificus* isolates CMCP6 (DUF5.sub.Vv+) and M06-24/O (DUF5.sub.Vv−). FIG. **27**B illustrates representative immunoblots (n=2) of lysates from cells incubated with *V. vulnificus* as indicated and probed for Ras cleavage and ERK1/2 dephosphorylation. FIG. **27**C illustrates phase-contrast images and immunoblot detection of Ras from HCT116 and MDAMB-231 cells treated as indicated for 24 h. FIG. **27**D illustrates in-vitro processing of 10 mM rKRas with mutations as indicated.

(28) FIG. **28**A, FIG. **28**B, FIG. **28**C, and FIG. **28**D illustrate a schematic summary of yeast deletion screen. FIG. **28**A illustrates a diagram of pYC-C2 plasmid expressing DUF5Vv-C2 under control of the GAL1 galactose-inducible promoter. FIG. **28**B illustrates plating efficiency of *S. cerevisiae* InvSc2 expressing DUF5Vv-C2 (C2Vv) compared to yeast transformed with empty vector (EV) and the more toxic full-length DUF5Vv and actin crosslinking domain from *V. cholerae* (ACDVc), which eliminates the actin cytoskeleton (Geissler B, et al. *Mol Microbiol* 73, 858-868 (2009)). FIG. **28**C illustrates schematic showing the arrayed library of non-essential deletion strains transformed with pYC-C2, followed by selection on glucose to repress expression of DUF5Vv-C2. The resulting yeast colonies were patched onto galactose and raffinose to induce expression. FIG. **28**D illustrates plate 24 of the library, showing the initial screen yeast transformed with empty vector

(C) and strains selected for secondary screening by quantitative plating (circled).

(29) FIG. **29**A and FIG. **29**B illustrate that DUF5Vv inhibits ERK1/2 phosphorylation, but not p38. FIG. **29**A and FIG. **29**B illustrate representative immunoblots (n=2) of lysates from cells treated as indicated for 24 h. Red boxes highlight differences in phospho-p38 (pp38) and phospho-ERK1/2 (pERK1/2) levels. Note that Panel b is the same figure from which lanes were removed to align with other western blots in FIG. **24**.

(30) FIG. **30**. HeLa cells treated with DUF5Vv lack active (GTP-bound) Ras. Bar graph of relative detection of active GTP-bound Ras (all isoforms) by G-LISA. Failure to detect active Ras was ultimately explained by the complete absence of Ras detectable by the monoclonal RAS10 antibody provided with the assay kit.

(31) FIG. **31**. Ras inactivation by DUF5.sub.Vv occurs rapidly. Immunoblot of lysates from cells treated for time indicated. Control samples (first four lanes) were collected 30 minutes after intoxication.

(32) FIG. **32**. DUF5Vv specificity against GFP-tagged small GTPases. HEK 293T cells transfected to express small GTPases with N-terminal EGFP-fusion as indicated were either untreated (−) or intoxicated with LFNDUF5Vv in combination with PA (+) for 24 h, at which time cell lysates were probed with anti-EGFP antibody. For triplicate blots, GFP* and GFP-x bands were quantified by Image J 1.64 and percent cleavage determined as GFP*/(GFP*+GFP-x). For FIG. **26**, samples were normalized to untreated cells to account for closely sized non-specific bands or natural breakdown. Raw pixel data is shown in table.

(33) FIG. **33**. DUF5Vv specificity against GST-tagged small GTPases. In vitro processing of 10 μM purified small GTPases with N-terminal fusion of GST (GST-x) to two fragments (GST* and GTPase*) by 10 μM rDUF5Vv for 10 min. This extended FIG. shows representative data (n=3). Only the positive samples, HRas and Rap1A, are duplicated in FIG. **26**F.

(34) FIG. **34**A, FIG. **34**B, and FIG. **34**C illustrate HeLa cell rounding and lysis due to *V. vulnificus. V. vulnificus* MARTX toxins have distinct compositions dependent upon the strain isolate, as shown in FIG. **27**. Representative (n=3) phase images of cell rounding (FIG. **34**A) and LDH release (FIG. **34**B) induced after 60 min co-incubation of bacteria as indicated with HeLa cells, at which point cells were collected for detection of Ras and pERK in FIG. **27**.

(35) FIG. **34**C illustrates cell lysis over time after addition of bacteria. Note that after 3 h, even bacteria without rtxA1 induce cell lysis due to the vvhA-encoded cytolysin/hemolysin (Fan et al. *Infect Immun* 69, 5943-5948 (2001)). Error bars represent mean±standard deviation.

(36) FIG. **35**. Malignant cells are affected by DUF5Ah from *A. hydrophila*. Phase images and immunoblot detection of Ras from HCT116 and MDA-MB-231 treated as indicated for 24 h.

(37) FIG. **36**. Schematic illustration of MARTX toxin and processing steps.

(38) FIG. **37**A and FIG. **37**B illustrate that RRSP (DUF5) cleaves all Ras isoforms and oncogenic KRas. FIG. **37**A illustrates SDS-Page analysis of cleavage of recombinant KRas, recombinant HRas, and recombinant NRas in vitro by recombinant DUF5.sub.Vv. FIG. **37**B illustrates SDS-Page analysis of cleavage of wild-type (WT), recombinant KRas G12V, recombinant KRas G13D, and recombinant KRas Q61R by recombinant DUF5.sub.Vv.

(39) FIG. **38**. Mutation of putative active site residues eliminates RRSP activity. Suspected catalytic residues E321, H323, E351, H352, and H451 (cumulatively referred to as “TIKI” residues) were mutated to alanines. RRSP TIKI with the five aforementioned substitutions and recombinant KRas were purified and mixed at equimolar concentration (10 μM) for 30 minutes at 37° C. Cleavage was analyzed by SDS-Page analysis. No cleavage was observed.

(40) FIG. **39**A and FIG. **39**B illustrate that Glu/His Pair catalyzes RRSP activity. FIG. **39**A illustrates SDS-Page analysis. RRSP with alanine substitution for E321, H323, and E351 and recombinant KRas were purified and mixed at equimolar concentration (10 μM) for 30 minutes at 37° C. No cleavage was observed in the E351A variant. FIG. **39**B illustrates SDS-Page analysis. RRSP with alanine substitution for H352 and H451 and recombinant KRas were purified and

mixed at equimolar concentration (10 μM) for 30 minutes at 37° C. No cleavage was observed in the H451A variant.

(41) FIG. **40**. KRas(1-169) was used as substrate for RRSP wild type and variant proteins. KRas and RRSP protein were mixed at equimolar ratio (10 μM) and incubated at 37° C. for 30 minutes. Reaction products were analyzed by SDS-Page.

(42) FIG. **41**A and FIG. **41**B. Fluorescent Thermal Shift shows RRSP variants are structurally stable. FIG. **41**A illustrates a denaturation profile of each RRSP variant. FIG. **41**B illustrates melting temperature of each RRSP variant.

(43) FIG. **42**A and FIG. **42**B illustrate that RRSP is not a metalloprotease. FIG. **42**A illustrates SDS-Page analysis. RRSP and recombinant KRas were purified and mixed at equimolar concentration (10 μM) with varying concentrations of phenanthroline in DMSO for 30 minutes at 37° C. Cleavage was still observed. FIG. **42**B illustrates SDS-Page analysis. RRSP and recombinant KRas were purified and mixed at equimolar concentration (10 μM) with varying concentrations of EDTA for 30 minutes at 37° C. Cleavage was still observed.

(44) FIG. **43**A-**43**F: Assessment of RAS cleavage by RRSP-DT.sub.B in KRAS-mutant PDAC cell lines. (A-E) Representative western blots and quantification of uncleaved/cleaved RAS and pERK levels in (A) YAPC, (B) SUIT-2, (C) KP-1N, (D) KP-4 and (E) PANC-1 PDAC cell lines following treatment with increasing doses of RRSP-DTB as indicated or 10 nM of catalytically inactive mutant RRSP*-DT.sub.B for 24 hours. Bar plots represent mean±SD of three independent experiments (*p<0.05, **p<0.01, ****p<0.0001 versus control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test; n=3). Superimposed solid red dose-response curves were used to extrapolate the concentration of RRSP-DT.sub.B required to cleave 50% of RAS in the corresponding cell line, as reported in (F).

(45) FIG. **44**A-**44**E: Effect of RRSP-DT.sub.B on viability and proliferation of KRAS-mutant PDAC cell lines. (A) Representative image of crystal violet-stained YAPC, SUIT-2, KP-1N, KP-4 and PANC-1 cells following 72 hours of treatment with varying doses of RRSP-DT.sub.B as indicated and 10 nM of RRSP*-DT.sub.B for 72 hours. (B) Dose-response curves showing the effect of RRSP-DT.sub.B and RRSP*-DT.sub.B on viability of multiple PDAC cell lines following 72 hours of treatment and summary of extrapolated IC.sub.50 values. Results are expressed as mean±SEM (n=3). (C) Dose-response curves of RRSP-DT.sub.B and RRSP*-DT.sub.B on 3D cultures of KP-4 cells and representative pictures of corresponding spheroids. (D) Representative image of crystal violet-stained colonies from YAPC, SUIT-2, KP-1N, KP-4 and PANC-1 cells pretreated with RRSP-DT.sub.B and RRSP*-DT.sub.B for 72 h and replated at low density to form colonies over 14 days. (E) Quantification of colonies shown in (D) from three independent experiments. Results are expressed as means±SD (**p<0.01, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3).

(46) FIG. **45**A-**45**F: Evaluation of the in vivo activity of RRSP-DT.sub.B in KRAS-mutant PDAC PDXs. (A, B) Average growth curves of KRAS.sup.G12VPDAC PDX tumors from a female donor (A, PDX1) and a male donor (B, PDX2) in NSG mice treated with vehicle (saline), 0.1 mg/kg of RRSP-DT.sub.B and 0.1 mg/kg of RRSP*-DT.sub.B. Mice were injected intraperitoneally every day (weekends excluded) for four weeks followed by a drug-washout phase of 24 days (no treatment). Drug treatment was resumed on day 24 in three mice from PDX1 that received 0.1 mg/kg of RRSP-DT.sub.B q.o.d during the first week and q.d. during the second week following washout (A). In PDX2, mice received 0.1 mg/kg of RRSP-DT.sub.B q.d. for four additional weeks after drug washout (B). Dotted lines indicate the average tumor volume on the first day of treatment. In PDX1, grey data point in FIG. **3**B and grey bar in FIG. **3**C are from a mouse that was euthanized earlier because of excessive tumor burden compare to its weight. (C, D) Representative images of PDX1 (C) and PDX2 (D) tumors at day 29 and corresponding column scatter plots showing individual tumor volumes. Data are means±SEM (n=6 mice per group; **p<0.01, ****p<0.0001; one-way ANOVA followed by Tukey's multiple comparison test, n=6). (E, F)

Waterfall plots depicting tumor regression as percentage change in tumor size from baseline for individual mice.
(47) FIG. **46**A-**46**H: Histological and immunohistochemical analysis of PDAC PDX tumors following treatment with RRSP-DT.sub.B. (A, B) Representative images of saline, RRSP*-DT.sub.B and RRSP-DT.sub.B-treated PDAC PDX1 (A) and PDX2 (B) tumors stained with H&E and Masson's trichrome stain as well as CK-19, Ki-67 and pERK antibodies (scale bar=250 μm). (C-E) Bar plots represent cell positivity to CK-19, Ki-67 and pERK antibodies expressed as the percentage of immunoreactive cells in the entire tumor sections in PDX1 and (F-H) PDX2. Data are means±SEM (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001 versus saline; #p<0.05, ##p<0.01, ###p<0.001 versus RRSP*-DT.sub.B; one-way ANOVA followed by Tukey's multiple comparison test, n=3).
(48) FIG. **47**A: Stability of RRSP-DT.sub.B in sera from immunocompetent mice. (A) Densitometric quantification of total RAS levels in KRAS-mutant YAPC and KP-4 cells treated with serum from mice that were previously injected with 0.1 mg/kg or 0.5 mg/kg of RRSP-DT.sub.B for 1 hour and 16 hours. Each bar represents the mean±SEM of RAS levels from 5 different mice per group (*p<0.05, ***p<0.001 versus saline; one-way ANOVA followed by Dunnett's multiple comparison test, n=5).
(49) FIG. **48**A-**48**B. Assessment of HB-EGF levels in PDAC cell lines and crystal violet cytotoxicity assay quantification. (A) Representative western blot image of HB-EGF expression in six different KRAS-mutant PDAC cell lines. (B) PDAC cells treated with increasing doses of RRSP-DT.sub.B and 10 nM of RRSP*-DT.sub.B for 72 hours were fixed and stained with crystal violet, which was subsequently dissolved in 10% acetic acid and measured by absorbance at 590 nm. Bar plots represent mean±SD of three independent experiments (*p<0.05, **p<0.01, ****p<0.0001 versus control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test; n=3).
(50) FIG. **49**A-**49**B. Monitoring of mouse weight during PDX studies. (A) and (B) Mouse weight was recorded during PDX1 (A) and PDX2 (B) studies and plotted against time (days).
(51) FIG. **50**A-**50**B. Whole tumor sections from PDX1 and PDX2 studies. (A) and (B) Whole tumor sections from PDX1 (A) and PDX2 (B) studies stained with hematoxylin and eosin, Masson's trichrome staining, CK-19, Ki-67 and pERK antibodies from three mice per group (saline and RRSP*-DT.sub.B) and all six mice in the RRSP*-DT.sub.B group. Scale bar=10 mm.
(52) FIG. **51**A. Immunohistochemical analysis of total RAS levels in PDAC PDX tumors following treatment with RRSP-DT.sub.B. (A) Representative images of saline, RRSP*-DT.sub.B and RRSP-DT.sub.B-treated PDAC PDX1 and PDX2 tumors stained with an anti-RAS antibody (scale bar=250 μm) and corresponding jitter plots representing the mean DAB intensity of RAS antibody in the entire tumor sections. Data are means±SEM (*p<0.05, **p<0.01; one-way ANOVA, n=3).
(53) FIG. **52**A-**52**B. Immunohistochemical analysis of cleaved caspase 3 in PDAC PDX tumors and caspase 3/7 activity in PDAC cell lines following treatment with RRSP-DT.sub.B. (A) Representative images of saline, RRSP*-DT.sub.B and RRSP-DT.sub.B-treated PDAC PDX1 and PDX2 tumors stained with an anti-cleaved caspase 3 (CC3) antibody (scale bar=250 μm). A section from human tonsils was used as positive control for CC3. (B) Activity of caspase 3/7 in PDAC cell lines after 24- and 48-hour treatment with RRSP*-DT.sub.B and RRSP-DT.sub.B at the indicated concentrations. Data are means±SEM (*p<0.05, **p<0.01, ***p<0.001 vs corresponding vehicle control; one-way ANOVA, n=3).
(54) FIG. **53**A-**53**B. Western blot images of RAS levels in YAPC and KP-4 cell lines treated with sera from mice previously exposed to RRSP-DT.sub.B. (A) and (B) Western blots showing total RAS levels in YAPC (A) and KP-4 (B) cells treated for 24 hours with serum from immunocompetent mice treated with RRSP-DT.sub.B at 0.1 mg/kg or 0.5 mg/kg for 1 hour and 16 hours. Five mice for each treatment group were employed. Vinculin was used as loading control.
(55) FIG. **54**A-**54**E. Engineered diphtheria toxin delivery machinery enables intracellular delivery

of RRSP. Schematic of chimeric fusions of RRSP to (A) the amino terminus of non-toxic, full-length diphtheria toxin (RRSP-DTa-DTT-DTR), (B) to autoprocessing cysteine protease domain (CPD) of MARTX toxin from *Vibrio vulnificus* (RRSP-CPD-DTa-DTT-DTR), and (C) to DTT-DTR without the DTa domain (RRSP-DTT-DTR or RRSP-DT.sub.B) and corresponding immunoblots against total RAS of lysates prepared from HCT-116 cells treated with RRSP-DT variants for 24 hours. (D) Immunoblots showing that the catalytically-dead RRSP*-DT.sub.B mutant and the translocation-deficient control (RRSP-DT.sub.B(ΔDTT)) do not affect RAS levels in HCT-116 cells. (E) Schematic diagram illustrating transport and intracellular trafficking of RRSP fused to the translocation T and receptor R binding domains of diphtheria toxin fragment B (DT.sub.B) via a receptor-mediated endocytic mechanism. Once in the cytosol, RRSP cleaves RAS resulting in downregulation of the MAPK/ERK signaling cascade. Empty cell and vesicles were taken from Servier Medical Art database (smart.seriver.com). HB-EGF, human heparin-binding epidermal growth factor-like growth factor.

(56) FIG. **55**A-**55**E. RAS processing by RRSP leads to reduced viability and proliferation of a triple-negative KRAS.sup.WT breast cancer cell line. (A). Representative western blot and densitometric analysis of uncleaved RAS and phosphorylated ERK (hereafter pERK) levels in MDA-MB-436 KRAS.sup.WT cells treated with increasing doses of RRSP-DT.sub.B for 1 and 24 hours. The catalytically-inactive RRSP*-DT.sub.B mutant was used as negative control at 10 nM and vinculin as gel loading control. Results are expressed as means±SD of four independent experiments (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=4). (B). Fitted dose-response curve of RRSP-DT.sub.B in MDA-MB-436 cells following 72 hours of treatment. Results are expressed as mean±SEM (n=3). (C). Representative images of crystal violet staining of MDA-MB-436 cells treated with RRSP-DT.sub.B or RRSP*-DT.sub.B as indicated for 72 hours. (D). Representative images and quantitative analysis of crystal violet-stained colonies from MDA-MB-436 cells pre-treated with RRSP-DT.sub.B and RRSP*-DT.sub.B at the indicated concentrations for 72 hours and replated at 2,500 cells/well to form colonies over 10 days. Results are expressed as means±SD of three independent experiments (*p<0.05, **p<0.01 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (E) Bright field images of MDA-MB-436 cells treated with RRSP-DT.sub.B and RRSP*-DT.sub.B at the indicated doses for 48 hours (scale bar=50 μM) and corresponding cell rounding quantification (***p<0.001, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3).

(57) FIG. **56**A-**56**E. RRSP halts tumor growth in a KRAS.sup.WT TNBC xenograft in vivo. (A). Tumor growth curve of vehicle, RRSP-DT.sub.B and RRSP*-DT.sub.B-treated athymic nu/nu female mice bearing MDA-MB-436-derived tumors. Mice received 0.1 mg/kg of RRSP-DT.sub.B and 0.1 mg/kg of RRSP*-DT.sub.B every other day (weekends excluded) via IP injection. (B). Column scatter plots showing individual tumor volumes at the end of the treatment schedule for MDA-MB-436 xenografts. Horizontal dashed line indicates the average tumor volume (baseline) on the first day of treatment (140 mm.sup.3). Data are means±SEM (n=5 mice per group; *p<0.05, **p<0.01). (C). Representative images of MDA-MB-436 tumors at the experimental endpoint. (D). Representative images of immunoreactivity to total RAS in sections from MDA-MB-436 tumors and (E) corresponding quantification of DAB optical density (**p<0.01, ***p<0.001, one-way ANOVA followed by Tukey's multiple comparison test, n=3; scale bar=100 μM).

(58) FIG. **57**A-**57**C. Comprehensive analysis of the effect of RRSP on the NCI-60 cancer cell line panel. (A). Bar plot showing the percent growth inhibition of RRSP-DT.sub.B on the NCI-60 panel. Cancer cell lines were ranked in descending order based on their growth inhibition % value. The presence of two dots on the bars indicate that two replicates were performed per each cell line and bars represent means. No dots indicate that only one replicate was available. (B). Spectrum of missense and nonsense mutations in KRAS, HRAS, NRAS, BRAF and EGFR genes in the 60 cell

lines of the panel. (C). Violin plot showing the median percent growth inhibition of cell lines treated with RRSP-DT.sub.B and grouped per tumor type in descending order. Only tumor types that had at least 5 cell lines per group were plotted in the graph.

(59) FIG. **58**A-**58**H. RRSP inhibits cell viability, proliferation and tumor growth in a KRAS.sup.G13D-dependent TNBC xenograft in vivo. (A). Representative western blot and densitometric analysis of uncleaved RAS and pERK levels in MDA-MB-231 KRAS.sup.G13D cells treated with increasing doses of RRSP-DT.sub.B and with 10 nM of RRSP*-DT.sub.B. Results are expressed as means±SD of four independent experiments (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=4). (B). Fitted dose-response curve of RRSP-DT.sub.B in MDA-MB-231 cells following 72 hours of treatment. Results are expressed as means±SEM (n=3). (C). Representative images of crystal violet staining of MDA-MB-231 cells treated with RRSP-DT.sub.B or RRSP*-DT.sub.B as indicated for 72 hours. (D). Representative images and quantitative analysis of crystal violet-stained colonies from MDA-MB-231 cells pre-treated with RRSP-DT.sub.B and RRSP*-DT.sub.B at the indicated concentrations for 72 hours and replated at 2,500 cells/well to form colonies over 10 days. Results are expressed as means±SD of three independent experiments (**p<0.01 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (E) Bright field images of MDA-MB-231 cells treated with RRSP-DT.sub.B and RRSP*-DT.sub.B at the indicated concentrations for 48 hours (scale bar=50 μM) and corresponding cell rounding quantification (***p<0.001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (F). Tumor growth curve of vehicle, RRSP-DT.sub.B and RRSP*-DT.sub.B-treated athymic nu/nu female mice bearing MDA-MB-231-derived tumors. Mice received 0.1 mg/kg of RRSP-DT.sub.B and 0.1 mg/kg of RRSP*-DT.sub.B every day (weekends excluded). (G). Representative images of tumors at the experimental endpoint. (H). Column scatter plots showing individual tumor volumes at the end of the treatment schedule. Horizontal dashed line indicates the average tumor volume (baseline) on the first day of treatment (194 mm.sup.3). Empty points indicate tumors from mice that were sacrificed earlier because they exceeded the 1500 mm.sup.3 threshold. Data are means±SEM (n=5 mice per group). Statistical analysis between vehicle and treatment groups was performed using one-way ANOVA followed by Tukey's multiple comparison test (**p<0.01, ***p<0.001).

(60) FIG. **59**A-**59**H. Effect of RRSP on viability and proliferation of colorectal HCT-116 KRAS.sup.G13D cells in 2D and 3D cellular models. (A) Representative western blot and densitometric analysis of uncleaved RAS and phosphorylated ERK in HCT-116 cells treated with increasing doses of RRSP-DT.sub.B for 1 and 24 h. Catalytically-inactive RRSP*-DT.sub.B was used as negative control at 10 nM. Results are expressed as means±SD of three independent experiments (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (B) Fitted dose-response curve of RRSP-DT.sub.B in HCT-116 cells following 72 h of treatment. (C). Representative images of crystal violet staining of HCT-116 cells treated with RRSP-DT.sub.B or RRSP*-DT.sub.B as indicated for 72 h. (D) Representative images and quantitative analysis of crystal violet-stained colonies from HCT-116 cells pre-treated with RRSP-DT.sub.B or RRSP*-DT.sub.B at the indicated concentrations for 72 h and reseeded at 2,500 cells/well to form colonies over 10 days. Results are expressed as means±SD of three independent experiments (**p<0.01, ***p<0.001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (E). Bright field images of HCT-116 cells treated with RRSP-DT.sub.B and RRSP*-DT.sub.B at the indicated concentrations for 48 h and corresponding cell rounding quantification (*p<0.05, ***p<0.001, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (F). Fitted dose-response curves in HCT-116 spheroids following treatment with RRSP-DT.sub.B and RRSP*-DT.sub.B at the indicated time and

concentration. Results are expressed as means±SEM (n=4). (G) Representative bright field images of HCT-116 spheroids treated at the indicated time and concentrations with RRSP-DT.sub.B and RRSP*-DT.sub.B and quantitative analysis of spheroids' volume (scale bar=200 μm). Results are expressed as means±SD of three independent experiments (**p<0.01, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (H). Representative images of immunoreactivity to total RAS and pERK in sections from HCT-116 spheroids and corresponding quantification of DAB optical density via color deconvolution (*p<0.05, ***p<0.001, ****p<0.0001; one-way ANOVA followed by Tukey's multiple comparison test, n=3; scale bar=400 μM).

(61) FIG. **60**A-**60**D. RRSP slows tumor growth in a CRC xenograft in vivo. (A) Tumor growth curve of vehicle, RRSP-DT.sub.B and RRSP*-DT.sub.B-treated athymic nu/nu female mice bearing HCT-116-derived tumors. Mice received 0.1 mg/kg of RRSP-DT.sub.B and 0.1 mg/kg of RRSP*-DT.sub.B every day (1×/day) or twice per day (2×/day) weekends excluded. (B) Column scatter plots showing HCT-116 individual tumor volumes at the end of the treatment schedule. Horizontal dashed lines indicate the average tumor volume (baseline) on the first day of treatment (100 mm.sup.3). Data are means±SEM. In all groups, n=10 mice. In the saline group, one mouse was sacrificed on day 28 due to a too large tumor. Statistical analysis between vehicle and treatment groups was performed using one-way ANOVA followed by Tukey's multiple comparison test (**p<0.01, ***p<0.001, ****p<0.0001). (C). Representative IHC images of immunoreactivity to total RAS in sections from HCT-116 tumors and (D) corresponding quantification of DAB optical density (*p<0.05, one-way ANOVA followed by Tukey's multiple comparison test, n=3; scale bar=100 μM).

(62) FIG. **61**A-**61**C. Densitometric analysis showing the effect of various RRSP chimeras on total RAS levels in HCT-116 cells. Densitometric analysis of western blots from HCT-116 cells (KRAS.sup.G13D) treated with the indicated amount of various RRSP chimeras. (A) RRSP-DTa-DT.sub.B is RRSP fused to a detoxified mutant DT (K51E/E148K) (DTa) via a (Gly-Gly-Gly-Gly-Ser)X2 ((G4S)2) linker. (B) RRSP-CPD-DTa-DT.sub.B is RRSP fused to the cysteine protease domain (CPD) of the *V. vulnificus* MARTX toxin as in (A). (C) RRSP-DT.sub.B is RRSP fused directly to DT residues 186-535 via a (G4S)2 linker.

(63) FIG. **62**A-**62**I. Purification of RRSP-DT.sub.B and RRSP*-DT.sub.B proteins and validation of panRAS antibody for detection of uncleaved/cleaved RAS. (A, D) Schematic representation of RRSP-DT.sub.B (A) and RRSP*-DT.sub.B (D) plasmid constructs. (B, E). Coomassie gel images of RRSP-DT.sub.B (B) and RRSP*-DT.sub.B (E) proteins showing degree of protein purity after nickel affinity purification, Strep-purification, removal of the SUMO-tag and size exclusion chromatography (SEC). (C, F). SEC peaks of purified RRSP-DT.sub.B (C) and RRSP*-DT.sub.B (F). (G) Left, western blot indicating presence of the Strep-tag in the RRSP-DT.sub.B purified protein. Right, western blot showing the effect of Strep-tagged and untagged RRSP-DT.sub.B on RAS processing and phosphorylated ERK in HCT-116 cells. Cells were treated with 10 pM of RRSP-DT.sub.B for 1 and 24 hours. Strep-tagged RRSP*-DT.sub.B was also used as negative control. (H) Standard curve displaying efficiency in endotoxin removal from RRSP-DT.sub.B and RRSP*-DT.sub.B protein preparations for in vivo applications. Bar plots show residual endotoxin content (expressed in EU/ml) in both RRSP-DT.sub.B and RRSP*-DT.sub.B protein preparations. (I) Western blot image showing validation of the purified pan-RAS antibody used for detection of uncleaved/cleaved RAS. Recombinant HRAS, NRAS and KRAS were incubated with or without recombinant RRSP and samples were run on a SDS-PAGE gel followed by western blotting. The antibody recognized both cleaved and uncleaved bands of all RAS isoforms.

(64) FIG. **63**A-**63**C. HB-EFG protein levels and additional data on the effect of RRSP-DT.sub.B on viability of MDA-MB-436. (A) Western blot image and densitometry showing expression of the DT receptor HB-EGF across MDA-MB-436, MDA-MB-231, HCT-116 and Hs578T cell lines. (B) Dose-response curve of RRSP-DT.sub.B in MDA-MB-436 cells following 24 h of treatment.

Slopes of the dose-response curves for both RRSP-DT.sub.B and mutant RRSP*-DT.sub.B were not steep enough to retrieve an IC.sub.50. Results are expressed as means±SEM (n=3). (C) Additional two replicate images of crystal violet staining of MDA-MB-436 cells treated for 72 h with RRSP-DT.sub.B and RRSP*-DT.sub.B as indicated.
(65) FIG. **64**A-**64**D. Effect of RRSP-DT.sub.B on mouse embryonic fibroblasts (MEF) and in vivo Maximum Tolerated Dose (MTD) study. (A) Viability and (B) crystal violet staining of RAS-less KRAS.sup.WT MEF cells following 72 h of treatment with RRSP-DT.sub.B and RRSP*-DT.sub.B as indicated. (C) Body weight expressed as percentage of initial weight over time from the MTD study that was performed treating athymic female nu/nu nude mice at the indicated doses every day for two weeks (weekends excluded, F=female, M=male). (D) Kaplan-Meier plot showing overall survival of mice treated with RRSP-DT.sub.B and RRSP*-DT.sub.B.
(66) FIG. **65**A-**65**C. Whole tumor IHC images and analysis of total RAS and pERK levels from the MDA-MB-436 xenograft. (A) Sections of whole tumors from the MDA-MB-436 xenograft immunostained with a total RAS antibody (n=3; scale bar=200 μm). Mice were treated every other day (weekends excluded) at the indicated treatment conditions for about 4 weeks. (B) Sections of whole tumors from the same MDA-MB-436 xenograft immunostained with a pERK antibody (scale bar=100 μm). Bar plot shows quantification of pERK DAB signal (n=3). (C). Sections of whole tumors from a shorter, 2-week long MDA-MB-436 xenograft immunostained with a pERK antibody and quantification of DAB signal (*p<0.05, one-way ANOVA followed by Tukey's multiple comparison test, n=3; scale bar=200 μM).
(67) FIG. **66**A-**66**G. Summary of the effect of RRSP-DT.sub.B on an additional MDA-MB-436 xenograft. (A) Tumor growth curve of vehicle and RRSP-DT.sub.B-treated athymic nu/nu female mice bearing MDA-MB-436 tumors at the indicated doses and treatment schedule. EOD, every other day. ED, every day. (B) Representative images of MDA-MB-436 tumors at the experimental endpoint. (C) Column scatter plots showing individual tumor volumes at the end of the treatment schedule. Horizontal dashed lines indicate the baseline tumor volume on the first day of treatment, which corresponds to the average of tumor volumes at the indicated time (118 mm.sup.3). Data are means±SEM and n=5 mice in every group. Statistical analysis between vehicle and treatment groups was performed using one-way ANOVA followed by Tukey's multiple comparison test (***p<0.001, ****p<0.0001). (D) Representative IHC images of immunoreactivity to total RAS in sections from MDA-MB-436 tumors and (E) corresponding quantification of DAB optical density (****p<0.0001, one-way ANOVA followed by Tukey's multiple comparison test, n=3; scale bar=100 μM). (F). Sections of whole tumors from the MDA-MB-436 xenograft immunostained with a total RAS antibody and (G) pERK antibody (n=3; scale bar=200 μm). Bar plot shows quantification of pERK DAB signal.
(68) FIG. **67**A-**67**C. Expression of HB-EGF and copy number alterations of KRAS, NRAS, BRAF and EGFR across several cell lines from the NCI-60 panel. (A). Heatmap of RNA-Seq expression z-scores for human HB-EGF in 53 out of 60 cell lines included in the NCI-60 panel. Data were retrieved from cBioPortal website for the study with id “cellline_nci60”. The color key indicates gene expression value (pink for upregulated and green for downregulated). (B). Bar plot showing the percent growth inhibition of RRSP-DTB for the 56 cell lines included in the analysis. Cancer cell lines were ranked in descending order based on their growth inhibition % value. The presence of two dots on the bars indicate that two replicates were performed per each cell line and bars represent means. No dots indicate that only one replicate was available. (C). Map of copy number alterations in KRAS, NRAS, BRAF and EGFR genes across the 53 cell lines from the NCI-60 panel. Red indicates gene amplification and blue gene deletions.
(69) FIG. **68**A-**68** D. Effect of RRSP-DT.sub.B on the TNBC Hs578T HRAS.sup.G12D cell line. (A). Representative western blot and densitometric analysis of uncleaved RAS and phosphorylated ERK in Hs578T HRAS.sup.G12D cells treated with increasing doses of RRSP-DT.sub.B for 1 and 24 h. The catalytically-inactive RRSP*-DT.sub.B mutant was used as negative control at 10 nM

and vinculin as gel loading control. Results are expressed as means±SD of three independent experiments (*p<0.05, ***p<0.001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3). (B and C) Fitted dose-response curve of RRSP-DT.sub.B in Hs578T cells following 24 (B) and (C) 72 h of treatment. Slopes of the dose-response curves for both RRSP-DT.sub.B and mutant RRSP*-DT.sub.B were not steep enough to retrieve an IC.sub.50. Results are expressed as means±SEM (n=3). (D). Images of crystal violet-stained plates in triplicate of Hs578T cells treated with RRSP-DT.sub.B and RRSP*-DT.sub.B as indicated for 72 h.

(70) FIG. **69**A-**69**D. Additional data on the effect of RRSP-DT.sub.B on viability of MDA-MB-231 cells and whole tumor IHC images and analysis of total RAS and phosphorylated ERK from the MDA-MB-231 xenograft. (A). Dose-response curve of RRSP-DT.sub.B in MDA-MB-231 cells following 24 h of treatment. Slopes of the dose-response curves for both RRSP-DT.sub.B and mutant RRSP*-DT.sub.B were not steep enough to retrieve an IC.sub.50. Results are expressed as means±SEM (n=3). (B) Additional two replicate images of crystal violet staining of MDA-MB-231 cells treated with RRSP-DT.sub.B and RRSP*-DT.sub.B as indicated for 72 h. (C) Sections of whole tumors from the MDA-MB-231 xenograft immunostained with a total RAS antibody (n=3; scale bar=200 μm). Mice were treated every day (weekends excluded) at the indicated treatment conditions for about 4 weeks. (D) Sections of whole tumors from the same MDA-MB-231 xenograft immunostained with a pERK antibody (scale bar=200 μm).

(71) FIG. **70**A-**70**C. Effect of RRSP-DT.sub.B on viability of CRC HCT-116 KRAS G13D cells after 24 hours of treatment and spheroid volume quantification. (A) Dose-response curve of RRSP-DT.sub.B in HCT-116 cells following 24 h of treatment. Slopes of the dose-response curves for both RRSP-DT.sub.B and mutant RRSP*-DT.sub.B were not steep enough to retrieve an IC.sub.50. Results are expressed as means±SEM (n=3). (B) Additional two replicate images of crystal violet staining of HCT-116 cells treated with RRSP-DT.sub.B or RRSP*-DT.sub.B as indicated for 72 h. (C). Quantitative analysis of spheroids' volumes. Results are expressed as means±SD of three independent experiments (**p<0.01, ****p<0.0001 versus corresponding control 0 nM; one-way ANOVA followed by Dunnett's multiple comparison test, n=3; d=day).

(72) FIG. **71**A-**71**B. Whole tumor IHC images and analysis of total RAS and phosphorylated ERK from the HCT-116 xenograft. (A) Sections of whole tumors from the same HCT-116 xenograft immunostained for phosphorylated ERK (scale bar=200 μm). Bar plot shows quantification of pERK DAB signal (n=3). (B) Sections of whole tumors from the HCT-116 xenograft immunostained with a total RAS antibody (n=3; scale bar=200 μm, left panel; scale bar=100 μm, right panel). Mice were treated every day (1×/day) and twice per day (2×/day) weekends excluded at the indicated treatment conditions for about 4 weeks.

(73) FIG. **72**. Selective targeting of RRSP-DT.sub.B to IL-2 expressing cancer cells via replacement of the DTR domain. Western blot image showing total RAS levels following treatment with RRSP-DTT-IL-2 of lymphoblast MOLT4 and Jurkat cells expressing high levels of IL-2 receptor and pancreatic CFPAC-I cells expressing none to low levels of the IL-2 receptor.

(74) FIG. **73**. Activation and Inactivation of RAS. Inactive RAS (GDP-bound) is activated by GEF by displacing GDP. Active RAS (GTP-bound) binds to downstream effectors to activate signaling. GAP molecules facilitate hydrolysis of GTP into GDP. Abbreviations: RAS; rat sarcoma. GDP; guanine nucleotide diphosphate. GTP; guanine nucleotide triphosphate. GAP; GTPase activating proteins. GEF; guanine exchange factors.

(75) FIG. **74**A-**74**F. LF.sub.NRRSP cleaves and inhibits all RAS isoforms and KRAS oncogenic mutants in RAS-dependent MEFs. A) Representative western blot analysis of LF.sub.NRRSP cleavage of RAS and inhibition of ERK in KRAS WT RAS-dependent MEFs after 24 h. Vinculin was used as a gel loading control. Protein detection for each immunoblot was analyzed on the same blot. Protein detection for each immunoblot was conducted on the same blot and cropped for each individual protein of interest. (B) Densitometric analysis of total percent RAS in indicated RAS-

dependent MEFs following LF.sub.NRRSP treatment for 24 h, n=3. (C,D) Densitometric analysis of fold change in pERK compared to PBS control after 24 h for RAS-dependent MEF cell lines indicated; n=3. (E) Brightfield images of KRAS WT RAS-dependent MEFs treated with either PA alone or in combination with LF.sub.NRRSP or LF.sub.NRRSP* at indicated timepoints. (F) Cell growth over time was monitored with time lapse video microscopy and quantified using Nikon Elements. Values shows relative growth inhibition in RAS-dependent MEFs compared to PA control at 96 h following treatment with either LF.sub.NRRSP or LF.sub.NRRSP*. Results for all panels are expressed as mean±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001 versus PA control as determined through one-way ANOVA followed by Dunnett's multiple comparison test).

(76) FIG. **75**A-**75**H. Growth inhibition of RAS-dependent MEFs cell lines. (A-H) Growth inhibition observed in of RAS-dependent MEF cell lines at indicted timepoints following treatment with PA alone or in combination with LF.sub.NRRSP or LF.sub.NRRSP*; n=3. Results are expressed as ±SEM

(77) FIG. **76**A-**76**H. RRSP-DT.sub.B growth inhibition in CRC cell lines. (A) Cell line panel of KRAS mutant CRC cells used in this study. (B-E) Cell growth over time was monitored by time lapse video microscopy and cell confluency was quantified using Nikon Elements. Fitted dose response curve of RRSP-DT.sub.B in CRC cell lines show relative growth compared to PBS control after 24 h. Results are displayed as mean±SEM, n=4. (F) Representative western blot analysis of RAS cleavage and ERK inhibition in CRC cell lines treated with either RRSP-DT.sub.B or catalytically inactive mutant (labeled by CI) after 24 h. All concentrations are expressed in nanomolar. In all cell lines, vinculin was used as gel loading control except SW620 cells in which aTubulin was used. Protein detection for each immunoblot was conducted on the same blot and cropped for each individual protein of interest. (G,H) Densitometric analysis of fold change in percent total RAS and pERK compared to PBS control after 24 h in CRC cell lines; n=3. IC50 concentrations for HCT-116 and SW1463 can be found in FIGS. **2**B and **2**C. For cell lines where IC50 values could not be extrapolated (SW620 and GP5d) RRSP-DT.sub.B was used at 0.1 nM. Results are expressed as means±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001 versus PBS control as determined through one-way ANOVA followed by Dunnett's multiple comparison test.

(78) FIG. **77**A-**77**J. Growth inhibition in RRSP-DT.sub.B treated colon cancer cell lines (A). Western blot analysis of HB-EGF receptor (DT receptor) from untreated colon cancer cell line lysates. Vinculin was used as loading control. (B-I) Fitted dose response curve of RRSP-DT.sub.B in colon cancer cell lines after 48 and 72 hours. Results are displayed as mean±SEM, n=4. (J) Representative brightfield images of colon cancer cell lines treated with either RRSP-DT.sub.B or RRSP*-DT.sub.B (0.1 nM) after 24 hours.

(79) FIG. **78**A-**78**E. RRSP-DT.sub.B decreases cell viability in highly sensitive cell lines but causes irreversible growth inhibition in lower susceptible cell lines (A) Relative cellular metabolic activity quantified using CellTiterGlo assay after 72-h treatment with 10 nM RRSP-DT.sub.B compared to PBS control in CRC cell lines. (B) Relative apoptosis quantified using Caspase-Glo 3/7 assay after 48-h treatment with either 1 or 10 nM RRSP-DT.sub.B compared to PBS control in CRC cell lines. (C) Representative images of crystal violet-stained colonies from RRSP less sensitive cell lines pretreated with 10 nM RRSP-DT.sub.B for 48 h and replated at low seeding density to form colonies over 14 days. (D) Quantitative analysis of crystal-violet stained colonies from less sensitive RRSP cell lines from three independent experiments. Results are expressed as means±SEM of three independent experiments (E) Measured cell senescence activity in RRSP less sensitive cell lines treated with 10 nM RRSP-DT.sub.B for 48 h then incubated with SA-ß Gal Substrate for 1 h at 37° C., n=3. All results described above are expressed as mean±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001, ns=not significant versus PBS control as determined through one-way ANOVA followed by Dunnett's multiple comparison test).

(80) FIG. **79**. Heatmap of human phospho-kinase array in HCT-116 and SW620 cells treated RRSP-DT.sub.B treated samples. Densitometric analysis of phospho-kinase array depicted through a heatmap of relative phosphorylated proteins levels in response to 10 nM RRSP-DT.sub.B compared to PBS control in HCT-116 and SW620 cells after 24 hr.
(81) FIG. **80**A-**80**E. RRSP-DT.sub.B cleavage of RAS induces p27 expression in CRC cell lines (A) Human phosphokinase array blots of HCT-116 and SW620 cells treated with either PBS or RRSP-DT.sub.B (10 nM) for 24 h. (B) Densitometric analysis kinase array depicted through a heatmap of relative phosphorylated proteins levels in response to RRSP-DT.sub.B compared to PBS control in HCT-116 and SW620 cells, n=1. (C) Representative western blot images of p27 and phosphor-RB expression in CRC cell lines treated with either RRSP-DT.sub.B or RRSP*-DT.sub.B for 24 h. Protein detection for each immunoblot was conducted on the same blot and cropped for each individual protein of interest. (D, E) Densitometric analysis of fold change in p27 and phospho-RB compared to PBS after 24 h in RRSP-DT.sub.B treated CRC cell lines; n=3. αTubulin was used as gel loading control. Results are expressed as mean±SEM of three independent experiments (*P<0.05, **<0.01, ****<0.0001 versus PBS control as determined through one-way ANOVA followed by Dunnett's multiple comparison test).
(82) FIG. **81**A-**81**C. Total RB expression in CRC cell lines (A) Representative images of HCT-116 cells transfected with pGFP-RB after 24 hr (B) Western blot analysis of HCT-116 cells transfected with GFP tagged RB following treatment with either PBS, RRSP-DT.sub.B or RRSP*-DT.sub.B after 24 hours. Total RB was detected using anti-GFP primary antibody. αTubulin was used as a gel loading control. (C) Western blot analysis of p21 levels in GP5d cell treated with either PBS, RRSP-DT.sub.B or RRSP*-DT.sub.B after 24 hr.
(83) FIG. **82**A-**82**D. RRSP-DT.sub.B induces G1 cell cycle arrest in CRC cell lines (A-D) Cell cycle flow cytometry analysis of CRC cell lines treated with either PBS, RRSP-DT.sub.B or RRSP*-DT.sub.B (1 nM) for 24 h. Percentage of cells in each phase are an average of three independent experiments. Bar graphs depict percentage of cells in G1 phase for each treated sample; n=3. Results are expressed as mean±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001 versus PA control as determined through one-way ANOVA followed by Dunnett's multiple comparison test).
(84) FIG. **83**A-**83**D. Cell cycle analysis of CRC cell lines treated RRSP-DT.sub.B (A-D) Representative flow cytometry plots of CRC cell lines treated with either PBS, RRSP-DT.sub.B or RRSP*-DT.sub.B (1 nM) after 24 hours. Gating parameters were used to only collect single live cell populations.
(85) FIG. **84**A-**84**D. Cell cycle analysis of RRSP-DT.sub.B treated CRC cell lines (A-D) Quantitative analysis of cell cycle analysis of CRC cell lines treated with PBS, 1 nM RRSP-DT.sub.B or 1 nM RRSP*-DT.sub.B for 24 hours; n=3. Results are expressed as means±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001 versus PA control as determined through one-way ANOVA followed by Dunnett's multiple comparison test).
(86) FIG. **85**A-**85**G. RRSP-DT.sub.B growth inhibition in BRAF mutant cell lines (A) BRAF mutant cell lines used in this study. (B, C) Representative western blot analysis of RAS cleavage and ERK activity in BRAF mutant cell lines treated using LF.sub.NRRSP or RRSP-DT.sub.B system after 24 hours. Protein detection for each immunoblot was conducted on the same blot and cropped for each individual protein of interest. (D) Cell growth over time in BRAF-dependent MEFs treated with LF.sub.NRRSP or PA control was monitored by time lapse video microscopy and cell confluency was quantified using Nikon Elements. (E-G) Cell growth over time was monitored by time lapse video microscopy and cell confluency was quantified using Nikon Elements. Fitted dose response curve of RRSP-DT.sub.B in HT-29 cells show relative growth compared to PBS control at indicated timepoint. Results are displayed as mean±SEM, n=4. Results are expressed as means±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001 versus PBS control as determined through one-way ANOVA followed by Dunnett's

multiple comparison test).
(87) FIG. **86**A-**86**H. RRSP-DT.sub.B effect on survival and cell cycle in HT-29 cells (A) Luminescence in HT-29 cells treated with wither PBS or RRSP-DT.sub.B for 72 hours. (B) Representative images of crystal violet-stained colonies from HT-29 cells pretreated with 10 nM RRSP-DT.sub.B for 48 h and replated at low seeding density to form colonies over 14 days. Quantitative analysis of crystal-violet stained colonies from three independent experiments. Results are expressed as means±SEM of three independent experiments (C) Measured cell senescence activity in HT-29 cells treated with 10 nM RRSP-DT.sub.B for 48 h then incubated with SA-ß Gal Substrate for 1 h at 37° C., n=3. (D) Representative western blot images of p27, RB and phosopho-RB expression in HT-29 cells treated with either RRSP-DT.sub.B or RRSP*-DT.sub.B for 24 h. Protein detection for each immunoblot was conducted on the same blot and cropped for each individual protein of interest. (E,F) Densitometric analysis of fold change in p27 and phospho-RB compared to PBS after 24 h in RRSP-DT.sub.B treated HT-29 cells n=3. αTubulin was used as gel loading control. (G) (above) Representative flow cytometry plots of HT-29 cells treated with wither PBS, RRSP-DT.sub.B or RRSP*-DT.sub.B (1 nM) after 24 h. Gating parameters were used to only collect single live cell populations. (below) Cell cycle flow cytometry analysis of CRC cell lines treated with either PBS, RRSP-DT.sub.B or RRSP*-DT.sub.B (1 nM) for 24 h. (H) Quantitative analysis of cell cycle analysis of HT-29 cell lines treated with PBS, 1 nM RRSP-DT.sub.B or 1 nM RRSP*-DT.sub.B for 24 h; n=3. Results are expressed as means±SEM of three independent experiments. All results described above are expressed as mean±SEM of three independent experiments (*P<0.05, **P<0.01, ****P<0.0001, ns=not significant versus PBS control as determined through one-way ANOVA followed by Dunnett's multiple comparison test).
(88) FIG. **87**A-**87**B. MEK/ERK signaling regulates RAP1a expression in MEFs (A) Western blot analysis of pERK expression in BRAF-dependent MEFs following treatment with UO126 MEK inhibitor for 2 h. (B) mRNA transcript expression of RAP1a in BRAF-dependent MEFs treated with UO126 MEK inhibitor (10 μM) for 2 h, n=2.
DETAILED DESCRIPTION
(89) The present invention is described herein using several definitions, as set forth below and throughout the application.
(90) As used in this specification and the claims, the singular forms “a,” “an,” and “the” include plural forms unless the context clearly dictates otherwise. For example, the term “a protease” should be interpreted to mean “one or more proteases” unless the context clearly dictates otherwise. As used herein, the term “plurality” means “two or more.”
(91) As used herein, “about”, “approximately,” “substantially,” and “significantly” will be understood by persons of ordinary skill in the art and will vary to some extent on the context in which they are used. If there are uses of the term which are not clear to persons of ordinary skill in the art given the context in which it is used, “about” and “approximately” will mean up to plus or minus 10% of the particular term and “substantially” and “significantly” will mean more than plus or minus 10% of the particular term.
(92) As used herein, the terms “include” and “including” have the same meaning as the terms “comprise” and “comprising.” The terms “comprise” and “comprising” should be interpreted as being “open” transitional terms that permit the inclusion of additional components further to those components recited in the claims. The terms “consist” and “consisting of” should be interpreted as being “closed” transitional terms that do not permit the inclusion of additional components other than the components recited in the claims. The term “consisting essentially of” should be interpreted to be partially closed and allowing the inclusion only of additional components that do not fundamentally alter the nature of the claimed subject matter.
(93) As used herein, the term “subject” may be used interchangeably with the term “patient” or “individual” and may include an “animal” and in particular a “mammal.” Mammalian subjects may include humans and other primates, domestic animals, farm animals, and companion animals such

as dogs, cats, guinea pigs, rabbits, rats, mice, horses, cattle, cows, and the like.

(94) As used herein, the term “biological sample” should be interpreted to include bodily fluids (e.g., blood, serum, plasma, saliva, urine samples) and body tissue samples. Suitable tissue samples may include tissue samples from cancerous tissues and tumors.

(95) The disclosed methods, compositions, and kits may be utilized to treat a subject in need thereof. A “subject in need thereof” is intended to include a subject having or at risk for developing diseases and disorders such as cell proliferative diseases and disorders which may include cancer and hyperproliferative disorders. A subject in need thereof may include a subject having or at risk for developing any of adenocarcinoma, leukemia, lymphoma, melanoma, myeloma, sarcoma, and teratocarcinoma, (including cancers of the adrenal gland, bladder, bone, bone marrow, brain, breast, cervix, gall bladder, ganglia, gastrointestinal tract, heart, kidney, liver, lung, muscle, ovary, pancreas, parathyroid, prostate, skin, testis, thymus, and uterus).

(96) The bacterial toxins disclosed herein may include a Ras/Rap1-specific endopeptidase (RRSP or DUF5 protease), a homolog thereof, or a portion thereof comprising a subdomain thereof such as the C2A subdomain and/or the C2B subdomain of the RRSP (DUF5 protease). The terms Ras/Rap1-specific endopeptidase, RRSP and DUF5 protease are used herein interchangeable and refer to the protease of the present disclosure. The disclosed bacterial toxins may include polypeptides derived from a number of microorganisms, including, but not limited to, *Vibrio vulnificus, Vibrio harveyi, Vibrio ordalii, Vibrio cholerae, Vibrio splendidus, Moritella dasanensis, Aeromonas salmonicida, Aeromonas hydrophila, Photorhabdus temperata, Xenorhabdus nematophila, Photorhabdus luminescens, Photorhabdus asymbiotica, Yersinia kristensenii*, and *Pasteurella multocida.*

(97) As utilized herein, a protein, polypeptide, and peptide refer to a molecule comprising a chain of amino acid residues joined by amide linkages. The term “amino acid residue,” includes but is not limited to amino acid residues contained in the group consisting of alanine (Ala or A), cysteine (Cys or C), aspartic acid (Asp or D), glutamic acid (Glu or E), phenylalanine (Phe or F), glycine (Gly or G), histidine (His or H), isoleucine (Ile or I), lysine (Lys or K), leucine (Leu or L), methionine (Met or M), asparagine (Asn or N), proline (Pro or P), glutamine (Gln or Q), arginine (Arg or R), serine (Ser or S), threonine (Thr or T), valine (Val or V), tryptophan (Trp or W), and tyrosine (Tyr or Y) residues. The term “amino acid residue” also may include amino acid residues contained in the group consisting of homocysteine, 2-Aminoadipic acid, N-Ethylasparagine, 3-Aminoadipic acid, Hydroxylysine, β-alanine, β-Amino-propionic acid, allo-Hydroxylysine acid, 2-Aminobutyric acid, 3-Hydroxyproline, 4-Aminobutyric acid, 4-Hydroxyproline, piperidinic acid, 6-Aminocaproic acid, Isodesmosine, 2-Aminoheptanoic acid, allo-Isoleucine, 2-Aminoisobutyric acid, N-Methylglycine, sarcosine, 3-Aminoisobutyric acid, N-Methylisoleucine, 2-Aminopimelic acid, 6-N-Methyllysine, 2,4-Diaminobutyric acid, N-Methylvaline, Desmosine, Norvaline, 2,2′-Diaminopimelic acid, Norleucine, 2,3-Diaminopropionic acid, Ornithine, and N-Ethylglycine.

(98) The amino acid sequence of the RRSP (DUF5 protease) of *Vibrio vulnificus* is provided as SEQ ID NO:1, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:2 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:1 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Vibrio harveyi* is provided as SEQ ID NO:3, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:4 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:3 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Vibrio ordalii* is provided as SEQ ID NO:5, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:6 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:5 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Vibrio cholerae* is provided as SEQ ID NO:7, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:8 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:7 which is C-terminal to the C2A subdomain. The amino acid sequence of

the DUF5 protease of *Vibrio splendidus* is provided as SEQ ID NO:9, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:10 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:9 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Moritella dasanensis* is provided as SEQ ID NO:11, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:12 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:11 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Aeromonas salmonicida* is provided as SEQ ID NO:13, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:14 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:13 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Aeromonas hydrophila* is provided as SEQ ID NO:15, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:16 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:15 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Photorhabdus temperata* is provided as SEQ ID NO:17, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:18 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:17 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Xenorhabdus nematophila* is provided as SEQ ID NO:19, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:20 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:19 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Photorhabdus luminescens* is provided as SEQ ID NO:21, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:22 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:21 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Photorhabdus asymbiotica* is provided as SEQ ID NO:23, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:24 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:23 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease of *Yersinia kristensenii* is provided as SEQ ID NO:25, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:26 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:25 which is C-terminal to the C2A subdomain. The amino acid sequence of the DUF5 protease homolog of *Pasteurella multocida* is provided as SEQ ID NO:27, and the amino acid sequence of the C2A subdomain is provided as SEQ ID NO:28 and the amino acid sequence of the C2B subdomain is provided in SEQ ID NO:27 which is C-terminal to the C2A subdomain.
(99) The terms “amino acid” and “amino acid sequence” refer to an oligopeptide, peptide, polypeptide, or protein sequence (which terms may be used interchangeably), or a fragment of any of these, and to naturally occurring or synthetic molecules. Where “amino acid sequence” is recited to refer to a sequence of a naturally occurring protein molecule, “amino acid sequence” and like terms are not meant to limit the amino acid sequence to the complete native amino acid sequence associated with the recited protein molecule.
(100) The amino acid sequences contemplated herein may include one or more amino acid substitutions relative to a reference amino acid sequence. For example, a variant polypeptide may include non-conservative and/or conservative amino acid substitutions relative to a reference polypeptide. “Conservative amino acid substitutions” are those substitutions that are predicted to interfere least with the properties of the reference polypeptide. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference protein. The following Table provides a list of exemplary conservative amino acid substitutions.
(101) TABLE-US-00001 Original Residue Conservative Substitution Ala Gly, Ser Arg His, Lys Asn Asp, Gin, His Asp Asn, Glu Cys Ala, Ser Gin Asn, Glu, His Glu Asp, Gin, His Gly Ala His Asn, Arg, Gin, Glu Ile Leu, Val Leu Ile, Val Lys Arg, Gin, Glu Met Leu, Ile Phe His, Met, Leu, Trp, Tyr Ser Cys, Thr Thr Ser, Val Trp Phe, Tyr Tyr His, Phe, Trp Val Ile, Leu, Thr

(102) Conservative amino acid substitutions generally maintain one or more of: (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain. Non-conservative amino acid substitutions generally do not maintain one or more of: (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.
(103) A "deletion" refers to a change in the amino acid sequence that results in the absence of one or more amino acid residues. A deletion removes at least 1, 2, 3, 4, 5, 10, 20, 50, 100, or 200 amino acids residues. A deletion may include an internal deletion or a terminal deletion (e.g., an N-terminal truncation or a C-terminal truncation of a reference polypeptide). A "variant," "mutant," or "derivative" of a reference polypeptide sequence may include a deletion relative to the reference polypeptide sequence.
(104) The words "insertion" and "addition" refer to changes in an amino acid sequence resulting in the addition of one or more amino acid residues. An insertion or addition may refer to 1, 2, 3, 4, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, or 200 amino acid residues or a range of amino acid residues bounded by any of these values (e.g., an insertion or addition of 5-10 amino acids). A "variant" of a reference polypeptide sequence may include an insertion or addition relative to the reference polypeptide sequence.
(105) A "fusion polypeptide" refers to a polypeptide, such as the bacterial toxins contemplated herein, comprising at the N-terminus, the C-terminus, or at both termini of its amino acid sequence a heterologous amino acid sequence, for example, an amino acid sequence that facilitates transport of the polypeptide into the cytosol of proliferating cells. Suitable polypeptides that allow translport into proliferating cells include bacterial toxins and elements known in the art. A "variant," "mutant," or "derivative" of a reference polypeptide sequence may include an fusion polypeptide comprising the reference polypeptide fused to a heterologous polypeptide.
(106) A "fragment" is a portion of an amino acid sequence which is identical in sequence to but shorter in length than a reference sequence. A fragment may comprise up to the entire length of the reference sequence, minus at least one amino acid residue. For example, a fragment may comprise from 5 to 1000 contiguous amino acid residues of a reference polypeptide. In some embodiments, a fragment may comprise at least 5, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, 150, 250, or 500 contiguous amino acid residues of a reference polypeptide; or a fragment may comprise no more than 5, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, 150, 250, or 500 contiguous amino acid residues of a reference polypeptide; or a fragment may comprise a range of contiguous amino acid residues of a reference polypeptide bounded by any of these values (e.g., 40-80 contiguous amino acid residues). Fragments may be preferentially selected from certain regions of a molecule. The term "at least a fragment" encompasses the full length polypeptide. A "variant," "mutant," or "derivative" of a reference polypeptide sequence may include a fragment of the reference polypeptide sequence.
(107) "Homology" refers to sequence similarity or, interchangeably, sequence identity, between two or more polypeptide sequences. Homology, sequence similarity, and percentage sequence identity may be determined using methods in the art and described herein.
(108) The phrases "percent identity" and "% identity," as applied to polypeptide sequences, refer to the percentage of residue matches between at least two polypeptide sequences aligned using a standardized algorithm. Methods of polypeptide sequence alignment are well-known. Some alignment methods take into account conservative amino acid substitutions. Such conservative substitutions, explained in more detail above, generally preserve the charge and hydrophobicity at the site of substitution, thus preserving the structure (and therefore function) of the polypeptide. Percent identity for amino acid sequences may be determined as understood in the art. (See, e.g., U.S. Pat. No. 7,396,664, which is incorporated herein by reference in its entirety). A suite of

commonly used and freely available sequence comparison algorithms is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S. F. et al. (1990) J. Mol. Biol. 215:403 410), which is available from several sources, including the NCBI, Bethesda, Md., at its website. The BLAST software suite includes various sequence analysis programs including “blastp,” that is used to align a known amino acid sequence with other amino acids sequences from a variety of databases.

(109) Percent identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

(110) A “variant,” “mutant,” or “derivative” of a particular polypeptide sequence may be defined as a polypeptide sequence having at least 20% sequence identity to the particular polypeptide sequence over a certain length of one of the polypeptide sequences using blastp with the “BLAST 2 Sequences” tool available at the National Center for Biotechnology Information's website. (See Tatiana A. Tatusova, Thomas L. Madden (1999), “Blast 2 sequences—a new tool for comparing protein and nucleotide sequences”, FEMS Microbiol Lett. 174:247-250). Such a pair of polypeptides may show, for example, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length of one of the polypeptides, or range of percentage identity bounded by any of these values (e.g., range of percentage identity of 80-99%).

(111) A “variant,” “mutant” or a “derivative” may have substantially the same functional activity as a reference polypeptide. For example, a variant, mutant, or derivative of a DUF5 protease or the C2A subdomain or the C2B subdomain thereof may function as a protease of a Ras protein, for example, and specifically cleave the Ras protein between a tyrosine at amino acid position 32 and an aspartic acid at amino acid position 33 of the amino acid sequence of the Ras protein.

(112) A protein, polypeptide, or peptide as contemplated herein may be further modified to include non-amino acid moieties. Modifications may include but are not limited to acylation (e.g., O-acylation (esters), N-acylation (amides), S-acylation (thioesters)), acetylation (e.g., the addition of an acetyl group, either at the N-terminus of the protein or at lysine residues), formylation lipoylation (e.g., attachment of a lipoate, a C8 functional group), myristoylation (e.g., attachment of myristate, a C14 saturated acid), palmitoylation (e.g., attachment of palmitate, a C16 saturated acid), alkylation (e.g., the addition of an alkyl group, such as an methyl at a lysine or arginine residue), isoprenylation or prenylation (e.g., the addition of an isoprenoid group such as farnesol or geranylgeraniol), amidation at C-terminus, glycosylation (e.g., the addition of a glycosyl group to either asparagine, hydroxylysine, serine, or threonine, resulting in a glycoprotein). Distinct from glycation, which is regarded as a nonenzymatic attachment of sugars, polysialylation (e.g., the addition of polysialic acid), glypiation (e.g., glycosylphosphatidylinositol (GPI) anchor formation, hydroxylation, iodination (e.g., of thyroid hormones), and phosphorylation (e.g., the addition of a phosphate group, usually to serine, tyrosine, threonine or histidine).

(113) Also contemplated herein are peptidomimetics of the disclosed proteins, polypeptides, and peptides. As disclosed herein, a peptidomimetic is an equivalent of a protein, polypeptide, or peptide characterized as retaining the polarity, three dimensional size and functionality (bioactivity) of the protein, polypeptide, or peptide equivalent but where the protein, polypeptide, or peptide bonds have been replaced (e.g., by more stable linkages which are more resistant to enzymatic degradation by hydrolytic enzymes). Generally, the bond which replaces the amide bond conserves many of the properties of the amide bond (e.g., conformation, steric bulk, electrostatic character,

and possibility for hydrogen bonding). A general discussion of prior art techniques for the design and synthesis of peptidomimetics is provided in “Drug Design and Development”, Chapter 14, Krogsgaard, Larsen, Liljefors and Madsen (Eds) 1996, Horwood Acad. Pub, the contents of which are incorporated herein by reference in their entirety. Suitable amide bond substitutes include the following groups: N-alkylation (Schmidt, R. et. al., Int. J. Peptide Protein Res., 1995, 46,47), retro-inverse amide (Chorev, M and Goodman, M., Acc. Chem. Res, 1993, 26, 266), thioamide (Sherman D. B. and Spatola, A. F. J. Am. Chem. Soc., 1990, 112, 433), thioester, phosphonate, ketomethylene (Hoffman, R. V. and Kim, H. O. J. Org. Chem., 1995, 60, 5107), hydroxymethylene, fluorovinyl (Allmendinger, T. et al., Tetrahydron Lett., 1990, 31, 7297), vinyl, methyleneamino (Sasaki, Y and Abe, J. Chem. Pharm. Bull. 1997 45, 13), methylenethio (Spatola, A. F., Methods Neurosci, 1993, 13, 19), alkane (Lavielle, S. et. al., Int. J. Peptide Protein Res., 1993, 42, 270) and sulfonamido (Luisi, G. et al. Tetrahedron Lett. 1993, 34, 2391), which all are incorporated herein by reference in their entireties. Contemplated herein are peptidomimetic equivalents of the disclosed therapeutic polypeptides comprising the amino acid sequence of a DUF5 protease C2A subdomain or the amino acid sequence of a DUF5 protease C2B subdomain.

(114) Also disclosed herein are polynucleotides, for example polynucleotide sequences that encode the polypeptides and proteins disclosed herein (e.g., DNA that encodes a polypeptide having the amino acid sequence of any of SEQ ID NOs:1-28 or DNA that encodes a polypeptide variant having an amino acid sequence with at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% sequence identity to any of SEQ ID NOs: 1-28.

(115) The terms “polynucleotide,” “polynucleotide sequence,” “nucleic acid” and “nucleic acid sequence” refer to a nucleotide, oligonucleotide, polynucleotide (which terms may be used interchangeably), or any fragment thereof. These phrases also refer to DNA or RNA of genomic, natural, or synthetic origin (which may be single-stranded or double-stranded and may represent the sense or the antisense strand).

(116) Regarding polynucleotide sequences, the terms “percent identity” and “% identity” refer to the percentage of residue matches between at least two polynucleotide sequences aligned using a standardized algorithm. Such an algorithm may insert, in a standardized and reproducible way, gaps in the sequences being compared in order to optimize alignment between two sequences, and therefore achieve a more meaningful comparison of the two sequences. Percent identity for a nucleic acid sequence may be determined as understood in the art. (See, e.g., U.S. Pat. No. 7,396,664, which is incorporated herein by reference in its entirety). A suite of commonly used and freely available sequence comparison algorithms is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST), which is available from several sources, including the NCBI, Bethesda, Md., at its website. The BLAST software suite includes various sequence analysis programs including “blastn,” that is used to align a known polynucleotide sequence with other polynucleotide sequences from a variety of databases. Also available is a tool called “BLAST 2 Sequences” that is used for direct pairwise comparison of two nucleotide sequences. “BLAST 2 Sequences” can be accessed and used interactively at the NCBI website. The “BLAST 2 Sequences” tool can be used for both blastn and blastp (discussed above).

(117) Regarding polynucleotide sequences, percent identity may be measured over the length of an entire defined polynucleotide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined sequence, for instance, a fragment of at least 20, at least 30, at least 40, at least 50, at least 70, at least 100, or at least 200 contiguous nucleotides. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures, or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

(118) Regarding polynucleotide sequences, “variant,” “mutant,” or “derivative” may be defined as

a nucleic acid sequence having at least 50% sequence identity to the particular nucleic acid sequence over a certain length of one of the nucleic acid sequences using blastn with the "BLAST 2 Sequences" tool available at the National Center for Biotechnology Information's website. (See Tatiana A. Tatusova, Thomas L. Madden (1999), "Blast 2 sequences—a new tool for comparing protein and nucleotide sequences", FEMS Microbiol Lett. 174:247-250). Such a pair of nucleic acids may show, for example, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length.

(119) Nucleic acid sequences that do not show a high degree of identity may nevertheless encode similar amino acid sequences due to the degeneracy of the genetic code where multiple codons may encode for a single amino acid. It is understood that changes in a nucleic acid sequence can be made using this degeneracy to produce multiple nucleic acid sequences that all encode substantially the same protein. For example, polynucleotide sequences as contemplated herein may encode a protein and may be codon-optimized for expression in a particular host. In the art, codon usage frequency tables have been prepared for a number of host organisms including humans, mouse, rat, pig, *E. coli*, plants, and other host cells.

(120) A "recombinant nucleic acid" is a sequence that is not naturally occurring or has a sequence that is made by an artificial combination of two or more otherwise separated segments of sequence. This artificial combination is often accomplished by chemical synthesis or, more commonly, by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques known in the art. The term recombinant includes nucleic acids that have been altered solely by addition, substitution, or deletion of a portion of the nucleic acid. Frequently, a recombinant nucleic acid may include a nucleic acid sequence operably linked to a promoter sequence. Such a recombinant nucleic acid may be part of a vector that is used, for example, to transform a cell.

(121) The nucleic acids disclosed herein may be "substantially isolated or purified." The term "substantially isolated or purified" refers to a nucleic acid that is removed from its natural environment, and is at least 60% free, preferably at least 75% free, and more preferably at least 90% free, even more preferably at least 95% free from other components with which it is naturally associated.

(122) "Transformation" or "transfected" describes a process by which exogenous nucleic acid (e.g., DNA or RNA) is introduced into a recipient cell. Transformation or transfection may occur under natural or artificial conditions according to various methods well known in the art, and may rely on any known method for the insertion of foreign nucleic acid sequences into a prokaryotic or eukaryotic host cell. The method for transformation or transfection is selected based on the type of host cell being transformed and may include, but is not limited to, bacteriophage or viral infection or non-viral delivery. Methods of non-viral delivery of nucleic acids include lipofection, nucleofection, microinjection, electroporation, heat shock, particle bombardment, biolistics, virosomes, liposomes, immunoliposomes, polycation or lipid:nucleic acid conjugates, naked DNA, artificial virions, and agent-enhanced uptake of DNA. Lipofection is described in e.g., U.S. Pat. Nos. 5,049,386, 4,946,787; and 4,897,355) and lipofection reagents are sold commercially (e.g., Transfectam™ and Lipofectin™). Cationic and neutral lipids that are suitable for efficient receptor-recognition lipofection of polynucleotides include those of Felgner, WO 91/17424; WO 91/16024. Delivery can be to cells (e.g. in vitro or ex vivo administration) or target tissues (e.g. in vivo administration). The term "transformed cells" or "transfected cells" includes stably transformed or transfected cells in which the inserted DNA is capable of replication either as an autonomously replicating plasmid or as part of the host chromosome, as well as transiently transformed or transfected cells which express the inserted DNA or RNA for limited periods of time.

(123) The polynucleotide sequences contemplated herein may be present in expression vectors. For example, the vectors may comprise: (a) a polynucleotide encoding an ORF of a protein; (b) a

polynucleotide that expresses an RNA that directs RNA-mediated binding, nicking, and/or cleaving of a target DNA sequence; and both (a) and (b). The polynucleotide present in the vector may be operably linked to a prokaryotic or eukaryotic promoter. “Operably linked” refers to the situation in which a first nucleic acid sequence is placed in a functional relationship with a second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects the transcription or expression of the coding sequence. Operably linked DNA sequences may be in close proximity or contiguous and, where necessary to join two protein coding regions, in the same reading frame. Vectors contemplated herein may comprise a heterologous promoter (e.g., a eukaryotic or prokaryotic promoter) operably linked to a polynucleotide that encodes a protein. A “heterologous promoter” refers to a promoter that is not the native or endogenous promoter for the protein or RNA that is being expressed. For example, a heterologous promoter for a LAMP may include a eukaryotic promoter or a prokaryotic promoter that is not the native, endogenous promoter for the LAMP

(124) As used herein, “expression” refers to the process by which a polynucleotide is transcribed from a DNA template (such as into and mRNA or other RNA transcript) and/or the process by which a transcribed mRNA is subsequently translated into peptides, polypeptides, or proteins. Transcripts and encoded polypeptides may be collectively referred to as “gene product.” If the polynucleotide is derived from genomic DNA, expression may include splicing of the mRNA in a eukaryotic cell.

(125) The term “vector” refers to some means by which nucleic acid (e.g., DNA) can be introduced into a host organism or host tissue. There are various types of vectors including plasmid vector, bacteriophage vectors, cosmid vectors, bacterial vectors, and viral vectors. As used herein, a “vector” may refer to a recombinant nucleic acid that has been engineered to express a heterologous polypeptide (e.g., the fusion proteins disclosed herein). The recombinant nucleic acid typically includes cis-acting elements for expression of the heterologous polypeptide.

(126) Any of the conventional vectors used for expression in eukaryotic cells may be used for directly introducing DNA into a subject. Expression vectors containing regulatory elements from eukaryotic viruses may be used in eukaryotic expression vectors (e.g., vectors containing SV40, CMV, or retroviral promoters or enhancers). Exemplary vectors include those that express proteins under the direction of such promoters as the SV40 early promoter, SV40 later promoter, metallothionein promoter, human cytomegalovirus promoter, murine mammary tumor virus promoter, and Rous sarcoma virus promoter. Expression vectors as contemplated herein may include eukaryotic or prokaryotic control sequences that modulate expression of a heterologous protein (e.g. the fusion protein disclosed herein). Prokaryotic expression control sequences may include constitutive or inducible promoters (e.g., T3, T7, Lac, trp, or phoA), ribosome binding sites, or transcription terminators.

(127) The vectors contemplated herein may be introduced and propagated in a prokaryote, which may be used to amplify copies of a vector to be introduced into a eukaryotic cell or as an intermediate vector in the production of a vector to be introduced into a eukaryotic cell (e.g. amplifying a plasmid as part of a viral vector packaging system). A prokaryote may be used to amplify copies of a vector and express one or more nucleic acids, such as to provide a source of one or more proteins for delivery to a host cell or host organism. Expression of proteins in prokaryotes may be performed using *Escherichia coli* with vectors containing constitutive or inducible promoters directing the expression of either a protein or a fusion protein comprising a protein or a fragment thereof. Fusion vectors add a number of amino acids to a protein encoded therein, such as to the amino terminus of the recombinant protein. Such fusion vectors may serve one or more purposes, such as: (i) to increase expression of recombinant protein; (ii) to increase the solubility of the recombinant protein; (iii) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification (e.g., a His tag); (iv) to tag the recombinant protein for identification (e.g., such as Green fluorescence protein (GFP) or an antigen (e.g., HA) that can be

recognized by a labelled antibody); (v) to promote localization of the recombinant protein to a specific area of the cell (e.g., where the protein is fused (e.g., at its N-terminus or C-terminus) to a nuclear localization signal (NLS) which may include the NLS of SV40, nucleoplasmin, C-myc, M9 domain of hnRNP A1, or a synthetic NLS). The importance of neutral and acidic amino acids in NLS have been studied. (See Makkerh et al. (1996) *Curr Biol* 6(8):1025-1027). Often, in fusion expression vectors, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such enzymes, and their cognate recognition sequences, include Factor Xa, thrombin and enterokinase.
(128) The presently disclosed methods may include delivering one or more polynucleotides, such as or one or more vectors as described herein, one or more transcripts thereof, and/or one or proteins transcribed therefrom, to a host cell. Further contemplated are host cells produced by such methods, and organisms (such as animals, plants, or fungi) comprising or produced from such cells. The disclosed exosomes may be prepared by introducing vectors that express mRNA encoding a fusion protein and a cargo RNA as disclosed herein. Conventional viral and non-viral based gene transfer methods can be used to introduce nucleic acids in mammalian cells or target tissues. Non-viral vector delivery systems include DNA plasmids, RNA (e.g. a transcript of a vector described herein), naked nucleic acid, and nucleic acid complexed with a delivery vehicle, such as a liposome. Viral vector delivery systems include DNA and RNA viruses, which have either episomal or integrated genomes after delivery to the cell.
(129) In the methods contemplated herein, a host cell may be transiently or non-transiently transfected (i.e., stably transfected) with one or more vectors described herein. In some embodiments, a cell is transfected as it naturally occurs in a subject (i.e., in situ). In some embodiments, a cell that is transfected is taken from a subject (i.e., explanted). In some embodiments, the cell is derived from cells taken from a subject, such as a cell line. Suitable cells may include stem cells (e.g., embryonic stem cells and pluripotent stem cells). A cell transfected with one or more vectors described herein may be used to establish a new cell line comprising one or more vector-derived sequences. In the methods contemplated herein, a cell may be transiently transfected with the components of a system as described herein (such as by transient transfection of one or more vectors, or transfection with RNA), and modified through the activity of a complex, in order to establish a new cell line comprising cells containing the modification but lacking any other exogenous sequence.
(130) DUF5 Domain of Multifunctional, Autoprocessing RTX (MARTX) Toxin
(131) Bacterial toxins can inactive host cellular processes. Ras is a cellular protein for the host response to stress that is modified in many human cancers to promote cell survival. We discovered that the DUF5 domain of the *Vibrio vulnificus* multifunctional, autoprocessing RTX (MARTX) toxin is an endopeptidase that specifically cleaves Ras between residues Y32 and D33. We determined the crystal structure, the minimally active portion, and the specificity for Ras among the small GTPases. Ras proteins with mutations common in cancer are also cleaved. We also have utilized a system and carriers to deliver the domain to cells independent of the holotoxin by fusion to a bacterial toxin element, Anthrax toxin Lethal Factor N-terminus, which allowa for delivery to cells by Protective antigen. Homology sequence analysis demonstrates that at least 12 bacteria produce a protein with at least 24% percent identity and the proteins from *Aeromonas hydrophila* and *Photorhabdus asymbiotica* are shown also to cleave Ras. We propose this family of proteins could be engineered for delivery to cancer cells as potential therapeutic agents for carcinoma, targeting both tumors with normal Ras and those with modified Ras proteins. We propose this family of proteins also for use in biological research to specifically and rapidly knock down or remove Ras.
(132) Ras-Dependent Cancers
(133) Ras-activating mutations are frequently observed in cancer. (See Fernandez-Medarde et al.,

"Ras in Cancer and Developmental Diseases," March 2011, vol. 2, no. 3: 344-358; Johannes L. Bos. "Ras Oncogenes in Human Cancer: A Review," Cancer Research 49, 4682-4689, Sep. 1, 1989; Julian Downward "Targeting RAS signalling pathways in cancer therapy," Nature Reviews Cancer 3, 11-22 (January 2003); Schubbert et al., "Hyperactive Ras in developmental disorders and cancer," Nature Reviews Cancer 7, 295-308 (January 2007); and Baines et al., "Inhibition of Ras for cancer treatment: the search continues," Future Med. Chem. 2011 October; 3(14): 1787-1808; the contents of which are incorporated herein by reference in their entireties). Ras-activating mutations are observed in 95% of pancreatic cancers, 45% of colorectal cancers, and 35% of lung adenocarcinoma. The RAS oncogenes (HRAS, NRAS and KRAS comprising activating mutations present in codon 12, 13, or 61) comprise the most frequently mutated class of oncogenes in human cancers (33%), stimulating intensive effort in developing anti-Ras inhibitors for cancer treatment. (See Prior et al., "A comprehensive survey of Ras mutations in cancer," Cancer Research. 2012 May 15; 72(10): 2457-2467, the content of which is incorporated herein by reference in its entirety). Unfortunately, there are no drugs that target Ras directly or indirectly, and there are currently no effective therapies for Ras-dependent cancers.

(134) Targeted Delivery or Expression of Bacterial Toxins into the Cytosol of Proliferating Cells

(135) The bacterial toxins disclosed herein may be administered in order to treat cell proliferative diseases and disorders such as cancer. The bacterial toxins may be administered by transfecting cancer cells with a polynucleotide or a polynucleotide vector that expresses the bacterial toxins in the cancer cells. Alternatively, the bacterial toxins may be formulated for intracellular protein delivery using methods known in the art including the use of carriers, for example, bacterial toxin elements as carriers, more specifically for example, anthrax lethal toxin for targeted delivery of protein into cells. (See WO 2014031861 A1; WO2008/076939; and WO 2001/21656, the contents of which are incorporated herein by reference in their entireties). Other bacterial toxins and bacterial toxin elements are contemplated and known in the art to be able to target and deliver proteins into cells. Alternative approaches for targeted delivery of protein into cells are known in the art. (See, e.g., Weill et al., "A practical approach for intracellular protein delivery," Cytotechnology. 20089 January; 56(1) 41-48; Walev et al., "Delivery of proteins into living cells by reversible membrane permeabilization with steptolysin-O," PNAS, Mar. 13, 2001, vol. 98, no. 6, 3185-3190; Cronican et al., "Naturally supercharged human proteins (NSHPs)," Chemistry & Biology 18, 833-838, Jul. 29, 2011; Torchilin, "Intracellular deliver of protein and peptide therapeutics," Drug Discovery Today: Technologies, Protein Therapeutics, 2009; M. Grdisa "The Delivery of Biologically Active (Therapeutic) Peptides and Proteins into Cells," Cell-penetrating peptides (CPPS), Current Medicinal Chemistry, 2011. Vol. 18; Morris et al., "A Peptide Carrier for the Delivery of Biologically Active Proteins into Mammalian Cells: Application to the Delivery of Antibodies and Therapeutic Proteins," Cell Biology, Volume 204, Part 20A, Chapter 2, 2006; Futami et al. "Intracellular delivery of proteins into mammalian living cells by polyethylenimine-cationization," J Bioscience and Bioengineering, Vol. 99, Iss 2, Febr 2005 95-103; and Kurzawa et al., "PEP and CADY-mediated delivery of fluorescent peptides and proteins into living cells," Biochimica et a Biophysica Acta (BBA)—Biomembranes Vol. 1798, Issue 12, December 2010 2274-2285).

(136) Applications and Advantages of Disclosed Bacterial Toxins

(137) Uses of the bacterial proteases disclosed herein include, but are not limited to: (a) uses as toxin components in bacterial toxin-based therapeutics for cancer and other diseases requiring killing of cells, including but not limited to immunotoxins, tetramer-toxins, bacterial delivery of toxins, and nanoparticles and others; (b) uses as reagents to treat cells to knock down Ras during biological research by direct delivery to cell cytosol by any method including chemical, mechanical, or biological strategies; (c) specific delivery by Protective antigen of this family of proteins to cells when fused to Lethal Factor N terminus as therapeutics; (d) specific delivery by Protective antigen of this family of proteins to cells when fused to Lethal Factor N terminus as a

reagent during biological research; (e) treatment of biochemical reactions involving Ras to rapidly remove Ras from an in vitro reaction; (f) linkage of this family of proteins to an antibody or tetramer to create an immunotoxin specifically developed to delivery to cancerous cells or other conditions requiring killing of cells; and (g) delivery of this family of proteins to tumors or malignant cells by any strategy that delivers protein to cell for use a cancer therapeutic.

(138) Some advantages of using the disclosed DUF5 protease or related proteases for inactivated Ras include, but are not limited to: (a) the DUF5 protease permanently modifies Ras by nicking Ras at a site essential for function, a modification which is not reversible as are other modifications; (b) the DUF5 protease exhibits specificity for isoforms of Ras including isoforms found in cancerous cells; (c) the DUF5 protease has a natural lack of structure in vitro and is thus amenable to easy transfer into cells by processes that require folding and unfolding; and (d) the DUF5 protease can be delivered to cells via fusion to anthrax toxin lethal factor (LF) in the presence of protective antigen (PA).

(139) Pharmaceutical Compositions

(140) The compositions disclosed herein may include pharmaceutical compositions comprising the presently disclosed bacterial toxins (RRSP) and formulated for administration to a subject in need thereof. Such compositions can be formulated and/or administered in dosages and by techniques well known to those skilled in the medical arts taking into consideration such factors as the age, sex, weight, and condition of the particular subject, and the route of administration.

(141) The compositions may include pharmaceutical solutions comprising carriers, diluents, excipients, and surfactants, as known in the art. Further, the compositions may include preservatives (e.g., anti-microbial or anti-bacterial agents such as benzalkonium chloride). The compositions also may include buffering agents (e.g., in order to maintain the pH of the composition between 6.5 and 7.5).

(142) The pharmaceutical compositions may be administered therapeutically. In therapeutic applications, the compositions are administered to a subject in an amount sufficient to elicit a therapeutic effect (e.g., a response which cures or at least partially arrests or slows symptoms and/or complications of disease (i.e., a “therapeutically effective dose”)).

EXAMPLES

(143) The following examples are illustrative and are not intended to limit the scope of the claimed subject matter.

Example 1—A Bacterial Toxin that is a Ras-Specific Protease

Background

(144) *Vibrio vulnificus* is a motile, Gram-negative, opportunistic human pathogen capable of causing severe gastrointestinal and wound infections, both of which can be fatal. Two major virulence factors have been identified associated with increased death during intestinal infection: the secreted cytolytic/hemolysin pore-forming toxin encoded by vvhA [3] and the multifunctional autoprocessing RTX (MARTX.sub.Vv) toxin encoded by gene rbcA1 [4-6]. However, results among different studies suggest that MARTX.sub.Vv is the most significant virulence factor of *V. vulnificus* [7].

(145) MARTX toxins are a recently described family of bacterial protein toxins originally characterized in *Vibrio cholerae*, but subsequently identified in many bacterial species [8][9, 10][6, 11][12]. These are large composite bacterial toxins that carry multiple effector domains that confer cellular toxicity after delivery by autoprocessing [9]. MARTX N- and C-termini repeats region are proposed to form a pore at the eukaryotic cell membrane for translocation of central “effector-domains” to the cytosol. Within the cytosol, the cysteine protease domain (CPD, covered by U.S. Pat. No. 8,257,946,B2) binds inositol hexakisphosphate and other inositol phosphate compounds, to initiate autoprocessing at leucine residues located in unstructured regions that link the “effector domains” [13-15]. The net result is release of the internal effector domains from the large protein toxin to the cytosol, where they are free to move throughout the cell to access cellular targets and to

exert their toxic effects (FIG. **1**).
(146) Despite the sequence conservation of the repeats regions and the CPD in proteins produces by different bacteria, each bacterial MARTX toxin carries a distinct set of effector domains and thus a distinct array of potential cytotoxic activities [8, 9]. Further, different isolates of the same species can produce MARTX toxins that deliver distinct repertoire of effectors [6, 10, 12].
(147) The first *V. vulnificus* MARTX toxin that was annotated was identified in the clinical isolates CMCP6 [8]. The central region of MARTX.sub.Vv CMCP6 (NP_759056.1) has five effector domains (FIG. **1**A). Domain of unknown function in the first position (DUF1) has no functional homologs in the database, but is found also in MARTX toxins from *Xenorhabdus bovienii* and *Xenorhabdus nematophila* [8, 9] The second effector domain is Rho-inactivation domain (RID). This domain has been demonstrated to stimulate cell rounding by inactivating cellular Rho GTPases dependent upon a catalytic cysteine residue [16, 17] The third effector domain has homology to the αβ-hydrolase (ABH) family of enzymes [8, 9] and has recently been shown to have phospholipase activity (Agarwal SN and Satchell, manuscript in preparation). The fourth effector domain is 30% identical to a domain found within the *Photorhabdus luminescens* Makes Caterpillar Floppy (MCF) toxins [8, 9]. This domain is associated with induction of apoptosis (Agarwal S G and Satchell, manuscript in preparation).
(148) DUF5 is the fifth effector domain in the toxin produced by *V. vulnificus* CMCP6 (DUF5.sub.Vv), but absent in other isolates. Our group demonstrated that an in-frame genetic mutation on the chromosome of CMCP6 to remove DUF5.sub.Vv from expressed MARTX.sub.Vv toxin results in a 54-fold reduced virulence, compared with the isogenic strain CMCP6 that expresses the full-length toxin. In addition, a strain that naturally lacks this domain was at least 10-fold less virulent than CMCP6 [6]. Our interest in this domain was rooted in this identification that the presence of DUF5.sub.Vv in the MARTX toxin of *V. vulnificus* is associated with the more highly virulent nature of *V. vulnificus* CMCP6 so we ventured to understand the molecular mechanism of action of this domain.
(149) Details on the Discovery of the Catalytic Activity of DUF5.sub.Vv and Related Proteins as Specific Endopeptidases for the Small GTPase Ras.
(150) DUF5.sub.Vv Represents a Family of DUF5-Like Proteins.
(151) The DUF5 domain of the *V. vulnificus* CMCP6 MARTX toxin (DUF5.sub.Vv) is found at amino acids G3579-L4089 based on Genbank sequence NP_759056.1. Domains with similarity to DUF5.sub.Vv are also found in MARTX toxins of at least 8 other bacterial species with amino acid identity varying from 43-98% identity.
(152) DUF5.sub.Vv Homologs in Other Bacteria (% Amino Acid Identity)
(153) TABLE-US-00002 Organisms with DUF5 homolog sequences % Identity *Vibrio ordalii* 97.8 *Vibrio choleras* 97.2 *Vibrio splendidus* 81.2 *Moritella dasanensis* 71.6 *Aeromonas salmonicida* 61.9 *Aeromonas hydrophila* 61.8 *Photorhabdus temperata* 58.8 *Xenorhabdus nematophila* 58.2 *Photorhabdus luminescences* 56.9 *Photorhabdus asymbiotica* 56.0 *Yersinia kristensenii* 42.5 *Pasteurella multocida* 24.4
(154) The domain is found also in *Photorhabdus* sp. as a single domain hypothetical protein with 56-59% amino acid sequence identity to DUF5.sub.Vv ([9] and search conducted for this document). DUF5.sub.Vv also has 24% amino acid sequence identity to a portion of the *Pasteurella multocida* toxin (PMT) [9].
(155) The solved structure of the C-terminus of PMT (PDB 2EBF) revealed three independent domains termed C1, C2, and C3 [18, 19]. The C3 domain is the catalytically active domain of PMT [20] and is not conserved in DUF5.sub.Vv. The C1 domain in DUF5.sub.Vv, PMT, and other bacterial toxins that have a homologous domain, has been shown to be a four helical bundled structure necessary for targeting toxin proteins to the cytosolic side of eukaryotic membranes [18, 21-24]. No function has been identified for the C2 domain of PMT. Transfection studies reveal this domain is not toxic when ectopically expressed in cells and bioinformatics comparing

DUF5.sub.Vv homologs suggest accumulated mutations in C2 may have rendered this domain inactive [39,40]. Thus, at the start of the project, there was no functional information regarding the activity of the C2 domain of DUF5.sub.Vv or any of its protein homologues.
(156) Structure of DUF5.sub.Vv.
(157) Recombinant DUF5.sub.Vv (rDUF5.sub.Vv; MARTX.sub.Vv Q3596-L4089 based on sequence NP_759056.1) was amplified from CMCP6 DNA and cloned into the expression vector pMCSG7 [25] to generate a fusion with a 6×HIS tag at its N-terminus for binding to a nickel column for affinity purification. The protein was expressed in *E. coli* and lysate prepared by sonication and centrifugation to recover the soluble fraction. rDUF5.sub.Vv was purified from the lysate by affinity chromatography using pre-packed GE Biosciences HisTrap FF resin and then by size exclusion chromatography using a pre-packed GE Biosciences Superdex S200 resin.
(158) This rDUF5.sub.Vv protein preparation was used for X-ray crystallography studies (FIG. **2**). rDUF5.sub.Vv structure was solved with an overall resolution of 3.4 Å. The overall structure of the protein aligns with the previously determined structure of PMT C1/C2 domains (RMSD=2.75) despite the fact that the two proteins share only 24% amino acid identity. The solved structure revealed that rDUF5.sub.Vv as predicted by secondary structure alignment is composed also of C1 (aa 3579-3669) and C2 domains (amino acids 3670-4089). The C2 domain could likewise be bisected into two subdomains: C2A (amino acids 3669-3855) and C2B (amino acids 3856-4089). Bioinformatics studies had also predicted two subdomains for C2 but predicted the active catalytic activity would be focused on C2B[39,40].
(159) The C2A Subdomain is the Cytotoxic Portion of DUF5.sub.Vv.
(160) To probe whether DUF5.sub.Vv has cytotoxic or cytopathic activity, the DNA sequence from *V. vulnificus* CMCP6 corresponding to DUF5.sub.Vv (amino acids 3579-4089) was amplified by PCR, cloned in the pEGFP-N3 (Clontech Laboratories Inc.) to generate a fusion with green fluorescent protein gene (egffi) and the resulting plasmid chemically transfected into HeLa cells. These studies showed rounding of cells that were expressing the EGFP fusion protein, but not control cells that were expressing EGFP alone. C2 also induced cell rounding when expressed in the eukaryotic yeast *Saccharomyces cerevisiae*. The minimal portion of DUF5.sub.Vv that demonstrated the cytopathic activity in HeLa cells was linked to the C2A domain by deletion analysis.
(161) To further demonstrate that DUF5.sub.Vv is toxic to cells, the DNA sequence from *V. vulnificus* CMCP6 corresponding to DUF5.sub.Vv (amino acids 3579-4089) was amplified by PCR, cloned into pRT24 (a modified version of pABII [42]) to generate a fusion with 6×His-tagged anthrax toxin lethal factor N-terminus (LF.sub.N) at the N-terminus. This protein LF.sub.NDUF5.sub.Vv can be delivered to the cytosol of cells by adding the purified protein to the cell culture media along with anthrax toxin protective antigen (PA), which is purified separately as a 6×His-tagged protein. The PA portion of the bipartitite anthrax toxin associates with the LF.sub.N portion of the fusion protein and LF.sub.NDUF5.sub.Vv is then translocated into the cell cytosol by PA, allowing for delivery of DUF5.sub.Vv to the cell cytosol independent of the remainder of the MARTX toxin. This intoxication system has been used for the study of many bacterial toxins and other proteins [16,41-44]. Several embodiments of the Lethal Factor/Protective antigen translocation system have been described (WO 2014031861 A1, WO 2001/21656 and WO2008/076939).
(162) Cells intoxicated with LF.sub.NDUF5.sub.Vv in combination with PA exhibited cell rounding, including HeLa cervical carcionoma cells, J774 macrophages, 293T fibroblasts, etc. Cells were not rounded by LF.sub.NDUF5.sub.Vv in the absence of PA or by PA in combination with purified LF.sub.N alone (without the DUF5.sub.Vv) The minimal portion of DUF5.sub.Vv essential for cytoxicity when delivered by LF.sub.N was mapped to MARTX.sub.Vv G3579-T3855 corresponding to the C1 domain plus C2A, as C1 is essential for toxin to reach the membrane after delivery though the PA pore.

(163) Similarly, it was found that cells treated with LF.sub.N fused to the DUF5 domain from the *Aeromonas hydrophila* MARTX toxin (aa 3041-3575 based on sequence strain 7966, from ATCC, GI: 117618727) also demonstrated cell rounding when delivered to cells and only when in combination with PA. Thus, despite having only 62% amino acid identity, these proteins seem to share a toxic mechanism.
(164) Discovery of DUF5.sub.Vv Targeting Ras.
(165) To further investigate the cytopathic function of DUF5.sub.Vv, a screen was conducted for suppressors in yeast that would permit growth of yeast when DUF5.sub.Vv C2 subdomain was ectopically expressed. This screen revealed>100 suppressor mutations that mapped to a plethora of cellular signaling pathways, enriched in pathways linked to cellular stress responses. Based on this finding, we investigated if the major transcription factor activated under conditions of cell stress in human epithelial cells—ERK1/2—would be affected by DUF5.sub.Vv accounting for the observed wide variety of downstream effects in yeast. Cells intoxicated with LF.sub.NDUF5.sub.Vv in the presence of PA were found to have reduced levels of phosphorylated ERK1/2 (pERK1/2) (FIG. **3**A, lower panels), despite having no difference in total levels of ERK1/2 (FIG. **3**A, upper panels). This result demonstrated that DUF5.sub.Vv does suppress the stress response pathways in cells controlled by ERK1/2.
(166) The relative concentration of active pERK1/2 in the cell is normally regulated by the Ras-Raf-MEK-ERK signaling cascade. At the top of the cascade, Ras is activated by conversion from an inactive, GDP-bound state to an active, GTP-bound state [26, 27]. As pERK1/2 levels were reduced, this led us to test if the activation state of Ras in LF.sub.NDUF5.sub.Vv intoxicated HeLa cells was affected. The G-LISA activation assay, commercially available from Cytoskeleton, Inc., specifically detects the GTP-bound activated form of all Ras isoforms dependent upon the final detection of Ras by the monoclonal antibody Ras10. This assay detected no active Ras-GTP in intoxicated cells (FIG. **3**B). Western blotting, using the Ras10 monoclonal antibody to detect the total amount of Ras within the HeLa whole cell lysate, showed that Ras was absent from lysates prepared from cells intoxicated with LF.sub.NDUF5.sub.Vv or LF.sub.NC1C2A exclusively when incubated in the presence of PA (FIG. **3**C). HeLa cells intoxicated for shorter time points than 24 hours revealed that loss of Ras detectable by the Ras10 monoclonal antibody occurred as early as 20 minutes from the time of exposure of cells to LF.sub.NDUF5.sub.Vv in the presence of PA (FIG. **4**).
(167) Following a strategy similar to DUF5.sub.Vv, we demonstrated that the HeLa cells intoxication with an LF.sub.N fusion of DUF5 from MARTX (LF.sub.NDUF.sub.Ah) (amino acid 3069-3570) also causes cell rounding after 24 hours. Western blot analysis was used to detect the total amount of Ras into the HeLa cell lysates demonstrated that Ras was undetectable into the HeLa cell lysate after intoxication with DUF5.sub.Ah (FIG. **5**), similar to DUF5.sub.Vv intoxicated cells. This demonstrates that DUF.sub.Ah has potentially a similar mechanism as DUF5.sub.Vv.
(168) Ras is Truncated in DUF5.sub.Vv Intoxicated Cells.
(169) The failure to detect all forms of Ras by the Ras10 antibody due to intoxication of cells by DUF5.sub.Vv and DUF5.sub.Ah was initially thought to be due to a covalent modification of Ras that disturbed the detection of Ras by the antibody. Other bacterial toxins are known to target Ras in this way. These include *Pseudomonas aeruginosa* ExoS that can ADP-ribosylate Ras (U.S. Pat. No. 5,599,665 A), but also modifies up to 20 other cellular proteins [28-30]. *Clostridium sordelli* TcsL or *Clostridium perfringens* TpeL are monoglucosyltransferases that can UDP-glucosylates Ras (Patent EP 0877622 B1), but also modifies many other small GTPase proteins like Rac [31-33] [45]. Similarly, *C. difficile* TcdA and TcdB show UDP-glucosylation modification of many small GTPases including Ras [45]. The recently revealed lack of specificity of these proteins has made them poor candidates for toxins that would attack Ras when developed as toxin therapeutics.
(170) As a first step to identify the modification in Ras that prevented detection with the Ras10 monoclonal antibody in cells intoxicated with DUF5.sub.Vv, HeLa cells were transiently

transfected to express the H-Ras isoform with a hemagglutinin (HA) tag (sequence YPYDVPDYA, SEQ ID NO:29) fused at the N-terminus for detection by the HA peptide monoclonal antibody. These cells were intoxicated for 24 hr LF.sub.NDUF5.sub.Vv in the presence of PA after which the HA-HRas protein was immunoprecipitated from cell lysate with using agarose beads conjugated with to the anti-HA peptide monoclonal antibody. The proteins specifically bound to the beads were eluted 3 M sodium thiocyanate solution and separated on an SDS-polyacrylamide gel. Coomassie brilliant blue staining of the gel revealed a 22 kDa protein band corresponding to HA-HRas for the unintoxicated Hela cells sample. However, for the intoxicated cells, an 18 kDa protein band was evident (FIG. **6**A). Subsequent analysis of the trypsin-digested excised protein band by mass spectrometry revealed that the 18 kDa band was HRas, but absent the N-terminus (FIG. **6**B). Western blot analysis, using anti-HA peptide monoclonal antibody and an anti-HRas polyclonal antibody specific for the C-terminus, confirmed that the 18 kDa band is HRas but truncated to remove the HA tag and the N-terminus of HRas (FIG. **6**C). This result indicated that DUF5.sub.Vv is either an endopeptidase or activates a previously unknown host cell endopeptidase that targets the N-terminus of H-Ras.
(171) In addition, cells were transfected with plasmids to express HA-tagged versions of KRas, NRas, and HRas. All 3 isoforms of Ras were susceptible to cleavage in vivo resulting in truncated proteins that are not detected by the anti-HA antibody, but are detectable by isoform specific antibodies that recognize the unique C-terminus of each of the isoforms (FIG. **7**).
(172) DUF5.sub.Vv is Itself an Endopeptidase that Targets Ras.
(173) To test if DUF5.sub.Vv is itself an endopeptidase that targets Ras isoforms rather than an activator of a host protease, gene sequences for KRas (KRas4B NP_004976.2), HRas (NP_001123914.1) and NRas (NP_002515.1) were cloned into pMCSG7 vector for *E. coli* expression with a 6×His tag at the N-terminus for nickel affinity purification. Recombinant rKRas and rHRas were purified from *E. coli* cell lysates using a pre-packed GE Biosciences His Trap FF column for single step NiNTA affinity chromatography. Recombinant NRas (rNRas) was expressed in inclusion bodies. The protein was therefore recovered from the insoluble fraction by suspension in buffer containing urea, purified by single step purification with NiNTA, and then rNRas refolded in the presence of excess GDP. rKRas, rHRas and rNRas were tested in vitro as substrate for rDUF5.sub.Vv (previously purified for crystallography studies described above) for an endopeptidase assay. The reaction products were analyzed by SDS-PAGE showed the cleavage of rKRas, rHRas, and rNRas by rDUF5.sub.Vv. (FIG. **8**). The cleavage of KRas was shown to occur regardless of the presence of guanosine nucleosides (FIG. **9**). The cleavage of rNRas was less efficient compared to rKRas and rHRas, but this was likely due to the requirement to refold the protein resulting in a mixed pool of proper and improper folded substrate rather than a preference for substrate as there was no difference in substrate specificity in vivo (FIG. **7**). Cleavage products for each reaction were analyzed by Edman degradation for N-terminal sequencing. The results revealed that DUF5 protein specifically cleaves KRas, HRas and NRas between residues Y32 and D33. (FIG. **10**). These two residues are in the middle of Switch I region of KRas. Overall, these results confirm that rDUF5.sub.Vv is itself an endopeptidase able to cleave all common isoforms of Ras in vitro without host cell cofactors.
(174) DUF5 Endopeptidase Activity in *Aeromonas hydrophila* and *Photorabdus asymbiotica*.
(175) As detailed above, DUF5.sub.Ah from the *A. hydrophila* MARTX toxin effector domain is 62% identical to DUF5.sub.Vv and induced similar phenotypes as DUF5.sub.Vv when delivered to cells in vivo. Gene sequences for DUF5.sub.Ah were cloned into pMCSG7 vector for *E. coli* expression and purified similarly to rDUF5.sub.Vv. The recombinant protein rDUF5.sub.Ah was able to cleave rKRas in the in vitro reaction (FIG. **11**) demonstrating that the same domain from a different MARTX toxin is also an endopeptidase for Ras. This result indicates these are representative members of the larger family of MARTX effectors from at least 8 MARTX toxin and that all DUF5 domains from MARTX toxins will have this activity.

(176) In addition to its presence in MARTX toxins, a hypothetical protein of *Photorhabdus* spp. (i.e. *P. asymbiotica* PAT3383 and *P. luminescens* Plu2400) has 56-59% similarity to DUF5.sub.Vv. In *Photorhabdus* spp., this hypothetical proteins is not linked to a MARTX toxin but instead is found as a stand-alone gene that encodes a 542-568 aa hypothetical protein. Recombinant PAT3383 (here known as DUF5.sub.Pa) was also successfully purified and shown to also cleave rKRas. N-terminal sequencing by Edman degradation of products excised from gel showed that all three DUF5 (DUF5.sub.Vv, DUF5.sub.Ah and DUF5.sub.Pa) cleave KRas between Y32 and D33. To our knowledge, none of the several DUF5 homologs identified has ever been characterized for its intrinsic function. DUF5.sub.Ah has been recently studied for its thermodynamic properties in the context on MARTX toxin unfolding and translocation [34].
(177) DUF5.sub.Vv Endopeptidase is Specific for Ras and does not Process Representative Members of Other Small GTPases.
(178) DUF5.sub.Vv specificity was further tested by examining cleavage of representative members of small GTPase family. Recombinant proteins for other fused Ras family members (Rit2, RalA and RheB2) and small GTPase from other Ras superfamily groups: Rab (Rab4A, Rab4B, Rab5A and Rab11A), Rho (RhoA, RhoB, RhoC, RhoG, Cdc42 and Rac1) and Ran. Each protein was individually expressed in *E. coli* fused to glutathione-S-transferase for purification on glutathione agarose. Cloning, expression and purification condition of this rGTPase library was previously reported [35]. The in vitro cleavage assay was performed incubating each purified rGST-GTPase with rDUF5.sub.Vv, rGST-HRas was used as positive control to demonstrate that the presence of GST does not interfere with the cleavage assay. The reaction products, analyzed by SDS-PAGE, showed that DUF5.sub.Vv could cleave only HRas. None of the other GTPase was cleaved by DUF5.sub.Vv (FIG. **12**). The overall results demonstrate that DUF5.sub.Vv is a novel Ras endopetidase for, which cleaves specifically KRas, HRas and NRas.
(179) DUF5 Endopeptidase Activity and Mutant KRas.
(180) In this application, we propose that the Ras-directed endopeptidase activity of DUF5.sub.Vv and homologous proteins with similar activity can be directed toward treatment of cancers. As DUF5.sub.Vv targets normal Ras to compromise the cell, it can be utilized in a vast array of cancers. However, a particular focus of this work could be to target cancers that result from mutation of Ras itself. To achieve this, cells that have Ras with amino acid substitutions must be shown to be susceptible to DUF5.sub.Vv.
(181) The cytotoxicity of DUF5.sub.Vv was tested in colorectal cancer cells (HCT116) and in breast cancer cells (MDA-MB-231). These two cells lines express, respectively, mutant KRas G12V and G13D. A dramatically morphology change was observed for HCT116 after 24 hours of intoxication with LF.sub.NDUF5.sub.Vv in the presence of PA (FIG. **13**A). The intoxicated cells showed a reduction in the number of cells and cell enlargement, suggesting swelling. In addition, the cells were observed to detach from the dish surface. MDA-MB-231 cells intoxicated with LF.sub.NDUF5.sub.Vv for 24 hours showed a more “typical” cell rounding phenotype, similar to that previously observed in HeLa cells (FIG. **13**B). With these experiments, we demonstrated the toxicity of DUF5.sub.Vv for cancer cells that are expressing mutant forms of KRas.
(182) As further evidence of its applicability to treatment of Ras cancers, recombinant mutant KRas G12V was cloned into pMCSG7 and expressed in *E. coli*. The purified rKRas G12V was incubated with rDUF5.sub.Vv to check its cleavability in vitro. The reaction products, analyzed on SDS-PAGE, showed that DUF5.sub.Vv is still able to cleave mutant KRas (G12V) (FIG. **14**).
(183) Benefits Over Other Technologies.
(184) Many bacterial toxins have been proposed for use in chemotherapy. Toxins that destroy the membrane, such as pore forming toxins have the potential to induce inflammation resulting in severe side effects. The advantage of this toxin over others is that it works from inside the cell to block normal cell survival pathways, thereby inducing loss of proliferation and normal non-inflammatory cell death.

(185) Unlike toxins that target such processes as protein translation, this toxin directly targets a central regulatory pathway that is normal altered in cancer cells to promote cell survival and is thus key to the survival of the cancer itself. Ras cancers are among the most difficult to treat cancers due to the mutations in Ras. By directly targeting Ras in these cells, we can remove the protein that is driving the survival of the cancer.

(186) A tripping point for some toxins (except those that form pores from the outside) is the ability to deliver to the cell cytosol where they can access target. We demonstrate that the DUF5 protein can be easily delivered to cells in an active form by the LF.sub.N-PA delivery system. This system has already been modified to directly target cancer cells. A problem with the LF.sub.N-PA delivery system, is that it is selective to translocate proteins that can rapidly unfold and spontaneously refold. We showed that this protein is able to cleave all molecules of Ras in cells at less then 30 minute after exposure indicating rapid translocation and delivery of active protein via the PA pore. Other delivery strategies will also require self-folding. We were able to purify this protein to homogeneity for the purpose of crystallography indicating that despite its plasticity, it is a stable protein for storage in vitro.

(187) The specificity for Ras is also a benefit. Unlike other toxins that target Ras, this protein does not as yet show any specificity outside of HRas, NRas, and KRas. It does not target other small GTPases, which is the case for the Clostridial toxins TcsL, Tpel, TcdA, and TcdB. It does not show evidence of having cellular substrates in a wide range of protein families such as *Pseudomonas* Exotoxin A. Finally, these other proteins covalently modify the substrate, which there is some evidence is reversible. By contrast, DUF5 irreversibly cleaves the Ras proteins and thus cannot be reversed by the cell. For diversity of immunogenicity and increasing efficacy and activity are at least three different family members that share this activity and these are representative of the families across a wide range of bacteria species.

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Example 2—Cytotoxicity of the *Vibrio vulnificus* MARTX Toxin Effector DUF5 is Linked to the C2A Subdomain

(189) Reference is made to Antic et al., *Proteins.* 2014 October; 82(10):2643-56, the content of which is incorporated herein by reference in its entirety.

(190) Abstract

(191) The multifunctional-autoprocessing repeats-in-toxin (MARTX) toxins are bacterial protein toxins that serve as delivery platforms for cytotoxic effector domains. The domain of unknown function in position 5 (DUF5) effector domain is present in at least six different species' MARTX toxins and as a hypothetical protein in *Photorhabdus* spp. Its presence in *Vibrio vulnificus* MARTX toxin increases potency of the toxin in mouse virulence studies, indicating DUF5 contributes to pathogenesis. In this work, DUF5 is shown to be cytotoxic when transiently expressed in HeLa cells. DUF5 localized to the plasma membrane dependent upon its C1 domain and the cells become rounded dependent upon its C2 domain. Both full-length DUF5 and the C2 domain caused growth inhibition when expressed in *Saccharomyces cerevisiae*. A structural model of DUF5 was generated based on the structure of *Pasteurella multocida* toxin facilitating localization of the cytotoxic activity to a 186 amino acid subdomain termed C2A. Within this subdomain, alanine scanning mutagenesis revealed aspartate-3721 and arginine-3841 as residues critical for cytotoxicity. These residues were also essential for HeLa cell intoxication when purified DUF5 fused to anthrax toxin lethal factor was delivered cytosolically. Thermal shift experiments indicated that these conserved residues are important to maintain protein structure, rather than for catalysis. The *Aeromonas hydrophila* MARTX toxin DUF5.sub.Ah domain was also cytotoxic, while the weakly conserved C1-C2 domains from *P. multocida* toxin were not. Overall, this study is the first demonstration that

DUF5 as found in MARTX toxins has cytotoxic activity that depends on conserved residues in the C2A subdomain.

Introduction

(192) Multifunctional-autoprocessing repeats-in-toxins (MARTX) toxins are large protein toxins (3500-5300 aa) secreted by Gram-negative bacteria.sup.1. These toxins carry from 1 to 5 protein effector domains, but also function as a delivery platform for transfer of these effector domains across the eukaryotic cell plasma membrane. These domains are then excised from the holotoxin by autoprocessing and released to the eukaryotic cell cytosol.sup.2-4 where they function as “effectors” freed from the translocation system of the toxin.sup.2-4. Among the various MARTX toxins of different mammalian, aquatic, and insect pathogens, a total of 10 different effector domains are carried by MARTX toxins, although the number and positional organization of the arrayed effectors vary across strains and species.sup.1. The effector domain repertoire of the toxins can be exchanged by uptake of exogenous DNA and incorporation of the new sequences and/or loss of old sequences by homologous recombination resulting in novel toxins in different strains of the same species.sup.5,6.

(193) Within the target cell, the effector domains are thought to each have cytopathic or cytotoxic activity such that the overall role of the toxin in the eukaryotic cell is the sum of the activities of the effectors it delivers. Thus, it is important to individually characterize the function of each effector using genetics, biochemistry, and cell biology approaches to understand how an effector exchange will affect bacterial pathogenesis.

(194) Among the 10 MARTX effector domains identified by sequence comparisons, only three have been functionally characterized.sup.1. The actin crosslinking domain (ACD) covalently links actin monomers via an isopeptide bond leading to actin cytoskeletal destruction.sup.7-10. The Rho GTPase inactivation domain (RID) disables the Rho regulatory pathway resulting in loss of active Rho and thereby to cytoskeleton depolymerization.sup.11,12. The ExoY domain is an adenylate cyclase.sup.13. The remaining seven MARTX toxin effector domains are uncharacterized but are often similar to domains of other large protein toxins.sup.1.

(195) One of the domains of unknown function is known as DUF5, indicating its presence in the 5th effector domain position of the *Vibrio vulnificus* strain CMCP6 MARTX toxin where it was first recognized.sup.14 (holotoxin diagrammed in FIG. **15**A). Within *V. vulnificus*, the presence of DUF5.sub.Vv increases the potency of the toxin during mouse infection resulting in a lower LD.sub.50 compared to an isogenic strain from which the effector domain was deleted or a naturally occurring strain that lost DUF5.sub.Vv via a homologous recombination events. Thus, DUF5.sub.Vv is a virulence factor that increases the pathogenicity of the strains that carry it as a domain within the MARTX toxin.

(196) DUF5.sub.Vv was initially recognized to have sequence similarity to *Pasteurella multocida* toxin (PMT), whose carboxyl-terminus is composed of three domains: C1, C2, and C315. The C1Pm subdomain from PMT is known to be a four helical bundled membrane localization domain (4HBM).sup.16. The conserved C1.sub.Vv subdomain from DUF5.sub.Vv has also been demonstrated to localize to the eukaryotic plasma cell membrane, where it binds anionic lipids via a basic-hydrophobicmotif.sup.12,17. Structural determination by nuclear magnetic resonance of the isolated C1Pm and C1.sub.Vv subdomains confirm both of these domains form a four helical bundle in solution.sup.18,19.

(197) However, none of the extensive characterization of PMT has revealed the function of its C2 domain. The PMT C3 domain is a deamidase enzyme with a catalytic cysteine residue that acts on the Gα subunits of trimeric G proteins.sup.20-23. It is notable that the sequence similarity of DUF5.sub.Vv with PMT is limited to the C1 and C2 domains and DUF5.sub.Vv does not share the C3 deamidase domain and thus DUF5.sub.Vv is not expected to have a similar activity (FIG. **15**A). DUF5 is present also within MARTX toxins of *Aeromonas hydrophila*, *Yersinia kristensenii*, *Vibrio splendidus*, and *Xenorhabdus nemotophila*1 and as the stand-alone hypothetical protein plu2400 in

Photorhabdus sp.,.sup.24 where it might be an effector with a distinct delivery mechanism such as Type III secretion or the Tc complex.sup.25.

(198) In this study, we initiated a de novo investigation on this protein of unknown function. We generated a structural model of DUF5.sub.Vv based on the structure of the PMT C-terminus.sup.15. We then show that ectopic expression of the domain in HeLa cells is cytotoxic. In *Saccharomyces cerevisiae*, expression of the DUF5.sub.Vv causes growth inhibition. The toxic effect in HeLa cells is mapped to a 186 amino acid C2A subdomain and shown to require an Asp and Arg residue. Overall, these studies mark our initial efforts to establish that DUF5 is a bona fide MARTX toxin effector.

Materials and Methods

(199) Cell Lines, Media, Reagents and Plasmids

(200) HeLa epithelial cells were grown at 37° C. with 5% CO.sub.2 in Dulbecco's Modified Eagle Medium (DMEM, Life Technologies Gibco) with 10% fetal calf serum (Gemini Bio-Products, West Sacramento, CA), 100 U/ml penicillin, and 1 μg/ml streptomycin. J774 macrophages, COS7 fibroblasts, and HEp-2 epithelial cells were grown in identical conditions. *E. coli* DH5a, TOP10 (Life Technologies Invitrogen) and BL21(DE3) were grown at 37° C. in Luria-Bertani (LB) liquid or agar medium containing either 100 μg/ml ampicillin or 50 μg/ml kanamycin as needed. *S. cerevisiae* strain InvSc1 (Invitrogen) was grown on YPD liquid or agar medium at 30° C. or commercial synthetic complete (SC-ura) supplemented with yeast nitrogen base (MP Biomedicals) as detailed below. Media components and common reagents were obtained from Sigma-Aldrich, Fisher, or VWR and common restriction enzymes and polymerases from New England Biolabs or Invitrogen. Custom DNA oligonucleotides and gBlocks were purchased from Integrated DNA Technologies (Coralville, IA). Plasmids were prepared either by alkaline lysis with precipitation in ethanol or purified using Epoch spin columns according to manufacturer's recommended protocol.

(201) Alignments and Structural Modeling

(202) Proteins with homology to DUF5.sub.Vv from strain CMCP6 were identified using BLASTP26 at the National Center for Biotechnology Information website. Amino acid sequences were trimmed to DUF5.sub.Vv homology region and aligned with CLUSTALW using MacVector 12.6.0. The DUF5.sub.Vv and DUF5.sub.Vv D3721A protein sequences were also aligned to the pdb database using HHpred27 and a pdb structural model built based on published PMT structure (pdb 2EBF15) using Modeller 28. Figures were generated from the structural model using MacPyMol.

(203) Construction of Plasmids for Ectopic Expression in HeLa Cells and Yeast

(204) DNA corresponding to coding sequence for amino acids 3579-4089 of the *V. vulnificus* rtxA1 gene (GI: 27366913; vv2_0479) was amplified from purified *V. vulnificus* CMCP6 chromosomal DNA using Pfx50 DNA polymerase (Invitrogen) and primers 1 and 2 (5′-gagctagcatgggtgataaaaccaaggtcgtggattta (SEQ ID NO:55), and 5-gccgtcgaccaaactgcccttgaacgtgatcttcggttt (SEQ ID NO:56)). The insert was digested with enzymes NheI and SalI and ligated into the similarly digested vector. For expression of the *P. multocida* toxA gene sequence corresponding to aa 573-1113 of PMT from strain 4533 (GI:149228008), a synthetic codon optimized double stranded DNA sequence was obtained from GenScript (Piscataway, NJ) in pUC57 and subcloned into pEGFP-N3 via the HindIII and BamHI sites.

(205) DNA sequences above were similarly amplified except with novel EcoRI and KpnI restriction sites incorporated into the oligonucleotides for transfer into yeast expression vector pYC2 NT/A (Invitrogen) using primers 6,7,8 and 11 (5′-aaggtaccgtttatcggtaagatgcaagttgcc (SEQ ID NO:30), 5′-agaattctcacaaactgcccttgaacgtgatc (SEQ ID NO:31), 5′-aaggtaccgggtgataaaaccaaggtcgtg (SEQ ID NO:32), and 5′-aaggtaccggatattgacgcttgggatcgt, SEQ ID NO:33).

(206) Ectopic Expression of EGFP Fusion Proteins in HeLa Cells

(207) Plasmids for transfection were prepared using the Qiagen Midi Prep kit. HeLa cells grown to

approximately 80% confluency were transfected using Fugene HD (Promega) and 2 μg plasmid DNA at a 4:1 ratio for 5 h after which fresh media was exchanged. For western blotting, cells were collected 24 h after transfection in 2×SDS-PAGE buffer, boiled for 5 min, and the proteins were separated by SDS-PAGE and transferred to nitrocellulose by the tank blot method. Western blotting was done as previously described17 using anti-GFP antibody conjugated to horseradish peroxidase (Milteny Biotec 130-091-833) at a 1:1000 dilution. SuperSignal West Pico Chemiluminescent Substrate and autoradiography was used for detection.

(208) For microscopy, HeLa cells were grown in 35 mm MatTek glass bottom microwell dishes to approximately 60% confluency and transfected as described above. Live cells were imaged 24 h after transfection by epifluorescence and differential interference contrast (DIC) microscopy at 200× using the Andor Spinning Disk confocal microscope. Images were overlayed using NIH ImageJ64 and assembled into figures using Adobe Photoshop CS6. Rounded cells were manually counted from at least 3 different transfections. Histograms of representative cells were plotted using GraphPad Prism 4.0 or 6.0.

(209) Purification of Proteins Fused to Anthrax Toxin Lethal Factor N-Terminus

(210) Plasmid vector pRT24 is a variant of pABII29 in which the coding sequence for amino acids 1-254 of anthrax toxin lethal factor (LF.sub.N) are expressed with an N-terminal His-tag under control of the T7 promoter. The plasmid was modified to replace the single BamH1 cloning site with an oligonucleotide that introduces the TEV cleavage site and ligation independent cloning site from pMCSG730. DUF5.sub.Vv DNA sequences were amplified with primers 3 and 4 (5′-tacttccaatccaatgctgataaaaccaaggtcgtggtcgattta (SEQ ID NO:58) and 5′-ttatccaatgtgaaagagcggtatttgcgccactcaa (SEQ ID NO:59)) and integrated into pRT24 by ligation-independent cloning.sup.30. A stop codon after the codon for Thr3765 was introduced to generate a sequence that would be truncated after C2A. Site-directed mutagenesis was then used to alter codons D3721 and R3841 to Ala as described above. DUF5 from *A. hydrophila* fused to LFN (LFN-DUF5.sub.Ah) was generated in the same manner as LFNDUF5.sub.Vv except using primers 14 and (5′-tacttccaatccaatgctccgggcaaaacggtggtgacg (SEQ ID NO:39), and 5′-ttatccacttccaatgctagacatcggcgtactcgacccgc (SEQ ID NO:40)) to amplify the sequence corresponding to the MARTX toxin aa 3069-3570 (GI: 117618727) from chromosomal DNA prepared from *A. hydrophila* 7966 obtained from the American Type Culture Collection.

(211) LFN and LFN fusion proteins were expressed in *E. coli* BL21 (DE3). Briefly, overnight cultures were diluted 1:100 in fresh LB containing the 100 μg/ml ampicillin and grown to OD600=1.0 at 37° C. before inducing the cultures with 1 mM isopropyl β-D-1-thiogalactopyranoside (IPTG) for 4 h at 32° C., except for LF.sub.NDUF5.sub.Vv expression which was induced with 0.5 mM IPTG and cells were grown at 30° C. *E. coli* cells were harvested by centrifugation at 10,100×g and resuspended in 100 ml Urea Buffer A (500 mM NaCl, 20 mM Tris pH 7.4, 5 mM imidazole and 8 M urea with addition of protease inhibitor tablets (cOmplete, EDTA-free purchased Roche Applied Sciences). Resuspended cells were sonicated using a Branson Sonifier model 102C for 7 min at 50% amplitude with the standard disruptor. Crude lysates were centrifuged at 17,600×g for 30 min to remove particulates and the remainder of the lysate was filtered across a PALL Acrodisc 0.45μ syringe filter. Lysate was loaded onto a 1 ml GE Healthcare HisTrap column using the AKTA purifier protein purification system (GE Healthcare). Column was washed with 5 ml Urea Buffer A with 10 mM imidazole, followed by 5 ml 50 mM imidazole buffer to remove contaminating proteins. His-tagged LFN proteins were eluted using an imidazole gradient from 50 to 250 mM. Peak fractions corresponding to the protein of interest were collected, pooled, and dialyzed to remove imidazole into a buffer containing 500 mM NaCl, 20 mM Tris, and 2 M urea, pH 7.4. Proteins were further purified by gel exclusion chromatography in the same buffer using a 16×100 Superdex 200 column (GE Healthcare). Purified proteins were concentrated using Millipore Amicon Ultra 30K spin concentrators and glycerol was added so that the final buffer was 300 mM NaCl 12 mM Tris pH 7.4, 1.2M urea, 20% glycerol. Protein concentration was

determined using the NanoDrop ND1000, and purity was estimated using SDS-PAGE. Proteins were stored at −80° C. until used.
(212) Protective antigen (PA) was purified from the soluble fraction of *E. coli* BL21(DE3). Cells were grown at 37° C. to OD600=0.8, then the culture was induced with 1 mM IPTG for 4 h at 30° C. Bacterial culture was harvested by centrifugation, then resuspended in 500 mM NaCl, 20 mM Tris, 5 mM imidazole, pH 8.0. Lysate was prepared as for LF.sub.N fusion proteins above except buffers did not contain urea. Sizing was performed as described above in 500 mM NaCl, 20 mM Tris pH 8.0 buffer.
(213) Intoxication of Mammalian Cells with LF.sub.N Fusion Proteins and PA
(214) All cell types were grown in 24 well tissue culture treated dishes (Falcon). 7 nM PA and 3 nM LF.sub.N-fusion proteins were added to 1 ml culture media overlaying the cells. Cells were incubated for 24 or 48 h at 37° C. in 5% CO.sub.2, after which cells were imaged at 100× by phase microscopy using a Nikon CoolPix 995 digital camera affixed to a Nikon TS Eclipse 100 microscope. For quantification, rounded cells were manually counted representing at least 3 independent experiments and results were graphed as histograms using GraphPad Prism 4.0 or 6.0.
(215) Assay for Cell Lysis
(216) Lactate dehydrogenase (LDH) release from intoxicated cells was determined using the Cytotox 96 Non-Radioactive Cytoxicity Assay (Promega). After intoxication, 50 μl of culture media was removed from each well, mixed with 50 μl of reaction reagent, and incubated at room temperature protected from light for 30 min. Upon addition of stop solution, absorbance was measured at 490 nm. For determination of total LDH, cells from the same wells were lysed by addition of Triton X-100 to the residual media to a final concentration of 0.1% and then sampled and assayed as described above to determine the maximum lysis value for each well. Percent cell lysis was calculated using the formula
(217) $\left(\frac{A490\text{media}}{(A490\text{media} + \text{cells})}\right)\text{exp} - \left(\frac{A490\text{media}}{(A490\text{media} + \text{cells})}\right)\text{untreated} \star 100.$
Assessment of Yeast Growth Inhibition
(218) *S. cerevisiae* strain InvSc1 was grown in YPD broth prior to transformation. Yeast cells were transformed using a PLATE solution method and transformants selected using SC agar medium without uracil, supplemented with glucose as previously described.sup.31. Transformed yeast cells were inoculated into liquid glucose synthetic complete medium (without uracil) and grown overnight at 30° C. The next day, cultures were centrifuged and washed three times with sterile water. Each sample was resuspended in water and OD.sub.600 was measured for each using Beckman Coulter DU530 Spectrophotometer. All samples were normalized to OD.sub.600=0.5 and then were 10-fold serially diluted. 5 μl of each dilution was spotted on solid agar selective medium (−uracil) with either 20 mg/ml glucose or 20 mg/ml galactose and 10 mg/ml raffinose. The plates were incubated at 30° C. for 3 days before growth was assessed and plates photographed using a digital camera. For growth cures, OD.sub.600 of overnight cultures was measured and inoculi were normalized to each other and then diluted into 50 ml of SC medium containing 20 mg/ml galactose and 10 mg/ml raffinose (-uracil) to induce expression from the plasmid. OD.sub.600 was measured every 2 h for 12 h to document growth patterns.
(219) Alanine Scanning Mutagenesis
(220) Site-directed mutagenesis to introduce an alanine or stop codon at locations noted in text was carried out using PfuTurbo DNA polymerase (Invitrogen) and custom oligonucleotides designed via Agilent PrimerDesign software. After amplification, DNA was treated with DpnI and transformed to *E. coli* TOP10. Isolated plasmids were sequenced to confirm gain of the desired mutation and to check for absence of unintended mutations during DNA amplification. Double mutant D3721R/R3841D in pYC-DUF5 plasmid was generated by cohesive end cloning of a synthetic DNA gBlock containing the R3841D mutation in exchange for the wild type sequence via flanking BamHI and AatII restriction enzyme sites (5′-
atctttatggtcgcgattgaagaagccaacggtaaacacgtaggtttgacggacatgatggttcgttgggccaatgaagaaccatacttg

gcaccgaagcatggttacaaaggcgaaacgccaagtgaccttggttttgatgcgaagtaccacgtagatctaggtgagc, SEQ ID NO:34). Purification of recombinant 6×HIS-tagged proteins for fluorescence thermal shift assays DNA corresponding to DUF5.sub.Vv was inserted into the overexpression vector pMSCG7 by ligation independent cloning using primers 12 and 13, (5′-tacttccaatccaatgctcaagagctgaaagaaagagcaaaag, SEQ ID NO:35 and 5′-tacttccaatccaatgctcaagagctgaaagaaagagcaaaag, SEQ ID NO:36). Additional mutations were generated by site directed mutagenesis. Plasmids were transformed into *E. coli* BL21 (DE3) for purification. Cells were grown to OD.sub.600=0.8 at 37° C. The temperature was reduced to 18° C. and protein expression induced by the addition of IPTG to a final concentration of 1 mM. Cells were grown overnight with shaking and then harvested by centrifugation. Bacteria were resuspended in a buffer containing 50 mM Tris (pH 8.3), 500 mM NaCl, 0.1% Triton X-100, and 5 mM β-mercaptoethanol and lysed by sonication. After centrifugation at 30,000×g for 30 min, the soluble lysate was filtered through a 0.22 μm membrane and loaded onto a 1 ml HisTrap column using the ÄKTA purifier protein purification system (GE Healthcare). After washes with 50 mM Tris, 500 mM NaCl, 50 mM Imidazole pH 8.3, the proteins were eluted in the same buffer with 500 mM imidazole. Proteins were further purified by gel filtration chromatography (Superdex 75 (16/60), GE Healthcare) in buffer containing 10 mM Tris-HCl, 500 mM NaCl, 5 mM β-mercaptoethanol, pH 8.3.
(221) Fluorescence Thermal Shift Assay
(222) The experiment was performed using a 96-well thin-wall PCR plate (Axigen). 20 μl reactions consisted of 2 μM protein in a solution of 5×SYPRO orange dye (Life Technologies), 0.1 mM HEPES, 150 mM NaCl, pH 7.5. Fluorescence intensity was monitored using the StepOnePlus™ Real-Time PCR Systems (Life Technologies) instrument. Samples were heated from 25° C. to 95° C. at a scan rate of 1° C./min. Tm values were extrapolated using Protein Thermal Shift™ Assay software (Life Technologies).
(223) Results
(224) DUF5.sub.Vv, but not C1C2Pm, is Cytotoxic when Ectopically Expressed in HeLa Cells
(225) To determine if DUF5 is a bona fide effector with cytotoxic effects on cells, the DNA sequence corresponding to *V. vulnificus* aa 3579-4089 (DUF5.sub.Vv) was amplified and cloned into ectopic expression vector pEGFP-N3 for expression of DUF5.sub.Vv as a fusion to EGFP under control of the CMV promoter. The plasmid was transformed into cultured HeLa cervical carcinoma epithelial cells and EGFP-positive cells were imaged after 24 hr. Cells expressing EGFP had a normal, cuboidal shape with less than 8% of cells rounded (FIG. **15**B). By contrast, 82% of cells ectopically expressing the DUF5.sub.Vv-EGFP fusion were small and rounded and many of the cells showed signs of blebbing indicating necrosis (FIG. **15**B,D). Some cells that had not yet fully rounded or necrosed showed DUF5.sub.Vv-EGFP localized to the cell periphery, consistent with the presence of the C1 plasma membrane localization domain (FIG. **15**C). Western blot detection of the DUF5.sub.Vv-EGFP fusion showed less total protein than detected for the EGFP-expressing control cells (FIG. **15**H), indicating that expression of this fusion protein was toxic to cells and many cells expressing the DUF5.sub.Vv-EGFP may have detached.
(226) DUF5.sub.Vv has 24% sequence identity with the C1-C2 domains of PMT (C1C2Pm) (FIG. **15**A). Since the toxA gene is carried on a bacteriophage with a low GC content (35% GC), a eukaryotic codon-optimized, synthetic copy of toxA sequences corresponding to C1C2Pm was obtained and expressed in cells generating a protein similar in size to DUF5.sub.Vv-EGFP (FIG. **15**H). Cells expressing C1C2Pm-EGFP appeared similar to EGFP-control expressing cells (FIG. **15**F). These results support previous data.sup.20,21,32,33 that all toxic activities of PMT are due to the C3 deamidase domain that is absent in DUF5.sub.Vv. Further, these data show that the cytotoxic activity of DUF5.sub.Vv may not be conserved in C1C2Pm, at least in HeLa cells.
(227) Cytotoxicity of DUF5.sub.Vv in HeLa Cells is Linked to the C2A Domain
(228) Despite the absence of functional conservation, C1C2Pm and DUF5.sub.Vv may share

structural conservation, although the function of the domains diverged. A structural model of DUF5.sub.Vv was generated based on the PMT structures.sup.15. Based on this model, the amino acids of DUF5.sub.Vv responding to the C1.sub.Vv and C2.sub.Vv domain were identified. Upon deletion of gene sequences for the C1.sub.Vv subdomain, the C2.sub.Vv-EGFP fusion is no longer localized to the cell periphery. Those cells highly expressing C2.sub.Vv-EGFP appear rounded, while low expressing cells remained normal (FIG. **16**D). These data are consistent with C2.sub.Vv being required for cytotoxicity and C1.sub.Vv being required for efficient delivery to the plasma membrane.
(229) In addition, as shown also by two recent bioinformatics studies.sup.34,35, the structural model showed that C2.sub.Vv could be split into two subdomains, C2A.sub.Vv and C2B.sub.Vv (FIG. **16**A). To determine if the cytotoxic activity of C2.sub.Vv is linked its C2A or C2B subdomain, DNA corresponding to the individual subdomains was cloned fused to egfp and expressed in HeLa cells. Cells ectopically expressing only C2A.sub.Vv-EGFP were highly necrotic, while cells expressing C2B alone appeared normal (FIG. **17**B-G) and produced EGFP-fusion protein detectable by western blotting (FIG. **17**H). However, due to the severe toxicity of C2A alone resulting in poor sample recovery, a corresponding fusion protein could not be detected by western blotting to confirm expression (FIG. **17**H).
(230) As an alternative verification of the cytotoxicity associated with C2A.sub.Vv, both full-length DUF5.sub.Vv and C1-C2A from *V. vulnificus* were purified fused to His-tagged *B. anthracis* LFN that is often used as a bioporter for toxin effectors in the absence of the holotoxin.sup.7,11,29,36. The purified proteins were insoluble in less than 2M urea, but nevertheless retained toxicity after delivery to cells by PA. The snap dilution out of urea in the tissue culture media likely allowed folding of the LFN domain and the protein then associated with PA for translocation and successful refolding of the DUF5.sub.Vv domain within the cytosol. Notably, both the full-length protein (FIG. **17**J) and the C1-C2A fragment (FIG. **17**K) resulted in rounding of cells confirming transfection studies that C2A is sufficient for cytotoxicity of DUF5.sub.Vv in HeLa cells. Furthermore, LFNDUF5.sub.Vv was cytotoxic to other mammalian cell types as well, including J774 macrophages, COST fibroblasts, and HEp-2 epithelial cells (Table 1).
(231) TABLE-US-00003 TABLE 1 Cell lines susceptible to DUF5V cytotoxicity.sup.a Rounding induced by LF.sub.NDUF5.sub.Vv? Cell Line +PA −PA HeLa human cervical carcinoma + − COS7 African green monkey fibroblast + − J774 murine macrophage + − Hep-2 human laryngeal epithelial + − .sup.aCells were intoxicated with LFNDUF5V in the presence (+) or absence (−) of PA. After 24 hr, rounding was observed by phase microscopy. Intoxication conditions were the same as reported in FIG. 17.
DUF5.sub.Ah from *A. hydrophila* is Also Cytotoxic
(232) As shown in FIG. **19**, proteins similar to DUF5.sub.Vv and PMT C1-C2 subdomains are found as uncharacterized proteins from other bacterial species. To further explore the possibility that these proteins comprise a novel functional family of cytotoxins, the DUF5-like effector domain from the *A. hydrophila* MARTX toxin was cloned in fusion with LFN and delivered to HeLa cells via PA. Protein purity was assessed by SDS-PAGE in panel 4C. After intoxication it was observed that HeLa cells were rounded similarly to what is seen with LFN-DUF5 (FIG. **18**A,B) indicating that this effector domain also has cytotoxic function. Furthermore, the rounding efficiency is similar between the two toxins (FIG. **18** D, E). Finally, neither toxin induced cell lysis when delivered to HeLa cells, at any of the concentrations tested (FIG. **18** F, G).
(233) Both Full-Length DUF5.sub.Vv and C2 Alone Cause Growth Inhibition when Expressed in Yeast
(234) To further explore the function of DUF5.sub.Vv, we tested if it would be toxic if expressed in *S. cerevisiae*. The gene sequence for DUF5.sub.Vv was cloned into the yeast expression vector pYC2/NTA placing the gene under the control of a galactose-inducible promoter. When transformed into yeast, the DUF5.sub.Vv-expressing yeast strain grew poorly under non-inducing

conditions and showed no growth under inducing conditions on either plates or broth culture (FIG. **20**A). Indeed, expression of DUF5.sub.Vv was more toxic than the MARTX ACD effector domain that has been previously studied in yeast (FIG. **20**A).sup.31. Toxicity was reduced by removal of the C1 MLD such that cells expressing C2 alone were viable under non-inducing conditions with 100- to 1000-fold reduced plating efficiency on galactose and no growth in broth culture (FIG. **20**A). The C1 MLD alone is not toxic when expressed in yeast (FIG. **20**A), as previously shown.sup.31,37.
(235) Distinct from studies in HeLa cells, yeast cells expressing C2A alone were viable when plated on galactose (FIG. **20**B). As an alternative verification for the essentiality of C2B in yeast, stop codons were introduced in the yeast expression plasmid at the codons for V3906 and G3948. Similar to expression of C2A alone, cells expressing proteins truncated within C2B also grew under inducing conditions on both plates and in broth (FIG. **20**C). Close examination of the plating efficiency of C2A compared to C2B indicates that expression of C2A alone may show a slight growth inhibition on plates or in broth but the effect is modest (FIG. **20**B).
(236) Overall, in yeast, distinct from HeLa cells, both C2A and C2B are required for full toxicity although some toxicity is exhibited by C2A alone. The additional requirement for C2B in yeast may reflect a modest difference in the stability of the protein in yeast.
(237) Identification of a C2A Inactivating Mutation by Alanine Scanning Mutagenesis in *S. Cerevisiae*
(238) Alanine scanning mutagenesis has proven to be a useful tool to identify critical residues for other of MARTX effector domains.sup.12,31,38. Alignment of DUF5 amino acid sequences from 5 MARTX toxins, Plu2400, and PMT showed that there are only 16 residues (8.5%) that are 100% identical across all proteins. If the potentially inactive PMT is excluded, 38 residues (20%) are identical across the remaining 6 effectors (FIG. **21**).
(239) To avoid severe toxicity associated with expression of full-length DUF5.sub.Vv in yeast, the plasmid for expression of C2 without the MLD was modified by site-directed mutagenesis targeting 65 total residues in C2A (FIG. **21**, indicated by asterisks), focusing predominantly on polar residues known to be important for catalysis of other bacterial toxins. Both conserved and non-conserved residues were changed to alanine codons. In addition, nine highly conserved residues in C2B were changed for alanine.
(240) *S. cerevisiae* transformed with mutagenized plasmids were recovered by growth on glucose and then tested for the ability to grown on galactose. Surprisingly, 73/74 of the mutations did not alter the growth inhibition exhibited by strains expression unaltered C2. The high frequency of mutations showing no relief of toxicity suggests that modest changes to overall structure are not sufficient to overcome the severe toxicity of this protein for yeast, even when the C1 MLD is absent.
(241) Only one mutant, D3721A found within the C2A domain was identified that facilitated growth of yeast expressing C2. By contrast, a more conservative substitution to glutamic acid did not restore growth to yeast (FIG. **21**A). D3721 is one of the 16 residues within C2A that is highly conserved in all the DUF5-like proteins, including PMT (FIG. **19**).
(242) As an independent verification of the importance of this residue to the function DUF5.sub.Vv, the mutation was transferred onto the full-length clone of DUF5.sub.Vv for expression in yeast. In this background, the mutation improved growth of yeast under non-inducing conditions to levels near vector control. Under inducing conditions, the plating efficiency was improved 100- to 1000-fold compared to expression of full-length DUF5.sub.Vv, although the growth inhibition was not alleviated as shown in liquid culture experiments (FIG. **21**B).
(243) Examination of the structural model of DUF5.sub.Vv (FIG. **16**A) showed that D3721 is present on helix 3 of C2A and makes polar contacts with R3841 on the final helix of C2A just before the start of C2B (FIG. **21**C,D). Previous change of R3841 to Ala as part of the screen indicated this residue was not essential in the context of C2. However, in the context of the full-

length DUF5.sub.Vv that includes the MLD, the phenotype of DUF5.sub.Vv R3841A is nearly identical to the phenotype of the D3721A resulting in an improved plating efficiency but poor growth in broth culture (FIG. **21**B). This residue is also among the 16 100% identical residues found within C2A (FIG. **19**). Combining the D3721A and R3841A resulted in a phenotype identical to that observed for either D3721A or R3841A substitutions alone and did not demonstrate an additive effect. Swapping the aspartate and arginine (D3721R/R3841D) did not improve growth either by plating or broth culture indicating that the potential bridge created by these residues is positional specific.

(244) D3721 and R3841 are Essential for Cytotoxicity

(245) As a final demonstration of the structural requirements for cytoxicity, the D3271A and R3841A mutations were introduced onto the recombinant overexpression plasmid for production of C1-C2A fused to LFN. Proteins were prepared for each mutant from insoluble pellets, urea was reduced to 1.2 M, and the unfolded proteins were delivered to cells by PA. Both mutants lost function in cytotoxicity compared to the similarly prepared unmodified LFNC1-C2A.sub.Vv protein (FIG. **22**A-E). Assessment of intoxication over time showed that cells treated with PA plus full length LFNDUF5.sub.Vv did not recover after 24 h intoxication and nearly 100% of cells remained rounded out to 48 hr. By contrast, ˜50% of cells initially intoxicated with PA plus LFNC1-C2A.sub.Vv recovered between 24 and 48 h and returned to normal shape. These data suggest either that C2B carries an additional cytopathic function that prevents recovery of the rounded cells or, more likely, that C2B stabilizes C2A such that the toxin avoids turnover in the cells after successfully inducing cell intoxication. In support of this possibility, fluorescence thermal shift experiments were conducted with full-length recombinant 6×His-DUF5.sub.Vv without fusion to LF.sub.N (FIG. **21**E). This recombinant 6×His-DUF5.sub.Vv has a half-maximal melting temperature (Tm) of 43.8° C., while the D3271A substitution lowers Tm by 6.0° C. to 37.8° C. The lower Tm indicates that D3271A causes a structural disturbance that can explain the reduced toxicity seen in yeast and HeLa cells indicating its interaction with R3841 may function to stabilize the protein structure rather than serve as a catalytic residue (FIG. **21**F). This is also consistent with the structural model of DUF5 where D3721 is located within the core of the protein, such that a mutation to alanine would cause a disturbance consistent with a drop in Tm and would also account for the higher initial fluorescence seen with DUF5 D3721A than wild type protein.

Discussion

(246) In this study, we undertook a structure-function approach to discover if the *V. vulnificus* MARTX toxin effector domain DUF5.sub.Vv is a cytotoxin accounting for its dramatic effect on virulence in mouse infection studies.sup.5. The C1.sub.Vv subdomain of this protein has been previously shown to localize to anionic membranes, but the function of the C2.sub.Vv subdomain at the membrane had not been previously investigated. Here, we demonstrate that DUF5.sub.Vv effector domain is cytotoxic to HeLa cells and to yeast resulting in growth inhibition. Further, the cytotoxic activity is localized to its C2A subdomain. In retrospect, mapping the cytotoxicity to the C2A subdomain is surprising because recent computer-based modeling studies of DUF5.sub.Vv and related proteins linked the C2B domain to the TIKI/TraB family of proteases leading to the proposal that C2B is a peptidase that functions in signaling.sup.34,35. However, we found that any putative protease activity associated with C2B would not contribute to cytotoxicity as complete removal of the subdomain from DUF5.sub.Vv-EGFP did not affect cytotoxicity after ectopic expression studies in HeLa cells and expression of C2B-EGFP did not cause any observable effect in HeLa cells. Further the computer-based analysis indicated that C2B residue H3902 would be essential for peptidase activity, but this residue was among those modified during expression in yeast that did not restore the ability of yeast to grow (FIG. **19**). These findings convincingly link the cytotoxic effect of DUF5.sub.Vv to its C2A subdomain; however, we cannot exclude that the C2B in addition to C2A could modify cell biological processes in manner that does not affect cell

viability or morphology during MARTX intoxication and that DUF5.sub.Vv itself is a multifunctional effector domain.

(247) The remainder of the study focused on identification of residues within C2A.sub.Vv that are essential for its cytotoxicity. Growth of yeast expressing C2.sub.Vv was used as a method to screen point mutations to identify those that would overcome the severe toxicity in yeast, a highly stringent phenotype generally indicative of an essential residue. The screen revealed a single essential Asp that initially was considered as a possible catalytic residue. However, the absence of additional residues in C2A.sub.Vv that would be predicted to form a catalytic site along with the finding that other highly conserved Gly, Pro, Tyr, Phe, Leu, and Ala are not essential suggests this subdomain functions by binding to a target protein rather than by covalent modification. The ability of the residue to tolerate substitution to the more structurally conservative glutamic acid also indicates this is not likely an aspartyl protease. We further found that the D3721A substitution reduced the Tm of the DUF5.sub.Vv indicating structural destabilization as opposed to loss of catalytic function.

(248) This stabilization may be due to its association with R3841 to retain optimal folding of the face that binds to the target protein or to serve as a switch to facilitate a change in the DUF5.sub.Vv structural conformation upon binding of C1.sub.Vv to the membrane (FIG. **21**D). The role as a switch in the context of membrane binding is particularly interesting since reduced toxicity due to R3841A was observed only in the context of the C1.sub.Vv membrane localization domain in both yeast and HeLa cells. The contact between D3721 and R3841 could affect the conformation at the interface between C2A.sub.Vv and C2B.sub.Vv since R3 841 that makes polar contacts with D3721 also contacts a S3986 in an unstructured loop of C2B. In other DUF5 homologues, the Ser is replaced by a Thr. Further, this Ser is absent from PMT, although Ser residues are localized nearby in this otherwise poorly conserved regions between PMT and DUF5.sub.Vv. Thus, it is intriguing to speculate that C2B.sub.Vv could function as a stabilization subdomain for C2A.sub.Vv with D3721 and R3841 functioning as part of the conformational switch to open up a binding site for the cellular target of C2A (FIG. **23**).

(249) A final component not addressed in this study is the biochemical mechanism or activity of DUF5.sub.Vv and DUF5.sub.Ah. While two residues, D3721 and R3841 were found to be essential for rounding of mammalian cells by DUF5.sub.Vv, this discovery does not as yet inform the biochemical or cell biological activity that results in cell rounding. This is particularly true since residues shown to be essential for DUF5.sub.Vv (D3721 and R3841) and conserved in DUF5.sub.Ah (D3215 and R33e5) are also conserved in PMT (as D720 and R861). Given that PMT is not able to round cells similar to DUF5.sub.Vv and DUF5Ah, we can only speculate that surrounding residues not conserved in PMT also contribute to the appropriate structure for DUF5.sub.Vv and DUF5.sub.Ah allowing these proteins but not PMT to properly interact with cellular components. Despite not yet directly demonstrating the biochemical or cell biological activity of the MARTX DUF5 effector domains, this study has provided numerous useful tools and reagents for these on-going studies but likewise reveals how identification of the cellular target could potentially be problematic. We found that the cytotoxicity is associated with C2A. However, this subdomain is highly toxic when ectopically overexpressed, which presents difficulties in identifying the target protein by common affinity precipitation techniques. A catalytically inactive variant is often highly useful to trap targets by affinity precipitation methods, but we found that the only inactive substitution also affects structural integrity and likely no longer binds its target in vivo. Our findings here that yeast is also affected by DUF5.sub.Vv does open the possibility that yeast-based genetic approaches could be very helpful to identify the target and these studies are currently ongoing.

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Example 3—Site-Specific Processing of Ras and Rap1 Switch I by a MARTX Toxin Effector Domain

(251) Reference is made Antic, I., et al., Site-specific processing of Ras and Rap1 Switch I by a MARTX toxin effector domain. Nat Commun, 2015. 6: p. 7396, the content of which is incorporate herein by reference in its entirety.

(252) Abstract

(253) Ras (Rat sarcoma) protein is a central regulator of cell growth and proliferation. Mutations in the RAS gene are known to occur in human cancers and have been shown to contribute to carcinogenesis. In this study, we show that the multifunctional-autoprocessing repeats-intoxin (MARTX) toxin-effector domain DUF5.sub.Vv from *Vibrio vulnificus* to be a site-specific endopeptidase that cleaves within the Switch 1 region of Ras and Rap1. DUF5.sub.Vv processing of Ras, which occurs both biochemically and in mammalian cell culture, inactivates ERK1/2, thereby inhibiting cell proliferation. The ability to cleave Ras and Rap1 is shared by DUF5.sub.Vv homologues found in other bacteria. In addition, DUF5.sub.Vv can cleave all Ras isoforms and KRas with mutations commonly implicated in malignancies. Therefore, we speculate that this new family of Ras/Rap1A-specific endopeptidases (RRSPs) has potential to inactivate both wild-type and mutant Ras proteins expressed in malignancies.

Introduction

(254) Rat sarcoma (Ras) oncoprotein is a small GTPase ubiquitous in eukaryotic cells and is a critical node that coordinates incoming signals and subsequently activates downstream target proteins. These targets include rapidly accelerated fibrosarcoma kinase (Raf), phosphatidylinositol-4,5-bisphosphate 3-kinase and mitogen-activated protein kinase (MAPK), which ultimately induces expression of genes directing cell proliferation, differentiation and survival. Regulation of Ras enzymatic activity is achieved by cycling between an inactive (GDP-bound) state and an active (GTP-bound) state. On activation, conformational changes in the Ras protein structure trigger Ras downstream signalling cascades by binding specific protein effectors.sup.1-4. Mutations in Ras proto-oncogenes are found in 9-30% of all human malignancies. In addition, Ras point mutations, which are observed at residues G12 and G13 in the P-loop and at Q61 in the Switch II region, are the most common mutations in human malignancies and are present in 98% of pancreatic ductal adenocarcinomas, 53% of colorectal adenocarcinomas and 32% of lung adenocarcinomas.sup.5-7. However, effective targeting of Ras has been very difficult and is considered a critical roadblock on the path towards generating new therapeutics against intractable human cancers.sup.8-12. Despite the potential of Ras proteins as therapeutic targets, there are no inhibitors for any of the three main human isoforms—HRas, KRas and NRas—or their constitutively activated mutant forms.sup.8-11.

(255) From a microbial pathogenesis perspective, activation of Ras is central to cellular detection of bacterial lipopolysaccharide and other pathogen-associated molecular patterns resulting in activation of innate immune defenses.sup.13. Although several bacterial toxins are known to target Ras by posttranslational modification to circumvent this important host response to infection, to date none have been shown to be highly specific for Ras.sup.14-16.

(256) Multifunctional-autoprocessing repeats-in-toxin (MARTX) toxins proteins are large composite-secreted bacterial protein toxins that translocate across the eukaryotic cell plasma membrane and deliver multiple cytopathic and cytotoxic effector proteins from a single holotoxin by autoprocessing.sup.17,18. In our previous work, we showed that the most highly virulent strains of the sepsis-causing pathogen *V. vulnificus* produce a 5,206-amino acid (aa) MARTX toxin with an extra effector domain termed DUF5.sub.Vv, for the domain of unknown function in the 5.sup.th position.sup.19. In fact, bacterial strains that produce a MARTX toxin with DUF5.sub.Vv are found to be 10- to 50-fold more virulent in mice than strains that produce a MARTX toxin without DUF5.sub.Vv (ref. 19). These data directly connect DUF5.sub.Vv with increased virulence during infection.

(257) The 509-aa DUF5.sub.Vv effector domain of the MARTX toxin was highly cytotoxic when ectopically expressed as a fusion to green fluorescent protein (GFP), resulting in rounding and shrinkage of cells.sup.20. Structural and functional bioinformatics studies have demonstrated that DUF5.sub.Vv is comprises two subdomains.sup.20,21. The amino-terminal C1 subdomain is a four-helix bundle that mediates localization to the plasma membrane by binding anionic phospholipids.sup.21,22. The carboxy-terminal C2 subdomain confers the cell rounding activity.sup.20. Moreover, DUF5.sub.Vv-C2 was found to inhibit growth when conditionally overexpressed in *Saccharomyces cerevisiae*.sup.20.

(258) In this study, we used a combination of genetic, cell biological and biochemical strategies to probe the mechanism of action of the C2 subdomain, to understand the connection of DUF5.sub.Vv to both cytotoxicity and increased virulence of the pathogen. We find that DUF5.sub.Vv site-specifically processes both Ras and the closely related small GTPase Rap1. Both proteins are critical for activation of the innate immune response during infection, which explains the crucial role of this effector domain in the increased virulence of *V. vulnificus* strains that have DUF5.sub.Vv As Ras is also important for cell proliferation in carcinogenesis, this enzyme could potentially be developed as a treatment for various types of tumours.

(259) Results

(260) DUF5.sub.Vv Causes ERK1/2 Dephosphorylation.

(261) Previously we showed that DUF5.sub.Vv-C2 is cytotoxic when ectopically expressed in cells.sup.20. As a strategy to identify molecular targets accounting for this cytotoxicity.sup.20, a genome-wide, arrayed, non-essential gene deletion library was screened for yeast strains that survived enforced expression of C2 (FIG. **28**). Of 4,709 yeast strains screened, 3.6% formed colonies on plates containing the inducer galactose, indicating that the yeast gene disruption suppressed C2-dependent growth inhibition. The hits were categorized based on information in the *Saccharomyces* Genome Database23. Eleven percent of the mutant yeast strains that overcame growth inhibition due to DUF5.sub.Vv-C2 expression harboured deletions in genes for transcription and/or translation. These mutations probably reduce DUF5.sub.Vv-C2 expression, accounting for suppression of growth inhibition. Twenty-four percent of the recovered yeast strains had defects affecting membrane or membrane proteins, possibly causing suppression of cytotoxicity due to the absence of the cellular target at the membrane (FIG. **24**A).
(262) Among the remaining hits, nearly half were connected to MAPKs or processes they regulate. Therefore, it was postulated that mammalian MAPK p38 and ERK1/2 could have altered activity during exposure of cells to DUF5.sub.Vv. We have previously demonstrated that the cytotoxic activity of DUF5.sub.Vv can be isolated away from the large MARTX by fusing DUF5.sub.Vv to the N terminus of anthrax toxin lethal factor (LFNDUF5.sub.Vv) and subsequently delivering the fusion protein to cells in culture using anthrax toxin protective antigen (PA20). Therefore, we used this system to test for changes in MAPK signalling dependent on exposure of cells to DUF5.sub.Vv.
(263) HeLa cervical carcinoma cells constitutively produce high levels of phospho-p38 and phospho-ERK1/2 (pERK1/2), making these cells an ideal model system to determine the underlying mechanism by which DUF5.sub.Vv interferes with MAPK signaling (FIG. **29**). For cells intoxicated with LF.sub.NDUF5.sub.Vv in combination with PA for 24 h, no change in levels of phosphop.sup.38 was observed (FIG. **29***a*). However, there was a marked absence of pERK1/2 in HeLa cells treated with LF.sub.NDUF5.sub.Vv+PA (FIG. **24**B and FIG. **29***a*). In addition, the first 276 aa of DUF5.sub.Vv, corresponding to the C1 membrane-targeting subdomain and the first 186 of C2 (C1C2A.sub.Vv), were sufficient to reduce pERK1/2 levels (FIG. **29***b*), consistent with previous results showing that C1C2A.sub.Vv is sufficient for cell rounding activity.sup.20. Thus, the yeast screen and subsequent studies in HeLa cells revealed that DUF5.sub.Vv modulates the activation state of ERK1/2 without affecting p38.
(264) Ras Depletion by DUF5.sub.Vv Inhibits Cell Division.
(265) Owing to its C1 membrane-targeting subdomain, DUF5.sub.Vv is exclusively present at the plasma membrane.sup.21; hence, inactivation of membrane localized Ras GTPases that control activation of ERK1/2 (refs 24,25) seemed a plausible mechanism for DUF5.sub.Vv dependent ERK1/2 dephosphorylation. Active Ras (GTP-bound) was probed using a G-LISA assay, where wells are coated with a Ras GTP-binding protein domain. Surprisingly, active Ras was undetectable in cell lysates intoxicated with LF.sub.NDUF5.sub.Vv+PA, suggesting that Ras was exclusively in the inactive, GDP-bound state (FIG. **30**). This result initially suggested that DUF5.sub.Vv affects levels of active Ras-GTP. However, additional control experiments revealed that Ras protein itself was undetectable in cell lysates, as measured by immunoblotting with a monoclonal anti-RAS10 antibody that detects all isoforms of Ras26, including KRas, HRas and NRas (FIG. **24***b*). This experiment shows that DUF5.sub.Vv directly targets the Ras protein rather than indirectly affecting its regulation.
(266) If Ras and pERK1/2 are truly absent from DUF5.sub.Vv-treated cells, proliferation should be inhibited in intoxicated samples. To measure disruption in cell proliferation due to the inhibition of the Ras-ERK pathway, the toxin was removed by washing, and treated cells were plated and resulting colonies counted after a 14-day incubation period. HeLa cells intoxicated for 24 h did not produce colonies even when plated at almost 70-fold higher seeding densities than control-treated cells (FIG. **24**C). Examination of ERK1/2 and Ras inactivation over time revealed that exposure of

cells to 3 nM LFNDUF5.sub.Vv for only 30 min was sufficient for nearly 100% inactivation (FIG. **24**D and FIG. **31**). In addition, exposure of cells to LFNDUF5.sub.Vv concentrations as low as 30 pM for 1 h was sufficient to significantly decrease cell proliferation (FIG. **24**E). Overall, these studies reveal that DUF5.sub.Vv directly targets Ras, resulting in loss of ERK1/2 phosphorylation and cell proliferation.
(267) Ras is Cleaved at the N Terminus in DUF5.sub.Vv-Treated Cells.
(268) Only a few bacteria are known to specifically target Ras as a strategy to circumvent the host response and all do so by covalent attachment of nucleotide-sugar moieties to critical residues.sup.14-16. To investigate whether the loss of detectable Ras protein levels was due to proteolysis and/or a posttranslational modification that would mask the antibody epitope, HeLa cells were transfected to ectopically express HRas with a haemagglutinin (HA)-tag on the N terminus (HA-HRas), so as to facilitate immunoprecipitation with anti-HA antibody-coupled beads. Analysis of proteins immunoprecipitated from LFNDUF5.sub.Vv+PA intoxicated cells revealed a Coomassie-stained band with a molecular weight B5 kDa smaller than the band observed in the untreated cells. Liquid chromatography—tandem mass spectrometry sequencing Q2 of tryptic peptides identified this protein as HRas, with no detection of the first three expected N-terminal peptides (FIG. **25**A).
(269) When the elution fraction was probed with anti-HA or anti-RAS10 monoclonal antibodies that detect the N terminus, a quantitative loss of the full-length protein from intoxicated cells was observed (FIG. **25**B, left panel). By contrast, an isoform-specific polyclonal antibody that detects the C terminus of HRas identified two bands of HRas: one representing the full-length HA-HRas and one B5 kDa smaller. We speculate this cleaved form of HRas was present in the immunoprecipation despite lacking the HA tag, because the HA-tagged fragment remained associated with the larger C-terminal fragment in the folded protein. This experiment suggested that DUF5.sub.Vv induces cleavage of Ras within the N terminus of the protein.
(270) To verify that Ras is processed and to determine which isoforms of Ras are affected, cells were transfected to express HA-tagged KRas, NRas or HRas. In cells treated with LFNDUF5.sub.Vv+PA, western blot analysis of whole-cell lysates showed that all three isoforms were cleaved at the N terminus. The anti-HA and RAS10 monoclonal antibodies directed against the N terminus did not detect KRas, NRas or HRas in treated cells, whereas isotype-specific antibodies directed against the C terminus detected the smaller processed forms (FIG. **25**C). A reduction in the total protein detected by the isoform-specific antibodies was also observed. This suggests that subsequent to processing, the cleaved forms are degraded, especially for HA-NRas and HA-HRas. These data show that Ras isoforms are not modified by addition of moieties but are instead severed near the N terminus, which is a novel mechanism for Ras inactivation.
(271) Recombinant DUF5.sub.Vv can Process all Ras Isoforms In Vitro.
(272) Two possible explanations of our results are that DUF5.sub.Vv activates a previously unknown cellular peptidase or functions as a Ras peptidase itself. To distinguish whether DUF5.sub.Vv directly catalyses proteolytic processing of Ras, recombinant 6×His-tagged DUF5.sub.Vv (rDUF5.sub.Vv) and Ras isoforms (r_Ras) were expressed in *Escherichia coli* and purified. When mixed together for an in-vitro reaction, rKRas was efficiently cleaved within 10 min in a concentration-dependent manner (FIG. **25**D). This reaction did not require addition of any other proteins or co-factors. rHRas and rNRas were likewise efficiently processed by purified rDUF5.sub.Vv (FIG. **25**E).
(273) N-terminal sequencing of KRas, HRas and NRas cleaved products revealed that all Ras isoforms were identically cleaved between Y32 and D33 (FIG. **25**F). These amino acids are found within the Ras Switch I region. Processing at this site would be expected to entirely abolish Ras signalling, as Y32 is required to orient and stabilize Switch I in the active (GTP-bound) state27. Cleavage within the Switch I region would further prevent the activation of downstream signalling cascades by disrupting the Ras effector protein interactions, thereby inhibiting activation of the

ERK1/2 transcriptional regulator and decreasing cell proliferation.sup.28-30.
(274) Other DUF5 Homologues Cleave Ras.
(275) Domains similar to DUF5.sub.Vv have been identified in other bacterial species (FIG. **26**A). To determine whether Ras processing is a conserved function among bacteria, the effector domain from the *Aeromonas hydrophila* MARTX toxin (rDUF5.sub.Ah) and a hypothetical effector protein from insect pathogen *Photorhabdus asymbiotica* (rDUF5.sub.Pa) were also purified and tested for proteolytic activity. Both proteins were found to cleave rKRas in vitro with cleavage occurring between Y32 and D33 (FIG. **26**B). As further validation, DUF5.sub.Ah was fused to LFN (LFNDUF5.sub.Ah). This protein induced both cytotoxicity and Ras cleavage in intoxicated cells when delivered to cells by PA (FIG. **26**C). Thus, DUF5 represents a new family of bacterial toxin effectors that catalyses site-specific processing of the Switch I region of all three major isoforms of Ras independently of any other cellular proteins.
(276) Rap1 is Also a Substrate for DUF5.sub.Vv.
(277) Other bacterial protein toxins are known to promiscuously target a wide range of small GTPases and other cellular proteins.sup.15. As the amino acid sequence of the Switch I region of Ras is well conserved across Ras subfamily members (FIG. **26**D), it was considered that DUF5.sub.Vv might also cleave other small GTPases. To test this, representative Ras subfamily small GTPases fused via their N termini to enhanced GFP (EGFP) were ectopically expressed in HEK 293T cells and anti-GFP antibody was used to detect the released N-terminal fragment. In cells treated with LF.sub.NDUF5.sub.Vv+PA, EGFP-HRas and EGFP-Rap1 were both cleaved with 480% efficiency. Processing of another Ras subfamily member, Rit2, was also detected in this assay, but with inconsistent efficiency, resulting in a large s.d. across multiple experiments (FIG. **26**E). This indicates that Rit2 may be a low-affinity substrate resulting in experimental variation dependent on the ratio of toxin to GFP-Rit2 in each cell or sample (FIG. **26**E). Other small EGFP-GTPases (RalA, RheB2, RhoB and Arf1) showed no cleavage, indicating they are not in-vivo substrates (FIG. **26**E and FIG. **32**).
(278) DUF5.sub.Vv specificity for Ras and Rap1 was further verified biochemically. Small Ras GTPases covering the diversity of Ras subfamilies were purified as substrates for in-vitro assay to assess whether rDUF5.sub.Vv could catalyse their cleavage. Among the 11 GTPases tested (FIG. **33**), only Rap1 was confirmed as a DUF5.sub.Vv substrate, with cleavage occurring after Y32 (FIG. **26**F), whereas Rit2 was not cleaved at all, confirming that in cells this is a low-affinity substrate (FIG. **33**). Other GTPases belonging to the Ras, Rho, Rab and Ran subfamilies were not processed (FIG. **33**). Thus, DUF5.sub.Vv is a specific protease that preferably cleaves Ras and Rap1 without cellular cofactors. The detection of Rap1 as an additional substrate is especially interesting for bacterial pathogenesis, as Rap1 activates ERK in response to bacterial components other than lipopolysaccharide and is critical for macrophage phagocytosis31,32.
(279) DUF5.sub.Vv Targets Ras During Bacterial Infection.
(280) Given the importance of Ras and Rap1 in the host response to bacterial infection, it is not surprising that DUF5.sub.Vv was previously shown to contribute to *V. vulnificus* virulence.sup.19. The strain CMCP6 produces a MARTX toxin that carries five effector domains, including DUF5.sub.Vv in the fifth position. By contrast, M06-24/O produces a toxin with only four effector domains (FIG. **27**A), having undergone a genetic recombination that resulted in an in-frame deletion of the DNA sequence for the DUF5.sub.Vv domain.sup.19,33. As a result of the loss of DUF5.sub.Vv, M06-24/O is tenfold less virulent than CMCP6 (ref. 19). The increased virulence of CMCP6 was found to be specifically due to DUF5.sub.Vv 19, even though both toxin forms induce cellular necrosis.sup.34,35 (FIG. **34**).
(281) To link this defect in virulence to Ras activation and demonstrate that Ras can be processed during normal toxin delivery, HeLa cells were co-cultured for 1 h with *V. vulnificus* and proteins in cell lysates were analysed by western blotting. Cells treated with wild-type bacteria producing full-length active MARTX toxin no longer showed detectable Ras or pERK1/2. This inactivation was

dependent on an intact rtxA1 toxin gene, as a null mutation in rtxA1 of *V. vulnificus* CMCP6 did not show loss of detectable Ras or pERK1/2. Further, co-culture of cells with *V. vulnificus* M06-24/O, which produces the MARTX toxin naturally missing DUF5.sub.Vv, did not affect Ras, linking this MARTX-dependent activity specifically to the DUF5.sub.Vv effector domain. Interestingly, cells treated with M06-24/O unexpectedly still showed a reduction of pERK1/2, revealing that these multifunctional toxins probably have redundant strategies to inactivate ERK during infection (FIG. **27**B).
(282) Oncogenic KRas is Processed by DUF5.sub.Vv.
(283) Point mutations resulting in constitutive activation of Ras have long been associated with many different types of adenocarcinomas.sup.5-7. The discovery of a novel bacterial toxin mechanism to halt cell proliferation through processing of Ras is not only important for understanding the function of bacterial toxins during infection but also presents an opportunity to potentially target Ras during carcinogenesis through delivery of DUF5. This strategy would be most successful if mutant forms of Ras found in cancer cells are also DUF5 substrates.
(284) When HCT116 colorectal carcinoma cells, which express KRas with a G13D mutation, were intoxicated with PA in combination with LFNDUF5.sub.Vv (FIG. **27**C) or LFNDUF5.sub.Ah (FIG. **35**), significant cell morphological changes were observed and Ras was undetectable by western blotting. Similar results were obtained with the breast cancer cell line MDA-MB-231 that likewise carries the KRas G13D mutation. This cell line also contains a G464V mutation in B-Raf36, an effector of both Ras and Rapt (ref 37), demonstrating that DUF5.sub.Vv can effectively intoxicate cells even if they have additional activating mutations downstream of Ras and Rap1.
(285) As further demonstration that DUF5.sub.Vv could be employed as a cancer treatment, rKRas was modified to carry three of the most common Ras mutations associated with tumorigenesis: G12V, G13D or Q61R7. All three mutant forms of KRas were confirmed as in vitro substrates for rDUF5.sub.Vv-dependent site-specific processing (FIG. **27**D). Thus, the ability of DUF5.sub.Vv to cleave KRas is unaffected by the most common RAS mutations. Overall, these data show that cells carrying constitutively active forms of Ras are not protected from DUF5.sub.Vv cytotoxicity and thus DUF5.sub.Vv is a valid candidate for use as an anti-tumour agent.
Discussion
(286) MARTX toxins are large bacterial toxins that carry multiple effector domains, each with a specific enzymatic activity. DUF5.sub.Vv, the extra effector domain of the MARTX toxin from the most virulent strains of the sepsis-causing pathogen *V. vulnificus*, was previously shown to be highly cytotoxic for mammalian cells, although the mechanism of this cytotoxicity was unknown.sup.20. In this work, we demonstrate that DUF5.sub.Vv is a representative member of a new family of bacterial toxin effectors that catalyse site-specific processing of the Switch I region of Ras and Rap1. Activated Ras or Rap1 would normally interact with downstream effectors such as c-Raf, to stimulate the phosphorylation of ERK1/2. In particular, Y32 in the Switch I region plays an important role in stabilizing the GTP-bound form of Ras and its interaction with the Raf kinases.sup.27. Thus, it is predicted that DUF5.sub.Vv cleavage between Y32 and D33 would destabilize the Switch I and presumably the interactions of Ras and Rap1 with their binding partners. As Ras and Rap1 form parallel pathways that relay signals from surface receptors and guanine nucleotide exchange factors to activate ERK1/2, disabling both small GTPases simultaneously nullifies all downstream signaling pathways.sup.38, resulting in the complete loss of pERK1/2 in DUF5.sub.Vv-treated cells. In the context of bacterial infection, this is important to inactivate innate immune responses, accounting for the direct linkage of this toxin effector domain to virulence of *V. vulnificus*. We propose that the DUF5 effector domain be renamed RRSP for Ras/Rap1-specific protease, acknowledging its site-specific processing of the Switch I region of Ras and Rap1.
(287) As small GTPases are responsible for regulating essential cell functions, many other bacterial protein toxins and effectors target GTPases by posttranslational modification or by manipulating

Q3 their function.sup.15. However, few of these toxins target Ras specifically, for example, *Pseudomonas aeruginosa* ExoS ADP ribosylates R41 of Ras and Rap.sup.39-41, and thereby directly inhibits phagocytosis in mice.sup.42. However, ExoS also has broad substrate recognition including other GTPases.sup.43 and other proteins such as moesin and vimentin.sup.16,44,45. Similarly, *Clostridium sordellii* lethal toxin TcsL (also known as LT) has been shown to glucosylate Ras at T35 in the Switch I.sup.46,47 resulting is cellular apoptosis.sup.48. In addition, TcsL UDP-glucosylates other small Ras, Rap, Ral, Rho and Rac GTPases with some specificity differences depending on strain.sup.49. Through a similar process, *Clostridium perfringens* large toxin TpeL modifies T35 of Ras and, to a lesser extent, Rap1 and possibly Rac1, except it preferentially uses UDP-Nacetylglucosamine as a sugar donor.sup.50,51.

(288) The unique feature of RRSP demonstrated here is its irreversible mechanism of action by cleaving rather than modifying Ras and Rap1. The biochemical basis for the specificity of RRSP for Ras and Rap1 should be explored further in the future. Although it is possible that the specificity is dictated by the conservation of the amino acid sequence in the Ras and Rap1 Switch I regions, it is more likely to be that recognition of the target is multifactorial depending on a multifaceted protein-protein interaction between RRSP and Ras or Rap1. This possibility is supported by studies of *Clostridium difficile* toxin TcdB recognition of RhoA as a substrate for glucosylation, which is mediated in part by specificity for target residue T37 in the Switch I region.sup.52, but also by Ser73 outside the Switch I.sup.53. In addition, amino acids of TcdB essential to discriminate substrate are found outside the catalytic site, further indicating that specificity of TcdB from Rho in not driven solely by the Switch I sequence.sup.54.

(289) In addition to protein-protein interactions, specificity of RRSP for Ras and Rap1 may include spatial localization to anionic membranes or specificity for the active or inactive state conformation when bound to GTP or GDP, respectively. However, in cells, we routinely observed 100% processing of all Ras isoforms in as little as 30 min and we also observed 100% cleavage of KRas G12V, G13D and Q61R in vitro, despite not controlling the GTP or GDP state using buffers. These data would seem to support the hypothesis that RRSP can target both active and inactive forms of Ras and thereby access both membrane and cytoplasmic pools of Ras. In addition, as the Switch I region undergoes structural changes with activation state, and both active and inactive forms of Ras seem to be substrates for RRSP, we suppose specificity is at least in part driven by protein-protein interaction outside the Switch I region and this will be explored in the future through detailed structural and binding studies.

(290) A critical question for bacterial infection is how the processing of Ras and Rap1 contributes to increased virulence. The MARTX toxin of *V. vulnificus* is known to play a role during infection both in paralysing phagocytic cells.sup.55 and in breaching the epithelial barrier to promote spread of the bacterium from the intestine to other organs.sup.56-58. Overall, small GTPases play a central role in the barrier function of epithelial layers such that loss of this control could contribute to bacterial spread across the intestinal barrier.sup.15. In particular, Ras and Rap1 are essential for sensing and signalling pathogen-associated molecular patterns and for regulating inflammatory responses of the host organism.sup.15,59. Ras and Rap1 function in response to bacterial components such as LPS and for macrophage phagocytosis, activating the ERK1/2 Q4 pathway cascade.sup.31,32; in the context of bacterial infection, inhibition of these cascades would slow down the host response to bacterial infection, such that *V. vulnificus* strains that carry this domain are more virulent.sup.19.

(291) A final impact of our discovery is the possibility that the RRSP effector domain could be deployed across the cell membrane to specifically target tumour cells using different delivery strategies. More than three decades after the discovery of Ras implication in cancer development, targeting Ras remains one of the hardest challenges of cancer research and drug discovery.sup.7. Here, we propose that proteins in this new RRSP effector family could be employed immediately as research tools, but in the future developed as new anti-cancer therapeutic agents. Of particular

immediate interest, re-engineered PA selectively targeting cancer cells could be used to deliver LFNDUF5 into cells to destroy Ras and thereby deregulate tumour growth and proliferation. This approach has already been validated in cell systems in which PA was fused to the epidermal growth factor for delivery of LFN-tethered cargo into cancer cells with upregulated expression of the epidermal growth factor receptor.sup.60. This system has also been proven with PA modified to bind to the HER2 receptor, a protein strongly upregulated in tumour cells, in particular breast cancer.sup.61. As alternative future approaches, RRSP effector domains could be fused to specific antibodies for use as an immunotoxin.sup.62, or expressed and delivered by *Salmonella* bacteria that home to solid tumours.sup.63. It could also be expressed by viruses engineered to specifically infect cancer cells.sup.64. The ability of RRSP to cleave both normal and mutant forms of Ras indicates that any developed reagent could be successful whether used for Ras cancers, non-Ras cancers, or other Ras-associated diseases.
(292) Methods
(293) General Molecular Biology Techniques.
(294) *E. coli* DH5a and TOP10 cells (Life Technologies) were grown at 37° C. in Luria-Bertani liquid or on agar medium supplemented with either 100 μg ml.sup.−1 ampicillin or 50 μg ml.sup. −1 kanamycin, as needed. Common reagents were obtained from Sigma-Aldrich, Fisher or VWR, and common restriction enzymes and polymerases were obtained from New England Biolabs or Life Technologies. Custom DNA oligonucleotides were purchased from Integrated DNA Technologies (Coralville, IA). Plasmids were prepared by alkaline lysis followed by precipitation in ethanol or purified using Epoch spin columns according to the manufacturer's recommended protocol. A Qiagen Midi Prep kit was used for preparation of plasmids used in yeast transformations. Plasmids were introduced into *E. coli* by electroporation and into HeLa cells by transfection using polyethylenimine (PEI).
(295) Yeast Non-Essential Gene Deletion Screen.
(296) The Life Technologies YKO yeast deletion library covering all non-essential genes was replicated from stocks at the Northwestern University High Throughput Analysis Laboratory using a Genetix QPixII Automatic colony picker. Each strain from the library was subsequently grown in 1 ml yeast extract peptone dextrose with addition of 50 ug ml 1 G418. After overnight growth at 30° C. with agitation, each strain was transformed with plasmid pYC-C2 using a PLATE solution method and transformants were selected on synthetic complete agar without uracil and with 2% glucose to repress DUF5.sub.Vv-C2 expression. Colonies were patched with toothpicks onto synthetic complete agar supplemented with 2% galactose and 1% raffinose to induce DUF5.sub.Vv-C2 expression. Initial positive selection was defined as yeast that formed a patch when grown on galactose. These were subsequently rescreened in a dilution plating assay as previously described.sup.20 and those with a plating efficiency comparable to a strain transformed with empty vector were considered validated hits. Identified strains were analysed and classified based on information in the Saccharomyces Genome Database website, last accessed on 25 Oct. 2014.
(297) Intoxication of Cells with Proteins Fused to LF.sub.N.
(298) HeLa, HCT116 and HEK293 cells were grown at 37° C. with 5% CO.sub.2 in DMEM medium (Life Technologies) with 10% fetal bovine serum (Gemini Bio-Products, West Sacramento, CA), 100 Uml 1 penicillin and 1 μg ml.sup.−1 streptomycin. Purification of LFN, LFNDUF5.sub.Vv and LFNDUF5.sub.Ah has been previously described20. PA purified as previously described.sup.65 was provided by Shivani Agarwal (Northwestern University). Cell lines were seeded overnight into tissue culture-treated dishes and flasks, except for HCT116 cells, which were seeded for 48 h. Before intoxication, the media was exchanged for fresh media and then 7 nM PA and 3 nM LFN-tagged toxins were added to the media and incubated for the times indicated in the legend at 37° C. with 5% CO.sub.2. Cells were imaged at ×10 at times indicated in the legend using a Nikon TS Eclipse 100 microscope equipped with a Nikon CoolPix 995 digital

camera or processed for western blotting or colony formation as detailed below.
(299) Western Blotting.
(300) A total of 2.5-5×10.sup.4 treated cells were washed with PBS, then resuspended in 120 ml of 2× Laemmli sample buffer and boiled for 10 min. Ten microlitres of lysate were separated by SDS-PAGE and transferred to nitrocellulose (Amersham) using the Bio Rad Trans-Blot Turbo system. Nitrocellulose membranes were blocked overnight at 4° C. in 5% (w/v) powdered milk diluted in Trisbuffered saline containing 0.001% Tween-20 (TBS-T). Immunodetection of proteins was conducted as previously described.sup.20, using primary antibodies purchased from Cell Signaling Technologies (p44/42 MAPK (ERK1/2) rabbit mAb 137F5 (1:1,000), phospho-p44/42 (ERK1/2) rabbit mAb 197G2 (1:1,000), p38 MAPK rabbit polyclonal 9212 (1:1,000) and phospho-p38 rabbit mAb 12F8 (1:1,000)), EMD Millipore (pan-Ras mouse mAb RAS10 (05-516, 1:1,000)), Thermo Scientific (HRas PAS-22392 (1:1,000), KRas PAS-27234 (1:1,000) and NRas PAS-28861 (1:1,000)) and Sigma-Aldrich (H6908 rabbit polyclonal (1:5,000), actin mouse mAb AC-40 (1:1,000) and Tubulin T6074, (1:10,000)). Antibody binding to proteins was detected using anti-mouse (1:5,000) or anti-rabbit (1:5,000) secondary antibodies conjugated to horseradish peroxidase from Jackson Immuno Research and developed using SuperSignal WestPico chemiluminescent reagents (Thermo Scientific) and X-ray film. For serial detection of proteins and detection of the actin-loading controls from the same nitrocellulose membrane, membranes were washed in TBS-T for 10 min and then stripped of antibody by washing the membrane for 10 min with stripping buffer (1.5% glycine, 1% Tween-20, 0.1% SDS). After two more 10-min washes with TBS-T, the membrane was re-probed for other proteins. Tubulin-loading controls were performed by cutting the membrane horizontally to separate the upper loading control portion containing tubulin from the lower portion containing the small Ras family GTPases. Uncropped western blottings are not shown but are provided herein but are provided in the Supplementary Material for Antic, I., et al., Site-specific processing of Ras and Rap1 Switch I by a MARTX toxin effector domain. Nat Commun, 2015. 6: p. 7396, which is incorporated herein by reference in its entirety.
(301) Ras G-LISA.
(302) Active (GTP-bound) Ras in intoxicated cells was measured using the Ras G-LISA activation colorimetric assay kit from Cytoskeleton, Inc. (Denver, CO). HeLa cells were seeded into 10-cm.sup.2 tissue culture-treated dishes and grown to ˜80% confluency, at which time the cells were intoxicated with LF.sub.N proteins in combination with PA for 24 h as described above. Cells were collected in the lysis buffer and total protein content was determined by the Precision Red assay using reagents supplied with the kit. The lysate was frozen in a dry ice-ethanol bath and stored at 80° C. Active Ras in each lysate was then determined according the manufacturer's protocol. This kit used the pan-Ras RAS10 mAb for detection of active Ras and this antibody was subsequently obtained directly from Millipore for western blotting detection of Ras as described above.
(303) Clonogenic Colony-Formation Assay.
(304) A total of 10.sup.5 HeLa cells were seeded into six-well dishes overnight, intoxicated with LFN protein as described above and assessed by a clonogenic colony-formation assay as described previously.sup.66. Briefly, cells were released from wells with 0.25% trypsin/EDTA (Sigma), counted in a hemocytometer and then diluted. The number of cells indicated was replated in fresh media in duplicate. After 14 days, cells were fixed with 70% ethanol and stained with 0.5% crystal violet, and colonies of more than 50 cells were counted. The surviving fraction was compared with cells treated with LFN+PA.
(305) Ectopic Expression of HA-Tagged Ras Isoforms.
(306) Plasmids for ectopic expression of HA-HRas (pcDNA3-HA-HRas wt, 14723) and HA-NRas (pCGN NRas wt, 39503) were obtained from Addgene (Cambridge, MA). Plasmids for overexpression of HA-KRas and HA-KRas G12V were obtained from Athanasios Vassilopoulos (Northwestern University). Plasmid DNA (2 mg) was mixed with 90 ml PEI diluted in incomplete DMEM media, vortexed 15 times and then incubated for 15 min at room temperature. Seven

hundred microlitres of complete DMEM were added into the plasmid-PEI mix and the whole volume was added to HeLa cells. After 24 h, cells were intoxicated as described above.
(307) Immunoprecipitation of HA-HRas and mass spectrometry.
(308) HeLa cells, either untreated or intoxicated with LF.sub.NDUF5.sub.Vv+PA as described above, were washed with cold PBS and then resuspended in RIPA buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.1% SDS, 0.5% sodium deoxycholate, 1% Triton and 'cOmplete' protease inhibitors). HeLa cell lysates were incubated with 50 ml of anti-HA agarose beads (Sigma) for 2 h at 4° C. under mild agitation. Beads were then washed five times with 500 ml of RIPA buffer and five times with 500 ml of washing buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl). Proteins bound to the beads were eluted with 3M sodium thiocyanate buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl). Elution fractions were analysed by SDS-PAGE followed by Coomassie staining or immunoblotting using anti-HA and isotype-specific anti-HRas antibody as described above. The smaller HRas band was excised from the gel, put in water and then frozen for shipping. Trypsin digestion followed by liquid chromatography—tandem mass spectrometry on the Thermo LTQ-FT Ultra spectrophotometer was conducted at the University of Illinois at Chicago Mass Spectrometry, Metabolomics and Proteomics Facility according to their standard protocols.
(309) Preparation of 6×his- or GST-Tagged Small GTPases.
(310) DNA sequences corresponding to KRas (KRas4B, NP_004976.2), HRas (NP_001123914.1) and Q5 NRas (NP_002515.1) genes were amplified from templates as described above, using primers designed for ligation-independent cloning, and the products were cloned into the pMCSG7 expression vector by ligation-independent cloning.sup.67. The G12V, G13D and Q61R mutations were introduced by site-directed mutagenesis using the pMCSG7-KRas vector as a template. Primers are listed in the Table 2 below:
(311) TABLE-US-00004 TABLE 2 Oligonucleotides Used in this Example DUF5 VV FWD TACTTCCAATCCAATGCTCAAGAGCTGAAAGAAAG AGCAAAAG (SEQ ID NO: 37) DUF5 VV REV TTATCCACTTCCAATGCTACAAACTGCCCTTGAAC GTG (SEQ ID NO: 38) DUF5 AH FWD TACTTCCAATCCAATGCTCCGGGCAAAACGGTGGT GACG (SEQ ID NO: 39) DUF5 AH REV TTATCCACTTCCAATGCTAGACATCGGCGTACTCG ACCCGC (SEQ ID NO: 40) DUF5 PA FWD TACTTCCAATCCAATGCTCCATTACTCCATGACCT CATCACC (SEQ ID NO: 41) DUF5 PA REV TTATCCACTTCCAATGCTACACATCATCATAACAC TTGCG (SEQ ID NO: 42) KRAS FWD TACTTCCAATCCAATGCTATGACTGAATATAAACT TGTGGTAGTTGGAGCTGG (SEQ ID NO: 43) KRAS REV TTATCCACTTCCAATGCTACATAATTACACACTTT GTCTTTGACTTCTTTTTCTTC (SEQ ID NO:44) HRAS FWD TACTTCCAATCCAATGCTATGACGGAATATAAGCT GGTGGTGGTG (SEQ ID NO: 45) HRAS REV TTATCCACTTCCAATGCTAGGAGAGCACACACTTG CAGCTC (SEQ ID NO: 46) NRAS FWD TACTTCCAATCCAATGCTATGACTGAGTACAAACT GGTGGTGG (SEQ ID NO: 47) NRAS REV TTATCCACTTCCAATGCTACATCACCACACATGGC AATCCC (SEQ ID NO: 48) EGFPC3-GST GCTTCGAATTCTGCACCCGGGTGGTCTGGTTCCGC FWD GTGGA (SEQ ID NO: 49) EGFPC3-GST CTAGATCCGGTGGATCCCCTCAGTGGTGGTGGTGG REV TGGTGC (SEQ ID NO: 50) KRAS_G13D TAGTTGGAGCTGGTGACGTAGGCAAGAGTGC FWD (SEQ ID NO: 51) KRAS_G13D GCACTCTTGCCTACGTCACCAGCTCCAACTA REV (SEQ ID NO: 52) KRAS_Q61R GATATTCTCGACACAGCAGGTAGAGAGGAGTACAG FWD TGCAATG (SEQ ID NO: 53) KRAS_Q61R CATTGCACTGTACTCCTCTCTACCTGCTGTGTCGA REV GAATATC (SEQ ID NO: 54)
(312) Plasmids were confirmed to be accurate by DNA sequencing and then transformed into *E.*

coli BL21(DE3). Cultures of *E. coli* were grown at 25° C. in Terrific Broth supplemented with 100 μg ml.sup.−1 ampicillin to an OD600 of 0.6-0.7 and then induced with 1 mM isopropyl-β-D-thiogalactoside and growth was continued at 18° C. for ˜18 h. Bacteria were harvested by centrifugation, re-suspended in buffer A1 (50 mM Tris pH 7.5, 500 mMNaCl, 10 mM MgCl2, 0.1% Triton X-100, 5 mM β-mercaptoethanol) and lysed by sonication. After centrifugation at 30,000 g for 30 min, the soluble lysate was loaded onto a 5-ml HisTrap column using the ÄKTA protein purification system (GE Healthcare). The column was washed with buffer B1 (10 mM Tris pH 7.5, 500 mM NaCl, 10 mM MgCl2, 50 mM imidazole) followed by elution in the same buffer with 500 mM imidazole (buffer C1). Proteins were further purified by size-exclusion chromatography (Superdex 200 (26/60), GE Healthcare) in buffer D1 (10 mM Tris-HCl pH 7.5, 500 mM NaCl, 10 mM MgCl2, 5 mMb-mercaptoethanol). GST-fusion GTPases were obtained from Seema Mattoo (Purdue University, IN), and expressed and purified as previously reported68.
(313) Preparation of 6×His-Tagged DUF5 Proteins.
(314) DNA sequences corresponding to DUF5.sub.Vv (*V. vulnificus* CMCP6—MARTX.sub.Vv Q3596-L4089, NP_759056.1), DUF5.sub.Ah (*A. hydrophila* ATCC7966—MARTXAh P3069-V3570—locus WP_011705266) and DUF5.sub.Pa (*P. asymbiotica* ATCC43949—P41-V532 locus WP_011705266) were amplified from their respective genomes using primers designed for ligation-independent cloning and the products were cloned into the pMCSG7 expression vector by ligation-independent cloning.sup.67. Primers are listed in Supplementary Table 1. Plasmids were confirmed to be accurate by DNA sequencing and then transformed into *E. coli* BL21(DE3). Cultures were grown in Terrific Broth supplemented with 100 μg ml.sup.−1 ampicillin at 37° C. until OD.sub.600=0.7-0.8 and then induced with 1 mM isopropyl-β-D-thiogalactoside at 18° C. for ˜18 h. Proteins were purified as described above for Ras proteins, except all buffers were adjusted to pH 8.3 instead of 7.5.
(315) In-Vitro Cleavage Assay and N-Terminal Sequencing.
(316) rKRas, rHRas, rNRas and GST-fused small GTPases were incubated with rDUF5 proteins at equimolar concentrations (10 mM) in 10 mM Tris pH 7.5, 500 mM NaCl, 10 mM MgCl2 at 37° C. for 10 min, unless otherwise indicated. Reactions were stopped by adding 6× Laemmli sample buffer and incubating the sample at 90° C. for 5 min. Proteins were separated on 18% SDS-polyacrylamide gels and visualized using Coomassie stain. Cleavage of Ras isoforms and GTPases was quantified from scanned gels using NIH Image J 1.64. To identify the cleavage site, proteins separated by 18% SDS-polyacrylamide were transferred onto a polyvinylidene difluoride membrane. After Coomassie staining, processed bands were excised from the membrane and sequenced on an ABI 494 Procise Protein Sequencer (Applied Biosystem) using automated Edman degradation at the Tufts University Core Facility.
(317) In-Vivo Cleavage Assay of Small GTPases.
(318) DNA sequences coding for HRas, Rap1A, Rit2, RalA, Rheb2A, RhoB and Arf1 were amplified from plasmids for overexpression of GST-GTPases as described above68. Products were inserted into pEGFP-C3 (Clontech) using SmaI and the Gibson Assembly Cloning Kit (NEB). HEK 293T cells were transfected with the resulting plasmids as described above. After 24 h, cells were intoxicated with LF.sub.N proteins and cleavage detected using monoclonal GFP-HRP antibody (Miltenyi Biotec) as described above. The amount of cleaved protein as a percent of total GFP protein was quantified from scanned gels using NIH Image J 1.64 and data were normalized to the pixels detected in the absence of intoxication.
(319) Bacterial Challenge of HeLa Cells.
(320) *V. vulnificus* rifampicin-resistant isolates of strains CMCP6, M06-24/O and CMCP6DrtxA1 (ref. 19) were grown at 30° C. in Luria-Bertani medium with 50 μg ml.sup.1 rifampicin. Overnight cultures were diluted 1:500 and grown at 30° C. with shaking until the OD.sub.600 reached 0.55-0.6. Bacteria from 1 ml were pelleted at 1,800 g for 4 min, washed once in PBS and then resuspended in 1 ml PBS. Media were exchanged over 5×10.sup.4 HeLa cells previously seeded in

12-well plates overnight for antibiotic-free media. *V. vulnificus* in PBS (multiplicity of infection=100) or an equal volume of buffer was added to media over cells and plates were centrifuged at 25° C. for 5 min at 500 g. After 60 min, cells were photographed as described above, to assess rounding before collection of lysate and western blotting of proteins in 15 ml of lysate as described above. In a separate set of experiments, cells in phenol red-free DMEM with 10% fetal bovine serum but no antibiotics were incubated up to 4 h. At 1-h intervals, 50 ml of supernatant were sampled and assayed for release of lactate dehydrogenase using the Cytotox 96 Non-Radioactive Cytotoxicity Assay (Promega), according to the manufacturer's protocol. Percent cell lysis was calculated as the lactate dehydrogenase release in the sample divided by a positive control lysed with 0.1% Triton X-100.

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Example 4—RRSP Exhibits Novel Proteolytic Activity

(322) Reference is made to the poster presentation entitled “RRSP Exhibits Novel Proteolytic Activity,” Matthew Lam, Marco Biancucci, and Karla J F Satchell, an Abstract of which was published on-line on Apr. 1, 2017. (See Lam et al., the FASEB Journal, Vol. 31, No. 1_supplement, April 2017, the content of which is incorporated herein by reference in its entirety).

(323) Title—RRSP Exhibits Novel Proteolytic Activity

(324) Abstract

(325) Rat sarcoma protein (Ras) is a protein involved in the transduction of signals necessary for cell survival and proliferation. Mutations in Ras can inhibit its enzymatic function and leave it constitutively active, resulting in uncontrollable cell proliferation, culminating in tumor growth. In addition, many bacterial protein toxins and effector domains target Ras GTPases to destroy eukaryotic cells and reduce cellular response against bacterial infection. The Multifunctional-Autoprocessing-Repeats-in-Toxins (MARTX) toxin is the primary virulence factor of *Vibrio vulnificus*, a bacterium that causes sepsis and death from wound infections or contaminated seafood. The domain in the 5th position of MARTX toxins produced by *V. vulnificus* has been shown to cleave between Y32 and D33 residues within the Switch I region of all Ras isoforms, inhibiting the Ras-MAPK pathway and subsequently cell proliferation.

(326) Identification of the catalytic site of this Ras/Rap1-specific endopeptidase (RRSP) was directed by bioinformatics suggesting that the C2B subdomain of RRSP is similar to the active sites of other enzymes such as bacterial erythromycin esterases EreA and EreB, and mammalian protein Tiki2. Despite these enzymes having different substrates, it was revealed that RRSP-C2B shares highly conserved residues that form the active sites of the otherwise distinct proteins. Consequently, a putative active site of RRSP composed of two conserved glutamate and three histidine residues could be modeled.

(327) Point mutations targeting these suspected catalytic residues were generated to assess the enzymatic activity of RRSP mutants. The purified mutants were then incubated with recombinant KRas. Alanine substitutions in three of the five conserved residues prevented in vitro processing of KRas only partially, suggesting that these residues play a more supportive role, such as substrate binding. However, mutations in the other two residues inhibited in vitro KRas processing entirely, and they were deemed as catalytic residues. Fluorescence thermal shift (FTS) was conducted to confirm that the structure of the RRSP mutant variants did not differ significantly from that of the wild type, indicating that the amino acid substitutions impacted enzyme activity and not the overall structural fold of the protein. Thus, the active site of RRSP is shown to be comprised of a pair of Glu/His residues and is most similar to that of the erythromycin esterase EreB. Unlike the metalloproteins EreA and Tiki2, treating RRSP with metal ions chelators such as EDTA and phenanthroline had no effect on substrate processing. Overall, these findings suggest that RRSP conserves a specific set of catalytic residues representative of a proposed erythromycin esterase-Tiki family, within which it has novel protease activity divergent from the metalloproteases of this family.

Background

(328) *Vibrio vulnificus* is a gram-negative bacteria commonly found in marine environments and is found as a contaminant of oysters and other shellfish. As such, *V. vulnificus* is an observed foodborne pathogen that causes gastroenteritis, wound infections, necrotizing fasciitis, and fatal septicemia in humans. Two two of the major virulence factors of *V. vulnificus* are a bacteria hemolysin and so-called “multifunctional autoprocessing repeats-in-toxin or “MARTX” toxin. A schematic illustration of the MARTX toxin and its processing steps is provided in FIG. **36**. Notably, deletions in rtxA1, which is the gene that encodes the MARTX toxin, or the DUF5 domain of MARTX specifically reduce the LD.sub.50 of the wild-type CMCP6 strain in mice by 2600× and 54×, respectively. (See Kwak I., et al., 2011).

(329) Using an in vitro cleavage assay, we observed that a recombinant DUF5.sub.Vv (rDUF5.sub.Vv) could cleave recombinant KRas (rKRas), HRas (rHRas), and NRas (rNRas) between Y32 and D33. (See FIG. **37**). In FIG. **37**A, rDUF5.sub.Vv was incubated with rKRas, rHRas, or rNRas at equimolar ratios (10 μM) and incubated at 37° C. for 30 minutes. Reaction products were analyzed by SDS-Page. The band labeled as r_Ras is uncleaved Ras protein, whereas the band labeled as r_Ras* is cleaved Ras protein. In FIG. **37**B, rKRas or or oncogenic variants thereof (i.e., G12V, G13D, and Q61R) were used as substrates for rDUF5.sub.Vv. rDUF5.sub.Vv was observed to cleave not only WT KRas but also oncogenic forms G12V, G13D, and Q61R, which are the most commonly found oncogenic mutations in KRas-implicated cancers. (See also Antic, I., et al., Site-specific processing of Ras and Rap1 Switch I by a MARTX toxin effector domain. Nat Commun, 2015. 6: p. 7396).

(330) We performed a bioinformatics analysis of DUF5.sub.Vv which suggested that catalytic residues for the Ras-protease activity lie in the C2B region (data not shown). The tertiary structures of members of the DUF399 and erythromycin esterase families had been solved, allowing a model of TIKI, which has a very similar primary structure, to be generated. (See Sanchez-Pulido, L. and C. P. Ponting, “Tiki, at the head of a new superfamily of enzymes,” Bioinformatics, 2013. 29(19): p. 2371-4; the content of which is incorporated herein by reference in its entirety). Furthermore, the active site of TIKI had been determined and could be mapped onto the model. (See id.). The pocket

of TIKI containing its catalytic residues shared structural similarities to a pocket on the C2B region of RRSP. We preformed an overlay of RRSP-C2B and BcR135 of the erythromycin esterase family which informed us of the putative catalytic residues of RRSP including E321, H323, E351, H352, and H451 (data not shown).

(331) We then assessed whether mutation of putative active site residues E321, H323, E351, H352, and H451 eliminates RRSP activity. Suspected catalytic residues E321, H323, E351, H352, and H451 (cumulatively referred to as "TIKI" residues) were mutated to alanines. Cleavage of recombinant KRas by recombinant WT RRSP and TIKI mutant RRSP then was performed in vitro and analyzed by SDS-Page analysis. RRSP with the five aforementioned substitutions and recombinant KRas were purified and mixed at equimolar concentration (10 μM) for 30 minutes at 37° C. No cleavage was observed. (See FIG. **38**).

(332) We next tested individual putative active site residues E321, H323, E351, H352, and H451 in RRSP. Each of E321, H323, E351, H352, and H451 were substituted with alanine in RRSP to create five variant forms of RRSP called E321A, H323A, E351A, H352A, and H451A. Recombinant forms of each of E321A, H323A, E351A, H352A, and H451A was synthesized and purified and mixed with recombinant KRas at equimolar concentration (10 μM) for 30 minutes at 37° C. Cleavage was assessed by SDS-Page analysis. No cleavage was observed in the E351A variant and the H451A variant. (See FIG. **39**). This suggests that E351 and H451 are required for the cleavage activity of RRSP for KRas.

(333) Next, using DALI server, we identified significant structural homologs with RRSP C2B domain (residues 277-508). In particular, secondary structure folding comprised between residues 303-474 of RRSP C2B showed similar topology with the bacterial type III effector protein HopBA1 of *Pseudomonas syringae* (PDB:5TO9), the erythromycin esterase (EraA)-like Bcr136 from *Bacillus cereus* (PDB:3BB5) and the ChaN heme-binding protein from *Campylobacter jejuni* (PDB:2G5G) (data not shown). Interestingly, we identified residues in RRSP C2B that were 100% conserved with the putative catalytic residues in HopBA1, Bcr136 and ChaN which included E321, H323, E351, and H451 (data not shown). Alanine substitution of E321 did not affected the catalytic activity of RRSP (see FIG. **40**) while RRSP H323A showed 50% reduction of Ras cleavage (see FIG. **40**). However, RRSP E351A and H451A did not process KRas (see FIG. **40**), suggesting their major role in the catalytic mechanism. RRSP H352 and R422 residues are in structural proximity of E351 and H451, and they were substituted with alanines residues to test their possible involvement with RRSP activity. Although RRSP R422A was still able to cleave Ras, RRSP H352A barely cleaved Ras (see FIG. **40**). Overall, these results demonstrate that RRSP E351 and H451 are putative catalytic residues, in accordance to HopBA1, Bcr136 and ChaN. However, H352 is present only in RRSP sequence suggesting an additional role of this residue, which could be involved in substrate recognition.

(334) We next tested whether the RRSP variants were structurally stable. FIG. **41** illustrates the fluorescent thermal shift for wild-type RRSP and RRSP variants are structurally stable. The denaturation profile of each RRSP variant does not vary greatly from that of the wild type, demonstrating that their respective mutations did not significantly alter tertiary structure. In addition, the melting temperature of each RRSP variant as determined by denaturation curves does not vary greatly from that of the wild type, again demonstrating that their respective mutations did not significantly alter tertiary structure.

(335) Finally, we tested whether RRSP is a metalloprotease. Recombinant RRSP and recombinant KRas were purified and mixed at equimolar concentration (10 μM) with varying concentrations of phenanthroline (which is a metal complexing reagent) in DMSO for 30 minutes at 37° C. Cleavage was still observed. (See FIG. **42**A). In addition, recombinant RRSP and recombinant KRas were purified and mixed at equimolar concentration (10 μM) with varying concentrations of EDTA for 30 minutes at 37° C. Cleavage was still observed. (See FIG. **42**A).

Conclusions

(336) We conclude that RRSP processing of Ras is catalyzed by a Glu/His pair in the C2B region of DUF5. However, RRSP is not a metalloprotease, but rather utilizes a mechanism of cleavage novel to its family of proteases.

Example 5—Proteolytic Pan-RAS Cleavage Leads to Tumor Regression in Patient-Derived Pancreatic Cancer Xenografts

(337) Reference is made Vidimas V., et al, Proteolytic pan-RAS cleavage leads to tumor regression in patient-derived pancreatic cancer xenografts. Mol. Cancer Ther. 2022, 5 (21), the content of which is incorporate herein by reference in its entirety.

(338) Abstract

(339) The lack of effective RAS inhibition represents a major unmet medical need in the treatment of pancreatic ductal adenocarcinoma (PDAC). Here, we investigate the anticancer activity of RRSP-DTB, an engineered biologic that cleaves the Switch I of all RAS isoforms, in KRAS-mutant PDAC cell lines and patient-derived xenografts (PDXs). We first demonstrate that RRSP-DTB effectively engages RAS and impacts downstream ERK signaling in multiple KRAS-mutant PDAC cell lines inhibiting cell proliferation at picomolar concentrations. We next tested RRSP-DTB in immunodeficient mice bearing KRAS-mutant PDAC PDXs. Treatment with RRSP-DTB led to ≥95% tumor regression after 29 days. Residual tumors exhibited disrupted tissue architecture, increased fibrosis and fewer proliferating cells compared to controls. Intratumoral levels of phospho-ERK were also significantly lower, indicating in vivo target engagement. Importantly, tumors that started to regrow without RRSP-DTB shrank when treatment resumed, demonstrating resistance to RRSP-DTB had not developed. Tracking persistence of the toxin activity following intraperitoneal injection showed that RRSP-DTB is active in sera from immunocompetent mice for at least one hour, but absent after 16 hours, justifying use of daily dosing. Overall, we report that RRSP-DTB strongly regresses hard-to-treat KRAS-mutant PDX models of pancreatic cancer, warranting further development of this pan-RAS biologic for the management of RAS-addicted tumors.

Introduction

(340) Pancreatic cancer is the 11th most commonly diagnosed cancer in the U.S. and the 3rd leading cause of cancer-related deaths, with an overall 5-year survival rate of 9%.sup.1. Poor prognosis is mainly attributed to late-stage clinical detection, early metastatic dissemination and lack of effective treatment options.sup.2, 3. Pancreatic ductal adenocarcinoma (PDAC) is the most common histologic subtype of pancreatic cancer, accounting for ˜90% of cases.sup.4, 5. PDAC genetic profiling has found that KRAS is mutated in ˜95% of patients, with NRAS and HRAS mutations accounting for <1% each and wild-type KRAS (KRASWT) for the remaining 5%.sup.6-8. Mutations in RAS genes (H-, N- and KRAS) result in RAS proteins being locked in the active GTP-bound state with consequent permanent activation of downstream RAS/MEK/ERK signaling, which controls cancer cell proliferation and survival.sup.9-11. Following decades of failures, inhibition of KRAS specific subtypes has had recent success.sup.12-16. Notably, sotorasib (LUMAKRAS™) was recently approved by the FDA as the first therapy for patients with KRASG12C-mutant non-small cell lung cancer.sup.17, 18.

(341) However, the usefulness of these KRASG12C inhibitors for PDAC is limited to ˜2% of patients.sup.19. Therefore, there is an urgent need for effective therapeutic strategies that target KRAS directly in PDAC.

(342) We recently demonstrated that the pan-RAS biologic RRSP-DT.sub.B inhibits tumor growth in xenograft models of triple-negative breast cancer (TNBC) and colorectal cancer (CRC) in mice via direct RAS cleavage.sup.20. RRSP-DTB is an engineered chimeric toxin comprised of the RAS/RAP1 specific endopeptidase (RRSP) from Gram-negative *Vibrio vulnificus* and the protein delivery machinery of diphtheria toxin (DTB). DTB delivers RRSP intracellularly via Heparin-binding EGF-like growth factor (HB-EGF)-mediated endocytosis in receptor-bearing cells only.sup.20. Human HB-EGF acts as the unique receptor of diphtheria toxin (DT) and is widely

expressed in epithelial tumors, including PDAC.sup.21. where it has been shown to cooperate with KRAS to promote KRAS-driven tumorigenesis.sup.22. Once released into the cytoplasm, RRSP cleaves RAS and RAP1 with high specificity within the Switch I region, which enables RAS binding to effectors and downstream signal transduction.sup.23, 24. All three RAS isoforms (H-, N- and KRAS) are cleaved by RRSP whether they are bound to GDP or GTP, so that the entire RAS cellular pool is inactivated.sup.25. Importantly, RRSP processes the main oncogenic RAS proteins with activating point mutations at residues G12, G13 and Q61 23. Cleavage of all RAS isoforms, mutant and wild-type, is an inherent advantage of RRSP and represents a potent strategy to reduce or regress growth in a broader spectrum of tumor types. The processing of RAS by RRSP-DTB leads to a variety of cellular outcomes including cell cycle arrest via upregulation of p27, apoptosis, or senescence. The differential cell fates were not correlated with the KRAS mutation, but rather, the mechanisms for inhibition of cell growth upon loss of RAS depend on signaling networks downstream of RAS.sup.26.

(343) Here, we use human KRAS-mutant PDAC cell lines and clinically relevant PDAC patient-derived xenografts (PDXs) to demonstrate preclinical effectiveness of the pan-RAS inhibitor RRSP-DT.sub.B against hard-to-treat PDAC. We also provide data on the stability of the toxin in the bloodstream following intraperitoneal (i.p.) injection, useful to support advanced RRSP-DT.sub.B therapeutic development.

Materials and Methods

(344) Protein Preparations, Chemicals and Cell Lines

(345) Purification of RRSP-DTB and catalytically inactive RRSP*-DTB proteins and endotoxin removal were performed as previously described.sup.20. All chemicals were from Sigma-Aldrich unless otherwise specified. Validated cell lines were kindly provided from the Reference Reagent Resource of the National Cancer Institute RAS Initiative at Frederick National Laboratory for Cancer Research (FNLCR). Cell lines at this resource are obtained by the Ras Initiative from collaborators or commercial sources and confirmed free of *Mycoplasma* using VenorGeM *Mycoplasma* Classic Endpoint PCR assay and are also subjected to short tandem repeat analysis using the AmpFLSTR Identifier PCR Amplification Kit to authenticate the cell lines, comparing the results to information located at https://web.expasy.org/cellosaurus/. Cells were cultured at 37° C. and 5% CO_2 atmosphere. YAPC and SUIT-2 were grown in RPMI-1640 (ATCC formulation) containing 10% fetal bovine serum (FBS; Gemini) and 1% penicillin/streptomycin (P/S; Invitrogen). KP-1N were grown in RPMI-1640 containing 5% FBS and 1% P/S. KP-4 were grown in DMEM-F12 with Glutamax (Gibco) containing 5% FBS and 1% P/S. PANC-1 were grown in DMEM (ATCC formulation) with 10% FBS and 1% P/S.

(346) Cytotoxicity, Viability, Clonogenic and Apoptosis Assays

(347) Cytotoxicity was assessed by staining cells with crystal violet. Briefly, 50,000 cells/well were cultured in 48-well plates and treated with RRSP-DTB and RRSP*-DTB for 72 hours. Cells were washed and crystal violet fixing/staining solution was added for 20 min at room temperature as previously described 20. Images of air-dried plates were acquired using a conventional desktop scanner.

(348) Cell viability was measured using CellTiter-Glo (Promega). 10,000 cells/well were grown in 96-well clear bottom white plates and treated with RRSP-DTB and RRSP*-DTB for 72 hours. CellTiter-Glo was then added to each well following the manufacturer's instructions and luminescence was recorded using a Tecan Safire2 plate reader. IC50 were calculated using the log(inhibitor) vs. response—variable slope (four parameters) function in Graphpad Prism v8.

(349) Cell proliferation was assessed by colony-formation assay. RRSP-DTB and RRSP*-DTB-treated cells for 72 hours were trypsinized, counted on a hemocytometer and replated in 48-well plates at 250 cells per well. Culture medium was replaced biweekly. On day 14, a crystal violet fixing/staining solution was added to the plates. Quantitative changes in cytotoxicity or clonogenicity were determined by solubilizing crystal violet-stained plates with 10% acetic acid

and measuring the absorbance at 590 nm using a plate reader (Tecan Safire2).
(350) Apoptosis was assessed by Caspase-Glo 3/7 assay (Promega). 10,000 cells/well were grown in 96-well clear bottom white plates and treated with 0.1 and 10 nM of RRSP-DTB and RRSP*-DTB as well as 1 μM of Staurosporine (positive control) for 24 and 48 hours. Caspase-Glo 3/7 was then added to each well following the manufacturer's instructions and luminescence was recorded using a Tecan Safire2 plate reader.
(351) SDS-PAGE and Western Blotting
(352) Protein extracts were prepared from cells by adding M-PER mammalian protein extraction reagent (Thermo Fisher Scientific) with protease and phosphatase inhibitors (Sigma-Aldrich). Protein content was measured using the Bio-Rad protein assay dye reagent concentrate (#5000006). Equal amounts of proteins were separated by SDS-PAGE followed by western blot analysis as previously described 24. Membranes were blotted using the following antibodies: anti-panRAS (Thermo Fisher Scientific Cat #MA1-012, RRID:AB 2536664), which recognizes RAS Switch I and thus detects only uncleaved RAS. Anti-mAb 4E8, a pan-RAS monoclonal antibody that recognizes both cleaved and uncleaved bands of all three RAS isoforms was prepared from a hybridoma cell line obtained from the National Cancer Institute RAS Initiative. The antibody preparation used in this study was previously validated 20. Anti-Phospho-p44/42 MAPK (pERK1/2; Cell Signaling Technology Cat #4377, RRID:AB_331775), anti-p44/42 MAPK (ERK1/2; Cell Signaling Technology Cat #4696, RRID:AB_390780), and anti-HB-EGF (Abcam, #ab185555). Anti-vinculin (Cell Signaling Technology Cat #13901, RRID:AB_2728768) was used for normalization. Secondary antibodies used were fluorescent-labeled IRDye 680RD goat anti-mouse (LI-COR Biosciences Cat #926-68070, RRID:AB_10956588) and IRDye 800CW goat anti-rabbit (LI-COR Biosciences Cat#926-32211, RRID:AB_621843). Blot images were acquired using the Odyssey Infrared Imaging System (LI COR Biosciences) and quantified by densitometry using NIH ImageJ software (ImageJ, RRID:SCR 003070). Percentage of uncleaved RAS was calculated as previously described.sup.24.
(353) Patient-Derived Xenograft (PDX) Studies
(354) Human PDAC biospecimens were obtained from the Pathology Core Facility (PCF) Biobank at Northwestern University. Collection of human tissue specimens and the PDX mouse protocol were approved by the Institutional Review Board and Institutional Animal Care and Use Committee (IACUC) at Northwestern University. All PDAC tumor models used have been fully tested and validated across the entire spectrum of creating, propagating, and characterizing the PDX models. The established PDAC tumor models are defined as the surgical collected fresh tumor tissue are subcutaneously implanted into NSG mice. This process is repeated two more times to ensure that the tumors can be propagated through at least three passages. Cryopreserved fragments are thawed and re-implanted to make sure that frozen tissue will regrow. At each passage, tumor tissue is compared to the original patient tumor on the basis of histopathology, immunohistochemistry for expression of clinically relevant biomarkers. Only PDX tumors that retain the histological characteristics of the patient tumor and to exclude development of lymphoma are considered confirmed. All tissues are routinely tested for *Mycoplasma* using the Universal *Mycoplasma* Detection Kit (American Type Culture Collection, Cat #30-1012K) for our studies. Molecular profiling for tumor genomic and transcriptomic characteristics and response to a standard-of-care combination therapy for PDAC PDX tumor are also used to establish the baseline characteristics of the model. At the time of this study, multiple KRASG12V PDXs were available from the repository and we selected two of them from patients that did not receive any adjuvant or neo-adjuvant therapy highlighting the different sex (female vs male) as a valuable variable. Although our repository included one KRASG12D PDX, it was from a patient that received multiple neo-adjuvant therapies (gemcitabine, oxaliplatin), tumor pathology/staging report was not available at the time of the study and therefore was not considered representative for this study. PDXs were established by the Center for Developmental Therapeutics (CDT) at Northwestern

University 27. Briefly, cryopreserved human PDAC biospecimens (passage 0, P0) were thawed, cut into small fragments, and two tissue fragments were xenografted into the dorsal region of two female NSG mice (P1). When tumor masses reached ˜1500 mm3 in size, tumors were excised, divided into equally sized specimens (˜2×2×2 mm) and implanted subcutaneously into the dorsal region of the appropriate number of female NSG mice (P2). Once tumors reached 150-200 mm3, mice underwent balanced randomization based on tumor size. Briefly, mice were grouped three per cage (two cages used for each treatment) so that differences in the median tumor size were minimized among groups prior to treatment administration. Next, mice were dosed i.p. with endotoxin-free 0.1 mg/kg of RRSP-DTB or RRSP*-DTB on a 5 days ON/2 days OFF schedule for four weeks. Control mice received endotoxin-free phosphate buffered saline (MilliporeSigma). After four weeks, all mice (except three mice from the RRSP-DTB-treatment group) were humanely euthanized, tumors excised, and fixed in 10% formalin overnight. The remaining three mice from the RRSP-DTB group were housed without treatment for three weeks and subsequently dosed with RRSP-DTB at the indicated treatment regimens. Tumor growth was monitored by digital caliper measurements of length (1) and width (w) of the tumors. Tumor volume was calculated using the following formula: volume (mm3)=(1×w2)/2. Mouse body weight was also measured regularly. Percentage change in tumor volume between day 1 and day 29 was calculated for each individual tumor as described in Patel et al..sup.28

(355) In Vivo Stability Study

(356) Five female C57BL/6 mice per group were injected i.p. with 0.1 mg/kg or 0.5 mg/kg of RRSP-DTB. Saline was used as vehicle control. After 1 and 16 hours, blood was collected from anesthetized mice via orbital bleeding, transferred to serum separator collection tubes (BD Biosciences, #365967), allowed to clot for at least 30 minutes and then centrifuged for 10 minutes at 10,000×g. The serum supernatant was aliquoted and stored at −80° C. until use. To determine whether RRSP-DTB was active in the blood after a single injection, serum collected from mice was diluted 1:100 dilution with saline, added directly to YAPC or KP-4 cultured pancreatic cells, and total levels of intracellular RAS cleaved after 24 hours was assessed by western blot as described above.

(357) Histology, Immunohistochemistry and Image Analysis

(358) Paraffin-embedding, sectioning, hematoxylin and eosin (H&E) and Masson's trichrome stainings as well as immunohistochemical staining of PDX specimens were performed by the Robert H. Lurie Comprehensive Cancer Center Pathology Core Facility. Tumor sections were stained with anti-Cytokeratin 19 ((CK-19), #ab76539; Abcam), anti-Ki-67 (#GA626; Dako), anti-cleaved caspase 3 ((CC3, #9661; Cell Signaling Technology), anti-pan-RAS ((RAS), #PA5-85947; Thermo Fisher Scientific) and anti-Phospho-p44/42 MAPK ((ERK1/2) (Thr202/Tyr204, (D13.14.4E) XP, #4370; Cell Signaling Technology) antibodies as described previously 20. Primary antibodies were detected using the appropriate secondary antibodies and 3,3′-diaminobenzidine (DAB) revelation (Dako).

(359) All slides were scanned using a Nanozoomer HT slide scanner (Hamamatsu). Quantification of positively immunostained cells was performed using customized Application Protocol Packages (APPs) within the VIS image analysis software (Visiopharm). Positive cells were counted in the whole tumor sections and expressed as the percentage ratio over the area of the whole section. Because the pattern of cytoplasmic RAS staining was diffuse, we quantified RAS signal intensity by color deconvolution using ImageJ (Fiji version) as previously described.sup.29.

(360) Statistical Analysis

(361) Graphpad Prism v.8 software was used for statistical analysis. Bar plots represent the mean of at least three independent experiments±the standard deviation (SD). Statistical significance was assessed using one-way analysis of variance (ANOVA) assuming normal distribution. Dunnett's multiple comparison post-test was employed to compare the mean of the control group to the mean of treatment groups. Tukey's multiple comparison test was used to compare the mean of each group

with the mean of every other group. Points in the fitted dose-response curve represent mean±standard error of the mean (SEM). Statistical analysis on fold-change data was performed after log transformation to obtain a more normalized distribution. For in vivo PDXs, data are reported as mean±SEM and one-way ANOVA was performed to assess differences among the treatment arms. Values of $p<0.05$ were considered statistically significant.

(362) Results

(363) RRSP-DT.sub.B Cleaves RAS in KRAS-Mutant PDAC Cell Lines

(364) For this study, we selected five different KRAS mutant PDAC cell lines, including YAPC (G12V), SUIT-2 (G12D), KP-1N (G12D), KP-4 (G12D) and PANC-1 (G12D). All five cell lines used expressed the DT receptor HB-EGF. However, an originally selected cell line HUP-T4 cells expressed low levels of HB-EGF, and subsequently was excluded from the study (FIG. **48**A).

(365) In order to assess intracellular delivery of RRSP via DT.sub.B and successful RAS target engagement, PDAC cells were treated with increasing concentrations of RRSP-DT.sub.B up to 10 nM and with 10 nM of catalytically inactive RRSP*-DT.sub.B for 24 hours. Western blotting of cell lysates showed that RRSP-DT.sub.B cleaved total RAS in all PDAC cell lines with similar picomolar potency (FIG. **43**A-E). The half-maximal inhibition concentration (IC.sub.50) values extrapolated from dose-response curves of uncleaved RAS normalized to vinculin showed that 50% of RAS cleavage was achieved with concentrations ranging between 10 and 280 pM (FIG. **43**F). The appearance of cleaved RAS also tracked with a significant reduction in levels of phosphorylated ERK (pERK) in all five cell lines (FIG. **43**A-F). Altogether, these data demonstrate successful DT.sub.B-mediated delivery of RRSP in PDAC cells and effective target engagement as shown by RAS cleavage and reduced pERK.

(366) RRSP-DT.sub.B Impacts Viability and Proliferation of KRAS-Mutant PDAC Cell Lines

(367) To examine the cytotoxic and/or growth inhibitory effects of RRSP-DT.sub.B, the treatment of PDAC cell lines was extended to 72 hours. A crystal violet cytotoxicity assay showed reduced confluency for YAPC and SUIT-2 cells indicating cytotoxicity, and moderate cell loss in KP-1N, KP-4 and PANC-1 cells compared to the control treatments indicating growth inhibition (FIG. **44**A and FIG. **48**B). Quantitative assessment of cell viability using CellTiter-Glo, which detects metabolic activity, confirmed that YAPC and SUIT-2 cells lost viability with IC.sub.50 in the picomolar range, while KP-1N, KP-4, and PANC-1 remained viable although with reduced proliferation (FIG. **44**B).

(368) KP-4 cells were found to generate structurally well-defined three-dimensional (3D) spheroids. When treated with RRSP-DT.sub.B, these spheroids not only showed a strong dose- and time-dependent reduction of size, but also of viability as measured by the 3D CellTiter-Glo assay (FIG. **45**C). These data show that the KRAS-mutant KP-4 cell line became more sensitive to RAS inhibition in 3D spheroid culture conditions compared to monolayers, consistent with previous findings.sup.30.

(369) The sensitivity of all cells to RRSP-DT.sub.B resulting in either cytotoxicity or irreversible growth inhibition was confirmed using a clonogenic assay. Following RRSP-DT.sub.B treatment for 72 hours, cells were harvested, counted, and replated at low density. At day 14, all cell lines failed to form colonies equivalent to the control treatment groups demonstrating an inability of residual cells to proliferate. Overall, these data reveal that RRSP-DT.sub.B strongly affects viability and proliferation of KRAS-mutant PDAC cell lines. Importantly, cell lines previously classified as KRAS-dependent were confirmed to be highly susceptible, and those classified as KRAS-independent (KP-1N, KP-4 and PANC-1).sup.31 were also sensitive to RRSP-DT.sub.B treatment.

(370) RRSP-DT.sub.B Leads to In Vivo Regression of KRAS-Mutant PDAC Xenografts

(371) Patient-derived xenografts (PDXs) are a powerful tool in translational cancer research because they retain biological characteristics of the parental tumor specimens and can better predict a patient response to an investigational drug than classic cell line-based in vivo models of

cancer.sup.32. We established two PDAC PDX models, one from a female donor (PDX1) and one from a male donor (PDX2), both harboring a KRAS G12V mutation, which accounts for 30% of KRAS mutations in PDAC.sup.27. NSG mice were engrafted with patient-derived PDAC tumors (eighteen mice per PDX group) and then six mice per group were treated i.p. with 0.1 mg/kg of RRSP-DT.sub.B or RRSP*-DT.sub.B once per day (q.d.) for 4 weeks (weekends excluded) or were mock-injected with saline. RAS inhibition by RRSP-DT.sub.B significantly reduced the growth of subcutaneously implanted KRAS-mutant PDX1 and PDX2 tumors, starting as soon as 8 days after treatment initiation (FIGS. **45**A and **45**B). After four weeks of treatment, on day 29, maximum growth inhibition was observed (FIGS. **45**C and **45**D). In both PDX1 and PDX2 studies, tumor regression, not just inhibition of tumor growth, was observed in all twelve mice treated with RRSP-DT.sub.B indicating successful treatment of pre-established tumors (FIGS. **45**E and **45**F). The reduction of tumor size was specifically due to the proteolytic action of RRSP, since the catalytically inactive RRSP*-DT.sub.B did not affect tumor growth.
(372) In order to determine whether tumor growth occurred in residual tumors after stopping the treatment, three mice from the RRSP-DT.sub.B group were left untreated from day 29 for about three weeks and tumor growth was regularly monitored. In both PDX1 and PDX2, we did not observe an increase in tumor size during the first 10 days (day 29-39) following RRSP-DT.sub.B withdrawal. Between day 39 and day 53, a slight increase in tumor size occurred in mice from PDX1 (FIG. **45**A) and PDX2 (FIG. **45**B), where the average tumor size reached the initial baseline. On day 53, treatment with RRSP-DT.sub.B was reinitiated. The treatment of PDX1 mice was originally de-escalated to every other day (q.o.d). As there was no reduction in tumor size after one week, RRSP-DT.sub.B treatment frequency was increased to five days per week during the second week. Increase in the treatment frequency ultimately led to a reduction in tumor size (FIG. **45**A).
(373) Since tumors from PDX2 had a higher growth rate compared to those from PDX1, mice were retreated with RRSP-DT.sub.B once per day (q.d.) for four weeks. At the end of this additional four-week treatment cycle, tumors reached a size comparable to that observed around day 39 (FIG. **45**B). Of importance, RRSP-DT.sub.B treatment was well tolerated in mice, as demonstrated by the lack of weight loss (FIG. **49**) across both treatment cycles.
(374) Altogether, these data demonstrate that RRSP-DT.sub.B treatment resulted in rapid in vivo regression of PDAC PDXs. Most importantly, PDAC tumors retained sensitivity to RRSP-DT.sub.B over time, did not develop resistance and remained sensitive for a second round of treatment.
(375) RRSP-DT.sub.B Treatment Results in Increased Fibrosis, Reduced Proliferation and Downregulation of pERK in In Vivo PDAC Tumors
(376) Histological and immunohistochemical analysis showed that the large PDX tumors resected from vehicle- and RRSP*-DT.sub.B-treated mice were characterized by small glandular components and extensive necrosis in the inner tumor core typical of clinical PDAC. By contrast, tumors from mice treated with RRSP-DT.sub.B showed absence of necrosis, reduction in the cell number, larger residual tumor cells and appearance of vacuolar structures indicative of tissue degeneration (FIG. **50**). In addition, RRSP-DT.sub.B-treated tumors were highly fibrotic (FIGS. **46**A and **46**B).
(377) Cytokeratin 19 (CK-19) is a characteristic ductal epithelial marker and poor prognostic factor in PDAC 33, 34 The percentage of cells in the tumors that stained positive for CK-19 was significantly lower in all tumors treated with RRSP-DT.sub.B compared to saline and RRSP*-DT.sub.B-treated mice (FIGS. **46**C and **46**F). This result indicated that fewer PDAC cells were present in the residual tumors. In addition, tumors from mice in the RRSP-DT.sub.B treatment group showed a marked reduction in proliferating Ki-67 positive cells, both after the first 4 weeks of treatment and also after the second treatment regimen that started at day 53 (FIGS. **46**D and **46**G).
(378) In addition, PDAC tumors from PDX1 and PDX2 excised at day 29 showed significantly

lower detection of pERK compared to saline and RRSP*-DT.sub.B-treated tumors (FIGS. **46**E and **46**H). In PDX1, we observed an increase in pERK levels on day 67 in tumors from mice that received RRSP-DT.sub.B every other day (q.o.d) during the first week and once per day (q.d.) during the second week following RRSP-DT.sub.B withdrawal. This tracked with a higher percentage of residual cancer cells positive to CK-19 and Ki-67 compared to day 29. By contrast, in mice from PDX2 that received the more aggressive second round of treatment, low levels of pERK were also observed in tumors resected on day 81. Analysis of total RAS levels showed significant reduction of RAS expression in sections from tumors treated with RRSP-DT.sub.B at day 67 in PDX1 and, although not significant, a similar trend was observed in PDX2 at day 29 (FIG. **51**). RAS levels in histological sections from PDX2 tumors at day 81 are not shown because of staining-related technical issues.

(379) Next, to investigate whether apoptosis was involved in RRSP-DT.sub.B-mediated anti-tumor activity, cleaved caspase 3 (CC3) expression was assessed in both PDX1 and PDX2. No changes in CC3 levels were detected in tumors treated with RRSP-DT.sub.B compared to controls. Similarly, activation of caspase 3/7 was not observed in four out of five PDAC cell lines tested. Only YAPC cells showed an increase of caspase 3/7 activity upon treatment with RRSP-DT.sub.B (FIG. **52**).

(380) Taken together, these results demonstrate that treatment of PDAC PDXs with RRSP-DT.sub.B leads to significant reduction in the percentage of cancer cells, increased fibrosis and decreased proliferation independently of apoptosis. Importantly, RRSP-DT.sub.B strongly reduces pERK levels, which is indicative of effective downstream RAS signaling downregulation.

(381) Evaluation of the In Vivo Stability of RRSP-DT.sub.B in Immunocompetent Mice

(382) An important aspect underlying the translatability of RRSP-DT.sub.B as an anticancer agent is to determine its in vivo stability. In order to do this, immunocompetent C57/BL6 mice were given a single i.p. injection of 0.1 or 0.5 mg/kg of RRSP-DT.sub.B. Serum from mice collected one and 16 hours after injection was diluted 1:100 and then was added to YAPC and KP-4 cells to test for recovery of cellular active RRSP-DT.sub.B from the bloodstream.

(383) We found that in sera from mice that received 0.5 mg/kg of RRSP-DT.sub.B for one hour, RRSP-DT.sub.B was active. Indeed, a remarkable reduction in RAS levels was observed in both YAPC and KP-4 cells treated with mouse sera. In KP-4 cells, reduced RAS levels were also detected at 0.1 mg/kg of RRSP-DT.sub.B. We did not observe a significant decrease of RAS levels in cells treated with sera from mice that received RRSP-DT.sub.B for 16 hours (FIG. **47**A and FIG. **53**), suggesting that RRSP-DT.sub.B that had been exposed to mouse serum for a longer time became less active and was not able to significantly reduce RAS levels as efficiently as at earlier times.

Discussion

(384) Aberrant KRAS signaling is considered the core hallmark of pancreatic cancer onset and progression.sup.35. Although significant advances have been made over recent years in the treatment of pancreatic cancer.sup.36, there are currently no KRAS inhibitors approved for PDAC, which still remains an urgent unmet medical need in oncology.

(385) We recently reported the therapeutic potential of biologic RRSP-DT.sub.B as a potent pan-RAS inhibitor that halts tumor growth of multiple TNBC and CRC xenografts harboring either wild-type or mutant RAS.sup.37. The NCI-60 human tumor cell line screening revealed that RRSP-DT.sub.B, in addition to its potency, has a broad spectrum of antitumor activity against several cancer types.sup.20. Because the NCI-60 screen does not include pancreatic cancer, we tested RRSP-DT.sub.B on a panel of five KRAS-mutant PDAC cell lines and demonstrated that picomolar concentrations of RRSP-DT.sub.B leads to growth inhibition and loss of cell proliferation, although only cells treated with the highest dose of 10 nM showed loss of metabolic activity. This was not completely unexpected since KP-1N, KP-4 and PANC-1 cells were previously classified as KRAS-independent by Singh and co-authors.sup.31. Interestingly, when KRAS-independent KP-4 cells were cultured as 3D spheroids, a stronger decrease in viability and

spheroids' size was observed compared to monolayers. We believe the greater sensitivity of KP-4 spheroids to RRSP-DT.sub.B might be due to the 3D suspension culture environment, thus corroborating previous findings showing that different culture systems have an impact on KRAS dependency.sup.30. These data confirm that PDAC cell lines previously listed as KRAS-dependent were extremely susceptible to RRSP-DT.sub.B. Most importantly, long-lasting growth inhibitory effects were seen in mutant KRAS-independent PDAC cell lines as well. Although the mechanisms underlying RRSP-DT.sub.B-induced growth inhibition in PDAC cell lines are still unknown, we have recently demonstrated that RRSP-DT.sub.B can lead to irreversible G1 cell cycle growth arrest via p27, induce a senescence-like phenotype and trigger apoptosis in a subset of colorectal cancer cell lines.sup.38. Our data show that apoptosis is not overall activated in PDAC tumors or cell lines following RRSP-DT.sub.B treatment. Although this was surprising, it was not completely unexpected since, in a bacterial pathogenesis context, toxins such as RRSP have evolved to hijack cellular pathways in order to avoid host innate responses and promote systemic spread without killing of the host. Overall, our data suggest that the main mechanisms underlying RRSP's antitumor activity involve cell cycle arrest and/or senescence rather than apoptosis.sup.26. In support of our findings, an independent study has recently shown that RRSP from *Photorhabdus asymbiotica*, which is 73% similar in amino acid sequence to RRSP from *Vibrio vulnificus*, also cleaves RAS.sup.24 and induces G1 cell cycle arrest via direct binding to CDK1.sup.39.
(386) Our data suggest that, unlike currently available KRAS inhibitors, RRSP-DT.sub.B has the potential to target undruggable KRAS in a wide spectrum of KRAS-mutant pancreatic cancers whether they rely or not on KRAS for survival. One of the inherent advantages of RRSP is that it cleaves all RAS isoforms, mutant and wild-type, thus representing a potent strategy to inhibit tumor growth. While one line of research has been focusing on targeting mutant KRAS only with molecules that enter all cells to ensure safety, our current efforts are directed to transporting RRSP preferentially into cancer cells while sparing normal cells, an approach previously found to be successful for other toxin-based targeting strategies.sup.40, 41, including the therapeutic Lumoxiti®, which is an FDA-approved treatment for hairy cell leukemia.sup.42.
(387) RRSP-DT.sub.B has been shown to halt tumor growth in mouse xenografts derived from TNBC and CRC cell lines.sup.37. Here, we report that administration of RRSP-DT.sub.B to NSG mice bearing KRAS-mutant PDAC PDXs not only slowed tumor growth, but also led to tumor regression as early as one week after treatment initiation. Following a drug washout period, tumors from both PDX1 and PDX2 started to slowly regrow. Most importantly, upon re-administration of RRSP-DT.sub.B, these tumors maintain sensitivity to the drug without becoming resistant, which is something rarely seen in monotherapy regimens and might be advantageous for future development of RRSP as an anticancer therapeutic. Lack of resistance could also be explained by the fact that at day 29 residual tumors are mainly composed by a dense network of collagen fibers with few interspersed CK-19-positive PDAC cancer cells that can be readily targeted by RRSP-DT.sub.B. PDAC tumors have extensive fibrosis, which is believed to contribute to their resistance to most therapies. We showed that RRSP-DT.sub.B increased intracellular fibrosis in vivo, which could be considered disadvantageous. However, contrary to expectations, experimental.sup.43, 44 and clinical.sup.45 studies have demonstrated that depletion of the stroma component of PDAC results in more aggressive tumors in mice and worse outcomes in patients. Therefore, rather that depleting the stroma, strategies that specifically target or modify the extracellular matrix (ECM) are preferred and currently being investigated in PDAC. Although we do not have a definitive answer to the question of whether RRSP-mediated increase in fibrosis is advantageous or disadvantageous following RRSP-DT.sub.B treatment, based on the available literature and our findings, we can speculate that it will be beneficial. In addition, we believe that because DT.sub.B has been shown to be extremely efficient at delivering RRSP in cells expressing the HB-EGF receptor, the non-cellular ECM component present in the tumor does not represent an obstacle for RRSP-DT.sub.B to reach the residual cancer cells interspersed within the ECM, effectively inhibiting their growth.

(388) In PDX1, because mice were subjected to a shorter RRSP-DT.sub.B treatment regimen following the drug-washout phase, we observed a larger number of positive CK-19 cells and increase pERK levels despite no increased proliferation. These findings have helped us improve RRSP-DT.sub.B treatment schedule. Indeed, a more aggressive dosing regimen as in PDX2 resulted in fewer CK-19-positive cells and reduced pERK expression comparable to those seen before RRSP-DT.sub.B washout phase. Moreover, the in vivo stability study showed that RRSP-DT.sub.B injected i.p. is delivered to the bloodstream in immunocompetent mice although with minimal activity persisting until 16 hours, supporting the more aggressive daily dosing schedule. Notably, despite the extended and more frequent in vivo administration of RRSP-DT.sub.B, no systemic toxicity was observed in mice.

(389) Altogether, our study provides compelling evidence that engineered RAS endopeptidase RRSP-DT.sub.B is highly efficacious against PDAC. Another as yet untested potential advantage of RRSP-DT.sub.B is that in addition to RAS, RRSP cleaves the metastasis-associated GTPase RAP1. Future studies in metastatic cancer models could be done to probe RRSP efficacy against metastatization and local invasion as an added advantage of RRSP in the treatment of PDAC,

(390) In recent years, protein degradation has become a promising therapeutic modality to target RAS. Besides RRSP, the bacterial protease toxin subtilisin was recently re-engineered to degrade RAS in vitro and degraded RAS also when ectopically expressed in mammalian cells.sup.46. In addition, several studies showed that degradation of endogenous KRAS can be achieved following ectopic expression in cancer cells of chimeric proteins that tether RAS-targeting peptides.sup.47-50 and monobodies.sup.51 to a E3 ligase that ultimately drives KRAS ubiquitination and proteasomal degradation. RAS degradation, unlike inhibition of the protein activity, is a strategy that may lead to prolonged inactivation of downstream RAS signaling bypassing the problem of intrinsic resistance of RAS proteins that have been covalently inhibited with small molecules. We likewise showed in 2015 that RRSP can cleave RAS when ectopically expressed resulting in loss of activity and reduced RAS levels.sup.23. In contrast to all other studies on engineered RAS degraders that depend on stable transfection and ectopic expression, RRSP-DT.sub.B is the only RAS degrader shown to cleave RAS when added exogenously to cells and to reduce tumors following injection in mice. Further, we found that the active toxin moves from the peritoneum to the bloodstream. Thus, RRSP-DT.sub.B is a first-in-class RAS degrader now shown herein to have potential to treat PDAC resulting in tumor regression. In contrast to small molecule inhibitors that target only one form of RAS.sup.52, treatment with RRSP-DT.sub.B showed no resistance in mice that received RRSP-DT.sub.B after a drug washout phase. Because of the unique way RRSP readily cleaves and degrades RAS proteins and the fact that, to our knowledge, RRSP-DT.sub.B is the most advanced and documented RAS biodegrader as for in vitro potency and in vivo activity, we believe that it is the leading player in the growing targeted RAS degradation space. Thus, further investigation of RRSP-DT.sub.B as a therapeutic tool for degrading RAS is warranted and optimization of the delivery modality is planned to achieve selective cancer cell receptor targeting.

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Innovation (N Y) 1.

Example 6—A Novel Engineered Chimeric Toxin that Cleaves Activated Mutant and Wild-Type RAS Inhibits Tumor Growth

(392) Reference is, Vidimar V. et al. An engineered chimeric toxin that cleaves activated mutant and wild-type RAS inhibits tumor growth. PNAS 2020, 117 (29)., which is incorporated in its entirety.

(393) Despite nearly four decades of effort, broad inhibition of onco-genic RAS using small-molecule approaches has proven to be a major challenge. Here we describe the development of a pan-RAS biologic inhibitor composed of the RAS-RAP1-specific endo-peptidase fused to the protein delivery machinery of diphtheria toxin. We show that this engineered chimeric toxin irreversibly cleaves and inactivates intracellular RAS at low picomolar concen-trations terminating downstream signaling in receptor-bearing cells. Furthermore, we demonstrate in vivo target engagement and reduction of tumor burden in three mouse xenograft models driven by either wild-type or mutant RAS. Intracellular delivery of a potent anti-RAS biologic through a receptor-mediated mecha-nism represents a promising approach to developing RAS thera-peutics against a broad array of cancers.

(394) Significance

(395) RAS oncoproteins have long been considered among the most elusive drug targets in cancer research. At issue is the lack of accessible drug binding sites and the extreme affinity for its GTP substrate. Covalent inhibitors against the KRAS G12C mutant have shown early clinical promise, however, targeting the other oncogenic RAS mutants across three RAS isoforms has proven challenging. Inhibition of activated wild-type RAS in the absence of canonical RAS mutations is also highly desirable in certain tumors. Here, we demonstrate delivery of an extremely potent pan-RAS and RAP1 cleaving enzyme in therapeutic quantities to specific receptor-bearing cells in vitro and in vivo. We aim to advance this approach to engineer the first targeted pan-RAS inhibitor for cancer therapy.

Introduction

(396) More than one-third of all human cancers harbor activating mutations in RAS oncogenes. Among the major isoforms, KRAS is the most frequently mutated oncogene, found in nearly 25% of malignancies and 85% of RAS-driven cancers (1-3). Notably, three of the four deadliest cancers (pancreatic, co-lorectal, and lung) exhibit a high frequency of KRAS mutations (2, 3). Moreover, NRAS and HRAS are also known oncogenic drivers in other neoplasms (2). Activating point mutations in RAS genes impair the intrinsic capacity of RAS proteins to hydrolyze OTP, thus locking them in a constitutively activated OTP-bound state. This leads to constitutive activation of downstream transduction-signaling networks, such as the RAF/MEK/ERK (MAPK) axis, which drives survival and uncontrolled proliferation (4-6). Even in the absence of gain-of-function mutations, RAS genes still play a major role in tumorigenesis due to hlper-activation of RAS-signaling pathways via overexpression of up-stream receptor tyrosine kinases (RTKs) and/or amplification of wild-type RAS, such as in head and neck squamous cell carcinoma (7), espphageal and gastic cancers (8), ovarian adenocarcinoma (9), and triple-negative breast cancer (TNBC) (10, 11).

(397) Due to their major role in a wide spectrum of cancers, RAS proteins have become a primary target for drug discovery, and extensive effort has been directed to the development of selec-tive RAS inhibitors (12). However, the high affinity for OTP and the absence of drug-accessible binding pockets have complicated efforts for decades, earning RAS the moniker “undruggable” (12-14). Nevertheless, recent success has been achieved by se-lectively targeting KRAS G12C with small molecules that co-valently bind to Cys12 in the KRAS Switch-II pocket (15-17), and clinical trials are underway to validate their effectiveness (18-20). However, these pharmacophores are specific for KRAS G12C and cannot be expanded to other mutants. Furthermore, KRAS.sup.G12c mutations account for only ˜11% of all KRAS mutations in cancer (21), detected mainly in lung

(14%), colorectal (5%), and pancreatic (1 to 3%) cancer (13, 15). Therefore, there remains an urgent need for a broadly applicable pan-RASinhibitor for use against all RAS-driven tumors, either mutation-dependent or -independent.
(398) Recently, we discovered a RAS/RAP1 specific endopeptidase (RRSP) from *Viblio vulnificus* that site-specifically cleaves RAS and its close homolog RAP1 between residues Y32 and D33 within the Switch I, a region crucial to RAS-mediated signal transduction (22). RRSP is highly specific for RAS and RAP1 and does not cleave other closely related GTPases (23). Importantly, RRSP cleaves all three major RAS isoforms, as well as oncogenic RAS with mutations at position 12, 13, and 61 (22). RRSP also targets both active (OTP-bound) and inactive (GDP-bound) RAS, resulting in destruction of the entire cellular RAS pool (23). By proteolytically cleaving the Switch I loop, RRSP prevents RAS from undergoing GDP-OTP exchange and binding the downstream effector kinase RAF, ultimately termi-nating ERK signaling in cells (24).
(399) Assessing the therapeutic potential of RRSP, however, is precluded by the fact that RRSP is a 56-kDa domain of a larger protein toxin that alone does not readily diffuse across biological membranes. Recently, we demonstrated that the translocation machinery of diphtheria toxin (DT) can be engineered to deliver a broad diversity of passenger proteins into target cells (25, 26). In this study, we exploit the receptor-targeting and membrane translocation properties of DT to deliver RRSP into targeted cancer cells.
(400) We created an engineered chimeric toxin that uses the DT translocation system and show that RRSP potently and irreversibly destroys RAS. We demonstrate the anticancer properties of this engineered chimeric toxin both in vitro and in vivo, providing proof-of-concept for the therapeutic development of RRSP as a pan-RAS inhibitor.
(401) Results
(402) Engineered Diphtheria Toxin Binding and Translocation Domain DT 8 Efficiently Delivers RRSP into Cells. DT consists of a catalytically active A fragment (DTA) and a B fragment (DTB) that includes both a receptor-binding domain (DTR) and a translocation do-main (DTT). DT.sub.B alone can bind its surface receptor, the heparin-binding epidermal growth factor-like growth factor (HB-EGF), and transfer a wide range of protein cargos to the cell cytosol via a receptor-mediated endocytic mechanism (25). The introduction of two amino acid substitutions (K51E and E148K) into DTA resulted in a nontoxic, catalytically inactive A fragment, herein referred to as DTa. To examine RRSP in-tracellular delivery, we expressed RRSP fused at the amino terminus of DTa with an intervening (G.sub.4S).sub.2 linker, a design previously used to deliver other protein cargos (25, 26). Western blotting with the RAS10 pan-RAS antibody showed that treatment with RRSP-(G.sub.4S).sub.2-DTa-DT.sub.B resulted in depletion of RAS in HCT-116 (KRAS.sup.G13D) cells at 1 nM (FIG. **54**A, **61**A).
(403) An intrinsic feature of the first-generation design is that RRSP is codelivered with the DTa domain. To generate chimeras that would deliver RRSP alone into cells without DTa, we added the autoprocessing cysteine protease domain (CPD) from *V. vulnificus* MARTX toxin that autocleaves and thereby releases its toxin effectors (i.e., RRSP) upon binding inositol hexakisphosphate in the host cytosol (27). When added to cells, RRSP-CPD-DTa-DT.sub.B improved RAS cleavage (partial inactivation at 100 pM) (FIG. **54**B and FIG. **61**B). In the third iteration design, we removed DTa entirely and appended RRSP via a (G4S).sub.2 linker to the native release machinery of DT that includes DIT and DTR (DT.sub.B) and named it RRSP-DTB. Indeed, cells treated with this much smaller protein showed complete loss of detectable RAS at concentrations as low as 3 pM (FIG. **54**C and FIG. **61**C). Control proteins with the catalytically inactive mutant RRSP.sub.H4030A (RRSP*-DT.sub.B) or with the DTT domain removed (RRSP-DT.sub.B[ΔDTT]) failed to deplete RAS from cells (FIG. **54**D). The uptake of RRSP-DT.sub.B, translocation to cytosol, and subsequent RAS cleavage are summarized in FIG. **54**E. Altogether, these results demonstrate that DT.sub.B is highly effective for intracellular translocation of RRSP across membranes to target RAS within cells. This final chimeric recombinant toxin was purified on a large scale, enabling the

experimental investigation of the anticancer potential of RRSP (FIG. **62**A-H).

(404) RAS Processing by RRSP Strongly Affects Viability and Proliferation of TNBC Cancer Cells with Activated Wild-Type RAS.

(405) We first investigated the anticancer potential of RRSP-DT.sub.B in TNBC, a devastating disease and the last frontier in breast cancer research (28). Although only ˜5% of breast cancers harbor RAS mutations (10), RAS proteins can be pathologically activated in TNBC via upregulation of RTKs and/or amplification of wild-type RAS (10, 29, 30). The basal-like MDA-MB-436 cell line features over-expression of KRAS.sup.wt and hyperactivation of ERK signaling (31). Protein levels of HB-EGF, the DT receptor, were assessed by Western blotting in MDA-MB-436 cells and additional cell lines used in this study as a correlation parameter between HB-EGF expression and cytotoxicity (FIG. **63**A). HB-EGF-expressing MDA-MB-436 cells were treated with increasing concentrations of RRSP-DT.sub.B for 1 and 24 h, and levels of intracellular RAS and phosphorylated ERK1/2 (pERK) were measured by Western blot. Treatment with RRSP-DT.sub.B resulted in complete cleavage of RAS at 10 nM after 1 h, while 0.01 nM RRSP-DT.sub.B was sufficient to completely cleave RAS after 24 h. At all concentrations, levels of pERK were reduced in lockstep with RAS cleavage (FIG. **55**). While pERK levels were markedly re-duced after 24 h, effects on cell viability reached a maximum at 72 h (IC50=0.005±0.001 nM) (FIG. **55**B and FIG. **63**B). In addition, crystal violet staining showed cell loss starting at 0.01 nM after a 72-h exposure to RRSP-DT.sub.B (FIG. **55**C and FIG. **56**C), while a colony formation assay showed that RRSP-DT.sub.B had a significant effect on cell proliferation at 0.01 nM and completely prevented colony fmmation at 10 nM over 10 d (FIG. **55**D). Phenotypically, RRSP-DT.sub.B caused significant cell rounding in MDA-1\ffi-436 cells at 0.01 nM (FIG. **55**E). Overall, RRSP-DT.sub.B causes highly potent cytotoxic effects via the efficient ablation of RAS signaling in a KRAS' cell line.

(406) RRSP Halts Tumor Growth of a KRAS' TNBC Xenograft

(407) Retaining the native receptor targeting of DT.sub.B to translocate RRSP is advantageous as a test system for mouse xenografts since DT is at least 1000-fold less potent on cells expressing murine HB-EGF than cells expressing human HB-EGF (32-34). Indeed, RRSP-DT.sub.B treatment did not affect the viability of mouse embryonic fibroblasts (MEFs) (FIG. **64**A-B). A maximum tolerated dose (MTD) study determined that 0.1 mg/kg administered intraperitoneally (IP) every day (q.d.) was well tolerated, while dosage at or above 0.5 mg/kg caused weight loss (FIG. **64**C-D). In the MDA-MB-436 xenograft, IP administration of RRSP-DT.sub.B at 0.1 mg/kg every other day (weekends excluded) for 4 weeks halted tumor growth (FIG. **56**A-B). Importantly, three of five mice treated with RRSP-DT.sub.B showed tumor regression by the end of the treatment schedule (FIG. **56**B). Most importantly, catalytically-inactive RRSP*-DT.sub.B did not affect tumor growth compared to vehicle (saline) control, demonstrating that tumor regression was due specifically to RRSP-mediated RAS processing activity. Indeed, RRSP-DT.sub.B dramatically reduced total RAS immunoreactivity in residual tumors, confirming effective RAS cleavage in vivo (FIG. **56**D-E and FIG. **65**A). Additional experimental groups using slight variations of the dosing strategy corroborated these findings (FIG. **66**A-F). Since ERK phosphorylation was limited to the outer rim of control MDA-MB-436 tumors excised from mice at 4 weeks (FIG. **65**B and FIG. **66**G), we assessed pERK levels at an earlier time point when the smaller size of large control tumors is more amenable to pERK detection. In tumors harvested at two weeks, we observed staining throughout the sections with high basal pERK levels in both saline and RRSP*-DT.sub.B-treated tumors and with significant reduction in pERK in the RRSP-DT.sub.B treatment group (FIG. **65**C). These results demonstrate that RRSP-DT.sub.B effectively engages and ablates RAS in vivo, resulting in decreased ERK activation and tumor regression.

(408) Assessment of Relative Susceptibility to RRSP-DT.sub.B Using the NCI-60 Panel

(409) We next screened RRSP-DT.sub.B against the National Cancer Institute NCI-60 human tumor cell line panel comprised of 60 cell lines representing nine different cancer types with

various genetic backgrounds, including RAS mutations. Growth inhibition caused by RRSP-DT.sub.B was measured by sulforhodamine B assay after 48 hours and reported in FIG. **57**A for the highest dose employed (13.5 nM). Fourteen cell lines were classified as “highly susceptible” to RRSP-DT.sub.B as they showed growth inhibition greater than or equal to 90%. Thirty-eight cell lines showed varying degrees of growth inhibition from 25 to 90% and were designated as “susceptible” to RRSP-DT.sub.B (FIG. **57**A).
(410) Generally, cell lines that carry KRAS missense mutations were among the most responsive to RRSP-DT.sub.B, while cell lines with mutations in BRAF (especially BRAF.sup.V600E) tended to be less responsive (FIG. **57**B). Mutations in HRAS, NRAS, or EGFR were not associated with a response pattern to RRSP-DT.sub.B treatment (FIG. **57**B). Notably, analysis of copy number alterations from exome data available for 53 out of 60 NCI cell lines showed amplification of KRAS and NRAS, as well as deletions in EGFR, in the cell lines most sensitive to RRSP-DT.sub.B (FIG. **67**B-C). Further, colon cancer cell lines as a group were the most sensitive to RRSP-DT.sub.B overall, followed by non-small cell lung cancer lines (FIG. **57**C).
(411) Eight cell lines in the NCI-60 screen showed<25% growth inhibition compared to mock treated and were categorized as “less susceptible”. In this group, growth of UACC-62 and MOLT-4 cells was not affected by RRSP-DT.sub.B. One requirement for RRSP-DT.sub.B cytotoxicity is expression of the DT receptor HB-EGF on the cell surface. Available HBEGF gene expression data (FIG. **67**A) showed that, while the correlation between RRSP-DT.sub.B sensitivity and HBEGF expression is not linear, many of the cell lines that responded to RRSP-DT.sub.B had higher expression of HBEGF. In addition, the TNBC Hs578T cell line (HRASG.sup.12D), categorized as less sensitive in the NCI-60 screen, was confirmed to have lower expression of HB-EGF protein and a moderate reduction in cell viability after 72 hours, although RAS cleavage and ERK dephosphorylation were detected at earlier time points (FIG. **63**A and FIG. **68**A-E). Overall, results from the NCI-60 screen indicate that most tumor types were sensitive to RRSP-DT.sub.B and cell lines with genomic abnormalities in RAS genes were markedly sensitive to RRSP-DT.sub.B treatment.
(412) RRSP Reduces Tumor Burden in a Mutant KRAS TNBC Xenograft Model
(413) The highly sensitive basal-like MDA-MB-231 TNBC cell line in the NCI-60 screen (FIG. **57**A) has a KRAS.sup.G13D mutation, is a KRAS-dependent cell line (31, 35), and was among the first cell lines characterized as sensitive to RRSP (22) (FIG. **57**A). Consistent with these findings, treatment of MDA-MB-231 cells with the engineered chimeric toxin RRSP-DT.sub.B cleaved RAS and reduced pERK levels to a similar extent as in MDA-MB-436 cells (FIG. **58**A). Also similar to MDA-MB-436 cells, cell viability (IC.sub.50=0.012±0.001 nM; FIG. **58**B-C and FIG. **69**A-B) and cell proliferation (FIG. **58**D) were strongly affected, and RRSP-DT.sub.B induced significant cell rounding in MDA-MB-231 cells starting at 0.1 nM (FIG. **58**E). As this cell line has a more rapid doubling time than MDA-MB-436 cells (36), mice with MDA-MB-231 xenografts were treated daily (5 days ON/2 days OFF) (FIG. **58**F). After 4 weeks, the RRSP-DT.sub.B treatment group had markedly smaller tumors than saline and RRSP*-DT.sub.B controls (FIG. **58**G-H). Moreover, the RRSP-DT.sub.B-treated tumors had a more focal/patchier staining pattern for RAS and pERK than tumors from saline and RRSP*-DT.sub.B treated mice, which instead exhibited a more diffuse staining pattern (FIG. **69**C-D). Indeed, MDA-MB-231 tumors (FIG. **58**G) were pale in color compared to the MDA-MB-436 tumors (FIG. **56**C) consistent with reports that MDA-MB-231 tumors exhibit low vascularization (37). This supports that MDA-MB-231 tumors in our study may be poorly vascularized, and tumor size reduction may be predominantly due to RRSP-DT.sub.B diffusion from the tumor periphery, resulting in the partial cleavage of RAS in the center of residual tumors. Even still, RRSP-DT.sub.B was highly effective in targeting TNBC MDA-MB-231 tumors resulting in a significant inhibition of tumor growth, thus confirming RRSP-DT.sub.B is effective against TNBC.
(414) RRSP-DT.sub.B Inhibits Cell Viability of Colorectal Cancer Cells in 2D Monolayers and 3D

Spheroids
(415) KRAS mutations are found in ˜50% of colorectal carcinomas (CRC), the third leading cause of cancer-related deaths, representing a major target population for anti-RAS therapy (38). Colon cancer cell lines were the most susceptible to RRSP-DT.sub.B in the NCI-60 screen. Here, we examined the anti-cancer potential of RRSP-DT.sub.B on the CRC cell line HCT-116 harboring a KRAS.sup.G13D mutation. Treatment of cells with RRSP-DT.sub.B at 10 pM led to RAS processing and reduced pERK levels at 10 pM after 24 hours (FIG. **59**A) and strongly decreased the viability of HCT-116 cells after 72 hours (IC.sub.50=0.0015±0.002 nM) (FIG. **59**B and FIG. **70**A). Crystal violet staining showed remarkable cell loss after treatment with RRSP-DT.sub.B as low as 10 pM (FIG. **59**C and FIG. **70**B). Clonogenic assays showed a significant reduction in HCT-116 colonies following treatment with 10 pM of RRSP-DT.sub.B and almost no colonies were found at 10 nM RRSP-DT.sub.B (FIG. **59**D) demonstrating complete loss of cell proliferation. These data agree with the RRSP-DT.sub.B-induced cell rounding phenotype observed with this and other sensitive cell lines (FIG. **59**E).
(416) Unlike two-dimensional cell monolayers, three-dimensional spheroids can recapitulate the architectural, microenvironmental and functional features of in vivo tumors, while retaining reproducibility and easy-to-use properties (39, 40). Effects on cell viability of spheroids from HCT-116 cells were observed after 3 days, with maximum effect observed after 7 days of treatment (IC.sub.50=0.085±0.015 nM) (FIG. **59**F). Bright field images and quantitative analysis showed a dose- and time-dependent reduction in spheroid size following RRSP-DT.sub.B treatment (FIG. **59**G and FIG. **70**C). Immunostaining of spheroid sections treated with 0.1 nM RRSP-DT.sub.B for 3 days showed that intracellular RAS was essentially absent and pERK undetectable (FIG. **5911**). Collectively, these results demonstrate that RRSP-DT.sub.B-dependent RAS ablation and subsequent loss of pERK is highly cytotoxic to CRC HCT-116 (KRAS.sup.G13D) cells in 2D monolayers and 3D spheroids.
(417) RRSP-DT.sub.B Exhibits Antitumor Activity in a Colorectal Cancer Xenograft Model
(418) Since HCT-116 cells are fast-growing cells (doubling time≤20 hours) that generate fast-growing tumors in vivo (41), we administered 0.1 mg/kg of RRSP-DT.sub.B to mice on a q.d. (1×/day) or twice per day (2×/day, b.i.d.) schedule (weekends excluded) for 4 weeks. Both dosing schedules resulted in significant tumor size reduction and tumor regression was observed in 2/10 mice in both RRSP-DT.sub.B treatment groups (FIG. **60**A-B). The residual tumors showed that RRSP-DT.sub.B effectively cleaved RAS, although only the b.i.d. dosing group achieved statistical significance (FIG. **60**C-D and FIG. **71**A). Quantitative analysis of pERK showed that the b.i.d. dosing group had a 2.5-fold reduction in pERK levels relative to controls and some tumors displayed focal staining patterns (FIG. **71**B). Overall, these data show that RRSP-DT.sub.B exhibited strong anti-tumor activity in a CRC xenograft model via irreversible inactivation of RAS.
Discussion
(419) Almost four decades ago, the discovery of RAS as the first human oncogene changed our understanding of cancer. Despite tremendous effort, the three RAS isoforms (KRAS, NRAS and HRAS) have been called “undruggable” and no direct therapies are currently in clinical use (1, 12-14). Nevertheless, promising results for small molecules that irreversibly bind the G12C mutant form of KRAS have led to ongoing phase I clinical trials to evaluate the efficacy and safety profile of AMG510, MRTX849 and ARS3248 (18-20, 42, 43). However, the KRAS.sup.G122C mutation is found in only a subpopulation of cancers, limiting the applicability of these compounds. Further, while RAS oncoproteins remain the main oncogenic drivers in RAS-addicted tumors, several studies have pinpointed the tumorigenic role of wild-type RAS proteins (44). Indeed, amplification of wild-type RAS genes or activation of wild-type RAS proteins via acute growth factor stimulation have been shown to sustain growth of multiple tumor types (7-10). Moreover, it has been previously reported that depletion of mutant RAS in heterozygous RAS cells can lead to overactivation of EGFR/RAS signaling from the remaining wild-type RAS (45). Therefore, there is

an urgent need for broadly applicable pan-RAS inhibitors that target not only the most common RAS mutants, but also wild-type RAS proteins aberrantly overactivated by mutation-independent mechanisms.

(420) Here, we describe a novel engineered chimeric toxin comprised of an endopeptidase from *V. vulnificus* that is highly specific for RAS and RAP1 and the protein translocation machinery of DT. This fusion protein mediates the endocytosis and cytosolic delivery of RRSP exclusively into HB-EGF receptor-bearing cells. Of note, HB-EGF is highly expressed in several human cancers, including gastric, ovarian and TNBC, and the non-toxic diphtheria toxoid CRM197 showed anti-tumor effects on TNBC xenografts (46). As such, we predicted TNBC could be highly susceptible to RRSP-DT.sub.B. We show that picomolar amounts of RRSP-DT.sub.B completely cleaved RAS proteins in these cells, eliminating RAS-dependent ERK phosphorylation. This resulted in a very potent effect on MDA-MB-436 cell viability (IC.sub.50=5 pM). Similarly, RRSP-DT.sub.B treatment led to a large reduction in tumor burden in MDA-MB-436 xenografts, with all mice showing significantly smaller tumors than control groups and 60% of mice showing tumor regression. Target engagement was confirmed by immunohistochemistry analysis of tumor sections showing complete depletion of RAS in residual tumors. We also observed low pERK staining in small dissected RRSP-DT.sub.B-treated tumors, although pERK levels in these specimens were not significantly different than control tumors that showed limited detection of pERK after four weeks. However, assessment of resected tumors from a shorter 14-day xenograft study revealed that RRSP-DT.sub.B significantly reduced levels of pERK. This effect was masked at four weeks likely due to technical factors. Specifically, a previous study reported that in large (>1 cm) glioma specimens, immunoreactivity of pERK was limited to the tumor periphery and was very low in the tumor core, as we also observed. This study suggested that due to time-dependent penetration of the formalin fixative, pERK state is lost in the deeper tissue cores by the time the fixative has permeated (47).

(421) The study of TNBC was extended then to demonstrate susceptibility of another TNBC cell line, MDA-MB-231. This cell line, which harbors an oncogenic mutations KRAS.sup.G13D and BRAF.sup.G464V, is characterized as KRAS-dependent (35) and is highly sensitive to RRSP (22). In xenograft studies using MDA-MB-231, RRSP-DT.sub.B induced a significant reduction in tumor size. Interestingly, the residual tumors did not show significant reduction in RAS or pERK, but instead revealed a patchy or focal distribution. Thus, the true reduction in RAS detection may be masked by the low vascularization of MDA-MB-231 tumors such that RRSP-DT.sub.B reduces overall tumor size by diffusion from the tumor periphery rather than infiltrating equally throughout the tumor center, as occurred with the MDA-MB-436 xenografts.

(422) We finally investigated the effects of RRSP-DT.sub.B treatment on the CRC cell line HCT-116 (KRAS.sup.G13D). As with the other cell lines tested, RRSP-DT.sub.B treatment resulted in complete RAS cleavage and pERK reduction in the low picomolar range, with a corresponding reduction in cell viability. Similar results were obtained in HCT-116 spheroids, including complete loss of RAS and pERK immunoreactivity. Notably, depletion of RAS and reduction in pERK were also observed in spheroids treated with the catalytically-inactive mutant RRSP*-DT.sub.B, but at a 10-fold higher dose. Previous binding studies showed that RRSP* retains its ability to bind recombinant KRAS, especially at high KRAS:RRSP* molar ratios (24). Therefore, although RRSP*-DT.sub.B does not cleave RAS, it might impact its activity when used at high doses and after longer exposure both in vitro and in vivo.

(423) As with the other tumor models, RRSP-DT.sub.B treatment resulted in significant reduction in tumor size in an HCT-116 xenograft study. Both q.d. and b.i.d. dosing strategies showed similar median reductions in tumor size, however the b.i.d. arm showed less variability represented by a small interquartile range. Effective target engagement was demonstrated by a reduction in total RAS levels in residual RRSP-DT.sub.B-treated tumors. We did not observe statistically significant differences between controls and RRSP-DT.sub.B-treated tumors when analyzing pERK

immunoreactivity. However, as noted above, large control tumors also showed poor immunostaining for pERK, and thus low values for control may have caused the lack of statistical significance in the small sample size when data are normalized. Even still, we observed reduced tumor size in all treatment groups and a decrease in pERK in the b.i.d. treatment group.

(424) In order to assess the broader applicability of RRSP-DT.sub.B and have a better understanding of its efficacy as an anti-cancer agent, RRSP-DT.sub.B was tested against the NCI-60 cell line panel. Results showed that RRSP-DT.sub.B inhibited cell growth in a wide variety of cell lines. Cell lines with missense mutations or amplifications in RAS genes were enriched in the highly responsive group. Moreover, although the expression levels of the DT receptor HB-EGF play an important role in determining RRSP-DT.sub.B susceptibility, some cancer cell lines that express less HB-EGF were still sensitive to RRSP-DT.sub.B, suggesting differential susceptibility of tumor cells to RAS/RAP1 inhibition. Thus, RRSP-DT.sub.B has the potential to be broadly applicable against many types of cancers, including both wild-type and mutant RAS tumors. In addition, cancer cell lines from colon and lung are particularly sensitive to RRSP-DT.sub.B. However, several tumor cell lines with no genetic defects in RAS or the MAPK pathway remain very sensitive to RRSP-DT.sub.B.

(425) In total, this study provides solid proof-of-concept that the effective, receptor-mediated, intracellular delivery of a potent anti-RAS biologic represents a promising new approach for the development of RAS-targeted therapeutics. We contend that the ability of RRSP to directly and irreversibly inactivate both wild-type and mutant RAS proteins represents an attractive mechanism compared to the current approach of targeting RAS mutants individually. However, a pan-RAS inhibitor is expected to induce dose-limiting toxicity due to the critical importance of RAS signaling in non-cancerous tissues. The ability of DT to deliver cargos exclusively to receptor-bearing cells provides a solution to this toxicity. While HB-EGF is upregulated on various tumor types, it is also widely expressed in humans, and may not represent the ideal receptor for tumor-targeting. In order to restrict the delivery of RRSP to tumor cells, the DT-based delivery platform described here can be re-targeted to various cell types by replacing the receptor-binding domain of DT with other binding moieties, such as antibody fragments or ligands. For instance, the recombinant immunotoxins Ontak and Tagraxofusp comprise wild-type DTA with DTR replaced by interleukin 2 (IL-2) or interleukin 3 (IL-3), respectively (48, 49). These drugs specifically target cells expressing the IL-2 receptor (IL-2R) or IL-3 receptor (IL-3R) and rely on the membrane translocating ability of DT to deliver the DTA domain into the cell cytosol, where it terminates protein synthesis leading to cell death. These immunotoxins are extremely potent and are indicated for treatment of cutaneous T-cell lymphoma (Ontak) and blastic plasmacytoid dendritic-cell neoplasm (Tagraxofusp) (48, 49). RRSP-DT.sub.B can be re-targeted in the same fashion, and RRSP-DTT-IL2 is able to efficiently cleave RAS in both MOLT-4 and Jurkat cell lines which express IL-2R, but not in CFPAC-I cells, which do not (FIG. **72**). In fact, MOLT-4 was amongst the least responsive cell lines in the NCI-60 screen, suggesting that even cancers currently in the low sensitivity group would be much more sensitive to RRSP-DT.sub.B following advanced engineering to target the chimeric toxin to alternative receptors appropriated for that cell type. Further study into engineered chimeric toxins, such as RRSP-DT.sub.B, could usher in the next-generation of anti-cancer biologics.

Materials and Methods

(426) Plasmids Design, Protein Purification and Endotoxin Removal

(427) DNA sequence corresponding to RRSP aa 3580-4089 was amplified from a plasmid containing the effector domain region cloned from *V. vulnificus* CMCP6 in vector pXL PCR-TOPO (1). The amplified gene was fused to different DT variants using the NEBuilder® HiFi DNA Assembly Cloning Kit (New England Biolabs Inc.). DNA sequence corresponding to CPD (nucleotides nt 12269-12894 of rtxA1) of MARTX toxin from *V. vulnificus* was codon-optimized and synthesized as a double-stranded DNA fragment (IDT) and inserted into the DT vector as

above (2). A point mutation was made in RRSP using QuikChange Lightning Multi Site-Directed Mutagenesis Kit (Agilent Technologies) to change His4030 CAT codon to Ala (GCT) to generate the catalytically inactive RRSP.sub.H4030A mutant (RRSP*-DT.sub.B). The final products were cloned into the Champion pET-SUMO expression system (Invitrogen). The different DT variants fused to RRSP were expressed as N-terminal His6-tagged and C-terminal StrepTagII-tagged proteins, transformed into either *E. coli* NiCo21(DE3) or BL21 (DE3) cells, and grown overnight in Luria-Bertani (LB) broth containing 50 μg/mL kanamycin. Next, overnight cultures were diluted 1:50 in fresh TB containing 35 μg/mL kanamycin and grown to OD.sub.600=0.8-1.0 at 37° C. Cultures were then induced with 1 mM isopropyl β-d-1-thiogalactopyranoside (IPTG) for 5 h at 25° C. Bacteria were pelleted by centrifugation at 10,000 rpm for 15 min and resuspended in 150 mL of Lysis Buffer (20 mM Tris-HCl pH 8.0, 500 mM NaCl, 20 mM imidazole, 2 mg/ml lysozyme and one tablet of EDTA-free protease inhibitor cocktail). Resuspended cells were sonicated using a Branson Digital Sonifier for 30 min (parameters: pulse ON 10 sec, pulse OFF 20 sec, 10 min, Amplitude 50%), the lysate spun down at 12,000×g for 30 min and then loaded onto a 5 ml His-Trap Crude FF column (GE Healthcare) using a ÄKTA Purifier protein purification system (GE Healthcare). His-tagged proteins were eluted with a buffer containing 20 mM Tris-HCl pH 8.0, 500 mM NaCl and 500 mM imidazole. Fractions corresponding to the protein peaks were collected, pooled and loaded onto a gravity column containing Strep-Tactin Superflow high capacity resin (#2-1208-025, Iba Lifesciences). Strep-tagged proteins were eluted using 20 mM Tris-HCl pH 8.0, 150 mM NaCl and 10 mM d-Desthiobiotin. The His-Sumo tag was then removed by adding 0.01 μg of Sumo protease per 100 μg of purified protein in 20 mM Tris-HCl pH 8.0, 150 mM NaCl and 1 mM dithiothreitol at 30° C. for 1 h. Next, proteins were further purified by size exclusion chromatography (SEC) using the HiLoad Superdex 16/600 200 prep grade column (GE Healthcare) run in 20 mM Tris-HCl pH 8.0 and 150 mM NaCl. Proteins were then dialyzed overnight using a ThermoFisher Slide-A-lyzer cutoff 20K in 20 mM Tris-HCl pH 8.0 and 150 mM NaCl and concentrated with a Millipore Amicon Ultra 30K spin concentrator. Finally, glycerol was added to the final protein solution containing 20 mM Tris-HCl pH 8.0, 150 mM NaCl, 10 mM imidazole and 8% glycerol. Protein purity was assessed by SDS-PAGE/Coomassie blue staining and concentration determined using NanoDrop ND1000 Spectrophotometer. Protein aliquots were flash frozen and stored at −80° C. until use. For in vivo applications, endotoxin was removed from all protein preparations using Pierce High-Capacity Endotoxin Removal Resin (#88270) and residual endotoxin was quantified using Pierce LAL Chromogenic Endotoxin Quantitation Kit (#88282) following manufacturer's instructions (Thermo Scientific).

(428) Cell Culture and Chemicals

(429) All cell lines were purchased from the American Type Culture Collection, except for the KRAS.sup.WT-expressing RAS-less MEF cells that were kindly provided by the RAS Initiative at Frederick National Laboratory for Cancer Research (FNLCR) (Designation RPZ26216, expressed transgene KRAS 4B WT) (3). Cells were cultured at 37° C. and 5% CO.sub.2 atmosphere. MDA-MB-436, MDA-MB-231 and HCT-116 were grown in DMEM-F12 with Glutamax (Gibco) containing 10% fetal bovine serum (FBS; Gemini) and 1% penicillin/streptomycin (P/S; Invitrogen). Hs578T were grown in DMEM containing 10% FBS and 1% P/S (complete DMEM). MEF cells were cultured in complete DMEM with 4 μg/ml of blasticidin. All chemicals, unless otherwise specified, were purchased for Sigma-Aldrich.

(430) Antibodies

(431) The anti-RAS 4E8 hybridoma cell line was kindly provided by the Frederick National Laboratory for Cancer Research (FNLCR). The antibody was purified by affinity chromatography as described in (4). Antibody validation was performed as follows. 10 μM of recombinant K-, N- or H-RAS proteins were incubated alone or together with 1 μM of recombinant RRSP for 5 minutes at room temperature and then 1 μg of each sample was run on an SDS-PAGE gel followed by western blotting. The purified antibody used at 1:2000 dilution specifically recognized both cleaved and

uncleaved bands of all three RAS isoforms (FIG. S**2**I). This antibody thus is a pan-RAS monoclonal, here designated as mAb 4E8.

(432) Some western blots and immunohistochemistry were performed with the commercially available anti-panRAS (Ras10, MA1-012; Thermo Fisher Scientific) antibody, that recognizes RAS Switch I and thus detects only uncleaved RAS. Other primary antibodies used are anti-Phospho-p44/42 MAPK (phosphorylated ERK1/2, Thr202/Tyr204, 197G2, Cell Signaling Technology #4377), anti-p44/42 MAPK (ERK1/2, L34F12, Cell Signaling Technology #4696), anti-HB-EGF (R&D Systems, #AF-259-NA;), and precision protein StrepTactin-HRP Conjugate rabbit Bio-Rad (#1610381) The anti-vinculin (Cell Signaling Technology #13901) was used for normalization. Secondary antibodies used were fluorescent-labeled IRDye 680RD goat anti-mouse (926-68070), IRDye 800CW goat anti-rabbit (925-322211) and IRDye 800CW donkey anti-goat (925-32214) from LI-COR Biosciences. Blot images were acquired using an Odyssey Infrared Imaging System (LI-COR Biosciences) and quantified by densitometry using NIH ImageJ software. Percentage of uncleaved RAS was calculated as previously described (5).

(433) SDS-PAGE and Western Blotting

(434) At the experimental endpoint, cells were washed once with ice-cold PBS and protein lysates were extracted using M-PER mammalian protein extraction reagent (Thermo Fisher Scientific) with protease and phosphatase inhibitors (Sigma-Aldrich). Protein concentration was quantified using Bio-Rad protein assay dye reagent concentrate (#5000006). Equal amounts of proteins were separated by SDS-PAGE followed by western blot analysis as previously described (5).

(435) Cell Viability, Proliferation and Growth Inhibition Assays

(436) For quantitative viability assays, 10,000 cells per well were cultured in 96-well clear bottom white plates in the corresponding complete growth medium and treated the next day with RRSP-DT.sub.B and RRSP*-DT.sub.B. At the end of treatments, CellTiter-Glo (Promega) was added to each well according to the manufacturer's manual and luminescence was acquired using a Tecan Safire2 plate reader. CellTiter-Glo was incompatible with assessment of MDA-MB-436 cell viability under the experimental conditions used. Indeed, despite the cells being very sensitive to RRSP-DT.sub.B, no change in luminescence were detected between untreated and treated cells presumably due to release of ATP to the media that complicates the CellTiter-Glo reaction. Therefore, for this cell line, viability was determined via crystal violet assay. Briefly, MDA-MB-436 cells (10,000 cells/well) were plated in 96-well plates and treated with RRSP-DT.sub.B and RRSP*-DT.sub.B. At the experimental endpoint, medium was carefully removed and cells gently washed with PBS. Crystal violet fixing/staining solution (0.05% (g/vol) crystal violet, 1% formaldehyde, 1% (v/v) methanol in phosphate buffered saline (PBS)) was then added to each well and the plate incubated at room temperature for 20 min. Next, wells were washed with tap water, air-dried, crystal violet solubilized using 10% acetic acid and absorbance recorded at 570 nm using a Tecan Safire2 plate reader. IC.sub.50 were calculated using the log(inhibitor) vs. response—variable slope (four parameters) function in Graphpad Prism. Crystal violet assay was also performed to visually assess viability of adherent cells using the same procedure described above but instead of solubilizing the dye, images of 6-well air-dried plates were acquired using a conventional desktop scanner. Cell proliferation was measured by colony-formation assay. Cells were treated with RRSP-DT.sub.B and 10 nM RRSP*-DTB. After 72 hours, cells were harvested by trypsinization, counted on a hemocytometer and replated in 6-well plates at 2,500 cells per well. Colony formation was monitored over 10 days period, during which medium was replaced every two days. On day 10, crystal violet was applied to visualize the colonies. The open source ColonyArea ImageJ plug-in was used for quantitative analysis of the area % covered by the stained colonies (6). Quantification of cell rounding was carried out by manually counting up to 200 rounded cells (with sharp edges and no protrusions) per image and means±SD of three independent experiments were plotted. Because all cells treated with 10 nM of RRSP-DT.sub.B appeared round, this condition was set as 100% cell roundness.

(437) An NCI-60 five dose growth inhibition screen was performed on a panel of 60 human tumor cell lines derived from nine different tumor types by the National Cancer Institute Developmental Therapeutics Program (NCI-DTP) and growth inhibition percentage was calculated in accordance with their standard protocol previously published (7). We normalized all data such that 0% growth inhibition corresponded to no change in the cell number relative to control after 48 hours of treatment with the high dose of RRSP-DT.sub.B (13.5 nM), while a drop of 100% corresponded to no cell growth relative to time zero (100% growth inhibition); values below 100% correspond to cell loss. Additional information about the screening methodology can be found on the NCI-DTP website (8).
(438) Maximum Tolerated Dose (MTD) and Xenograft Studies
(439) The MTD study was performed at Charles River Laboratories (Wilmington, MA). Both female and male athymic NU(NCr)-Foxn1.sup.nu mice were used (5 mice/group). Mice received increasing dosing of RRSP-DT.sub.B (0.004, 0.02, 0.1 and 0.5 mg/kg) and RRSP*-DT.sub.B (0.5 mg/kg) via IP injection on an everyday schedule—weekends excluded—for two weeks. Mouse weight was monitored on a regular basis until the end of the experiment. Mice were humanely euthanized after loss of 20% body weight and counted as non-survivors.
(440) MDA-MB-436 and MDA-MB-231 mouse xenografts were performed following our animal protocol, which was approved by the Institutional Animal Care and Use Committee (IACUC) at Northwestern University. Mice were maintained in Allentown cages in a sterile housing facility under controlled environmental conditions. 5-week old athymic NU(NCr)-Foxn1.sup.nu female mice were injected subcutaneously with 2.5×10.sup.6 cells in 100 μl/mouse of 50% 1× endotoxin-free PBS/50% Matrigel (Corning #356237) onto the right flank under anesthesia. On day 7, when tumors reached 100-200 mm.sup.3 in size, mice were randomized into groups of 5 and treatment started. For the MDA-MB-436 xenograft, control mice were administered IP with saline (endotoxin-free PBS) and treatment groups with 0.1 mg/kg of RRSP-DT.sub.B and 0.1 mg/kg of RRSP*-DT.sub.B on an every-other-day schedule (weekends excluded) for about 4 weeks. In a second experiment with the same cell line, mice were treated with 0.1 mg/kg of RRSP-DT.sub.B every other day (EOD) and 0.1 mg/kg of RRSP-DT.sub.B every day (ED) for about 4 weeks as well as with 0.25 mg/kg of RRSP-DT.sub.B for two weeks followed by 0.1 mg/kg RRSP-DT.sub.B for additional two weeks (weekends excluded). For the MDA-MB-231 xenograft, mice were treated with 0.1 mg/kg of RRSP-DT.sub.B and 0.1 mg/kg of RRSP*-DT.sub.B on an ED schedule (weekends excluded) for 4 weeks. Tumor size was measured twice a week using a digital caliper and tumor volume was calculated using the following formula: volume (mm.sup.3)=(1×w.sup.2)/2, where 1 is the length and w the width of the tumor. Mouse body weight was also measured twice a week.
(441) The HCT-116 xenograft study was performed at Charles River Laboratories (Wilmington, MA). 5×10.sup.6 cells in 100 μl PBS (no Matrigel) were inoculated into the right flank of athymic NU(NCr)-Foxn1.sup.nu female mice. When tumors reached an average size of 80-120 mm.sup.3, mice were randomized into 5 groups of 10 and treatment started. The first group received PBS (saline), the second 0.1 mg/kg of RRSP-DT.sub.B every day (1×/day), the third 0.1 mg/kg of RRSP-DT.sub.B twice per day (2×/day), the fourth 0.1 mg/kg of RRSP*-DT.sub.B 1×/day and the fifth 0.1 mg/kg of RRSP*-DT.sub.B 2×/day. Both tumor size and mouse body weight were measured biweekly. For all the in vivo experiments performed in this study, when tumors exceeded 1500 mm.sup.3, mice were sacrificed as per protocol. At the end of the treatment schedule, mice were euthanized, tumors excised, and fixed in 10% formalin overnight.
(442) Spheroid Formation, Viability and Image Analysis
(443) For tumor spheroids generation, a single cell suspension of HCT-116 cells was seeded at a concentration of 10,000 cells/well into 96-well ultra-low attachment plates (Corning #4520) in complete medium. Two days after cell seeding, treatments were added and viability assessed after 1, 3, 5 and 7 days using the Promega CellTiter-Glo 3D Cell Viability Assay following the

manufacturer's protocol. Spheroids were imaged at every time point with an EVOS XL Core imaging system and their volume analyzed using a freely available ImageJ plug-in as described in (9). Spheroids fixation and agarose-embedding prior to immunohistochemical analysis was performed as previously described (10).

(444) Immunohistochemistry of Spheroids and In Vivo Tumors

(445) Spheroids as well as in vivo tumors processing, paraffin-embedding, sectioning and immunohistochemical stainings were performed by the Robert H. Lurie Comprehensive Cancer Center's Pathology Core Facility. Anti-rabbit pan-Ras (#PA5-85947; Thermo Fisher Scientific) and anti-rabbit Phospho-p44/42 MAPK (Erk1/2) (Thr202/Tyr204, (D13.14.4E) XP Cell Signaling #4370) antibodies were used at 1:500 and 1:300 dilutions, respectively. Primary antibodies were detected using a standard anti-rabbit secondary antibody followed by 3,3′-diaminobenzidine (DAB) revelation (Dako). Quantification of immunohistochemical signal intensity was performed by color deconvolution using ImageJ (Fiji version) as previously described (10).

(446) Bioinformatic Analysis

(447) Bioinformatics analyses were performed using R version 3.5.2. NCI-60 mutation data were retrieved from Reinhold W C et al (11). For the A549/ATCC and MDA-MB-231/ATCC cell lines, mutations were obtained from the ATCC website. NCI-60 gene copy number alteration data and RNA expression z-scores were downloaded from cBioPortal website for the study with id "cellline_nci60" using the TCGAretriever R package from the cran R-project website. Bar plots, tile plots, and violin plots were built using ggplot2.

(448) Statistics

(449) Statistical analysis was performed using Graphpad Prism software v.8. Bar plots represent mean of at least three independent experiments±the standard deviation (SD). Statistical significance was determined using one-way analysis of variance (one-way ANOVA) assuming normal distribution. Dunnett's multiple comparison post-test was used to compare the mean of treatment groups to the mean of the control group and Tukey's multiple comparison test was used to compare the mean of each group with the mean of every other group. Points in the fitted dose-response curve are mean±standard error of the mean (SEM). Statistical analysis on fold change data was carried out after log transformation of the data to obtain a more normalized distribution. For in vivo xenografts, data are reported as mean±SEM and one-way ANOVA was performed to test for differences among the groups. Values of $p<0.05$ were considered statistically significant.

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Example 7—RAS/RAP1 Specific Protease (RRSP) Cleaves all Major Oncogenic Forms of KRAS and Inhibits Growth in KRAS/BRAF Mutant Colon Cancers

Introduction

(451) The oncoprotein Rat sarcoma GTPase (RAS) cycles between GTP-bound (active) and GDP-bound (inactive) states for activation of downstream effectors, each playing key roles in cell proliferation and survival [1, 2]. This process is highly reliant on GTPase activating proteins (GAPs) and guanine exchange factors (GEFs) for hydrolysis of GTP and nucleotide exchange of GDP to GTP, respectively (FIG. **73**) [3, 4]. Upon growth receptor stimulation, activated RAS recruits downstream effectors, including Rapidly Accelerated Fibrosarcoma (RAF) kinase and phosphatidylinositol-3-kinase (PI3K). These effectors subsequently activate signaling pathways responsible for cell growth and survival, including the mitogen-activated kinase to extracellular signal-regulated kinase (ERK) signaling pathway and Ak-thymoma kinase (also known as AKT) to mammalian target of rapamycin (mTOR) pathway, respectively [6, 7]

(452) Thirty percent of all human cancers contain mutations in RAS [1, 14]. Mutant RAS, paired with loss of function in tumor suppressor genes such as TP53 and APC, are sufficient to fully transform cells and drive tumorigenesis [14]. Nearly all RAS mutations occur as point mutations at G12, G13 or Q61, resulting in constitutive activation of RAS [1]. Among the major RAS isoforms (HRAS, NRAS, and KRAS), KRAS is the most frequently mutated isoform among all cancers (85%) followed by NRAS (11%) and HRAS (4%) [14]. RAS mutations are highly enriched specifically in three of the four most lethal cancers in the United States, including pancreatic adenocarcinoma (98%), colorectal adenocarcinoma (52%), and lung adenocarcinoma (32%)[1, 14].

(453) Although numerous studies support the advantages of targeting RAS to treat cancer, it remains an unsolved challenge in the clinic [64, 139-142]. Recent studies have taken advantage of biochemical properties of specific RAS mutants to develop selective small molecule inhibitors specific for highly oncogenic mutant forms of RAS. In particular, small molecules targeting KRAS G12C have been developed and are undergoing clinical trials [15, 16, 143]. Many of these agents have shown clinical success with one molecule receiving FDA accelerated approval earlier this year for treatment of KRAS G12C tumors in non-small cell lung carcinoma [91]. Despite this success, the strategy of selective inhibition has problems of being applicable to only a limited range of cancers integrated with personalized medicine and cannot be used to treat cancers that lack the specific mutation. To address this gap, new approaches are being developed to target RAS more broadly either with proteases that directly cleave RAS or with linkers that target RAS for cellular degradation [127, 136, 144-147].

(454) In line with this alternative strategy, our lab has identified a potent protease that cleaves RAS

called the Ras/Rap1-specific endopeptidase (RRSP). RRSP is a small domain of a large toxin secreted by the bacterium *Vibrio vulnificus* during host infection. *V. vulnificus* delivers RRSP into intestinal epithelial cells during host infection, where it targets all RAS isoforms and close homolog Ras-related protein 1 (RAP1). Through RAS inactivation, this bacterium suppresses the host immune response, thereby aiding systemic dissemination of the bacterium [129, 148]. Detailed structural and biochemical studies have shown that RRSP attacks the peptide bond between T32 and D33 in the Switch I region of both RAS and RAP1 [134]. As a result, RAS and RAP1 are unable to undergo GTP-GDP exchange or bind to their downstream effectors [131, 132]. Recently, RRSP engineered as a chimeric toxin for in vivo delivery was shown to significantly reduce breast and colon tumor growth in xenograft mouse models [136].

(455) The potential applicability of RRSP to a broader range of cancers was screened using the standardized National Cancer Institute (NCI) cancer cell panel [137]. Fourteen of 60 cell lines were classified as highly susceptible with cells undergoing cytotoxic effects. However, 38/60 of cell lines showed growth inhibition, but not cytotoxicity. Only 8/60 showed low or no susceptibility with cell growing near normal rates, possibly due to lack of the receptor for the engineered chimeric toxin [136]. The observed wide range of cell fates highlights that the cellular responses to total RAS cleavage has the potential to be quite variable across cancer cell lines. Here, we investigate how RRSP processing affects cell signaling and demonstrate that cleaving total RAS can have a variable impact on cancer cell growth and survival. Specifically, we demonstrate that RRSP can disrupt colorectal cancer (CRC) cell growth through multiple mechanisms, including loss of cell viability, cell cycle arrest, and senescence.

(456) Results

(457) RRSP Cleave and Inhibits Proliferation in RAS Wildtype and KRAS Mutant Cells

(458) RRSP was previously shown to specifically cleave HRAS, NRAS, and KRAS when the proteins were ectopically expressed in HeLa cells and recombinant RRSP was shown to process purified KRAS G12D, G13D, and Q61R in biochemical assays [134]. To get an even broader sense of RRSP effectiveness across different isoforms and mutants of RAS, we tested RRSP against the 'RAS-dependent' mouse embryonic fibroblast (MEF) cell line panel developed by Drosten et al. [6]. These isogenic cell lines have endogenous RAS genetically deleted from their genome and a single allele of a human RAS gene is integrated to rescue growth. For delivery of RRSP into mouse cells, we used the anthrax toxin-based delivery system wherein the anthrax toxin lethal factor N-terminus was fused with RRSP (LF.sub.NRRSP) or LF.sub.NRRSP with a catalytically inactivating H4030A amino acid substitution (here after referred to as LF.sub.NRRSP*). Intracellular delivery of RRSP (previously known as DUF5) by anthrax toxin protective antigen (PA) was previously demonstrated in several mammalian and mouse cell lines [130, 134, 135].

(459) In MEFs expressing human KRAS, HRAS, or NRAS, treatment with 3 nM LF.sub.NRRSP dramatically decreased intact full-length RAS levels with increased detection of cleaved RAS. For each isoform, LF.sub.NRRSP was found to cleave at least 80% of RAS after 24 h (FIG. **74**A). As expected, controls treated with PA alone or in combination with catalytically inactive LF.sub.NRRSP* showed no change of intact RAS protein levels (FIG. **74**A,B). We observed similar RRSP activity in MEF cell lines expressing oncogenic KRAS, including G12V, G12D, G12C, G13D, and Q61R. Amongst each of the mutant RAS alleles tested, we observed ˜25% of total RAS remaining following LF.sub.NRRSP treatment, with no significant loss of RAS in cells treated with alone (FIG. **74**B). The oncogenic RAS variants with the higher percentage of RAS remaining following LF.sub.NRRSP treatment were G13D and Q61R, although these differences were not statistically significant. Further, the total RAS remaining in each LF.sub.NRRSP-treated MEF cell line was not statistically significant between groups. In addition to cleavage of RAS, LF.sub.NRRSP treated cells showed significant decrease in phosphorylation of ERK when compared to cells treated with PA alone or with the catalytically inactive LF.sub.NRRSP* (FIG. **74**C, D).

(460) To test the impact of processing of different RAS isoforms on cell proliferation, RRSP-treated cells were tracked using time lapse imaging for four days. At early timepoints following treatment, LF.sub.NRRSP-induced a severe cell rounding that was not observed in PA alone and LF.sub.NRRSP* control treated cells (FIG. **74**E). This phenotype is consistent with previous studies with RRSP and is possibly linked to cleavage of RAP1, which regulates cytoskeletal dynamics [130, 134, 149]. LF.sub.NRRSP treated cells showed reduced confluency at both 48 and 96 h (FIG. **74**E) and continuous treatment for 96 h resulted in at least at least a 60% reduction in confluency for all RAS-dependent MEF cell lines compared to cells treated with either PA only or LF.sub.NRRSP* mutant controls (FIG. **74**F, FIG. **75**A-H).
(461) Altogether, these results in MEFs demonstrate that RRSP is equally able to cleave all isoforms of RAS and mutant KRAS to inhibit both ERK phosphorylation and cell proliferation in a defined system. Thus, the KRAS mutation does not likely solely account for differences in cancer cell fate upon treatment with RRSP. Instead, the differences more likely depend on processes downstream of RAS processing that could vary in different cell lines.
(462) RRSP Inhibits Proliferation and pERK Activation in CRC Cell Lines
(463) To probe the effect of processing of RAS on downstream signaling, we focused on four KRAS mutant CRC cell lines, each harboring different allelic mutations in KRAS (FIG. **76**A). Due to problems with variable expression of the anthrax toxin receptor on the selected human cancer cells, we switched to a recently described, highly potent RRSP chimeric toxin wherein RRSP is tethered to the translocation B fragment of diphtheria toxin (RRSP-DT.sub.B) [136]. Similar to the anthrax toxin system, RRSP-DT.sub.B binds to a human receptor heparin binding epidermal growth factor-like growth factor (HB-EGF), is endocytosed, and translocated into the cytosol across the vacuolar membrane. Expression of HB-EGF receptor was found to be similar between the selected CRC cell lines (FIG. **77**A).
(464) To examine RRSP growth sensitivities between the CRC cell lines, cells were treated with increasing concentration of RRSP-DT.sub.B or with catalytically inactive RRSP-DT.sub.B (RRSP*-DT.sub.B) and growth inhibition was monitored. HCT-116 cells showed the greatest impact of RRSP-DT.sub.B on cell confluency and the lowest IC.sub.50 (FIG. **76**B, **77**B,C), consistent with prior data using different methods that HCT-116 cells are highly susceptible to RRSP [136]. Cells treated with catalytically inactive RRSP*-DT.sub.B showed no difference, confirming the sensitivity was due to processing of RAS (FIG. **76**B, FIG. **77** B,C). SW1463 cells were also highly susceptible to RRSP-DT.sub.B but with a 12-fold higher IC.sub.50 compared to HCT-116 (FIG. **76**C, FIG. **77** D,E). Cell line SW620 was less susceptible to RRSP-DT.sub.B after 24 h with about a 40% growth inhibition compared with control at the highest dose tested of 10 nM (FIG. **76**D, FIG. **77**F,G). This result using a different method is consistent with prior results, which categorized SW620 as responding to RRSP by growth inhibition, although the percent inhibition here was less due to the earlier time point used for comparison [136]. Cell line GP5d was also less susceptible to RRSP, but also showed growth inhibition when compared to cells treated with the control (FIG. **76** D,E, FIG. **77** H,I). Across all of the cell lines, at least 80% of total RAS was cleaved by RRSP (FIG. **76** F,G). In addition, phosphorylation of ERK was significantly reduced compared to respective RRSP*-DT.sub.B treated samples (FIG. **76** F,H). We did observe some variability in detection of uncleaved RAS between cell lines, which can be attributed in part to cells sensing depleted pools of RAS and therefore upregulating expression. This is best observed in HCT-116 cells treated with RRSP-DT.sub.B where total uncleaved RAS protein levels increase above the levels of Phosphate buffered saline (PBS) control even as cleaved RAS accumulates (FIG. **76** F).
(465) The differences in growth following RRSP treatment further impacted long term survival. Using ATP as an indicator of cell viability, the CellTiterGlo Assay can quantitatively measure the presence of metabolically active cells through detection of luminescence signal, even if the cells fail to proliferate. In highly susceptible cell lines HCT-116 and SW1463 cells, treatment with

RRSP-DT.sub.B resulted in significantly decreased luminescence compared to mock treated controls after 72 h (FIG. **78**A). By contrast, GP5d and SW620 showed no difference in relative ATP levels after 72 h.

(466) To further understand the survival differences between RRSP-treated cell lines, we used the Caspase-Glo 3/7 Assay, which quantitatively measures caspase-3/7 activity using luminogenic caspase-3/7 substrate to indicate onset of apoptosis. In highly susceptible cell lines HCT-116 and SW1463, treatment with RRSP at both low (1 nM) and high (10 nM) concentrations of RRSP-DT.sub.B significantly increased luminescence compared to mock-treated controls after 48 h, suggesting onset of apoptosis (FIG. **78**B). This was not the case for GP5d cells where the signal was highly variable across replicate samples and even decreased in response to RRSP-DT.sub.B treatment. For SW620 cells, we observed RRSP-DT.sub.B treatment significantly decreased luminescence compared to mock-treated control suggesting a suppression of apoptosis. When treated for 48 h and reseeded at low cell densities SW620 and GP5d both showed a decrease in colony formation, suggesting that RRSP can induce a permanent non-proliferative state, even as cells maintain metabolic activity (FIG. **78** C,D). SW620 cells also showed significant increase activity of the enzyme β-galactosidase, a marker of senescence. This would support our earlier observation in which apoptosis was suppressed in RRSP-DT.sub.B-treated SW620 cells since senescence is known to counteract apoptosis pathway activation [150]. However, β-galactosidase activity of treated GP5d cells remained unchanged (FIG. **78** E). Altogether, these data demonstrate that RRSP activity results in induction of apoptosis in highly susceptible cell lines while, in less sensitive lines, the cells remain metabolically active, but are unable to proliferate and, in some cases, enter into senescence.

(467) RAS Cleavage Can Induce Upregulation of Cyclin-dependent Kiase Inhibitor p27 Hypophosphorylation of RB

(468) We next took advantage of the unique cell line specific effects on cell growth and survival to better understand the underlying mechanisms regulating cell fate following RAS inhibition. Cell lysates from treated or untreated HCT-116 (highly sensitive) and SW620 (less sensitive) were incubated overnight with nitrocellulose membranes containing capture antibodies towards 43 different phosphorylated proteins (FIG. **80**A). For RRSP-treated HCT-116 cells, there was increased phosphorylation observed for cell stress proteins such as p38a, p90 ribosomal S6 kinase (RSK1/2/3), and Jun-activated kinase (JNK) (FIG. **79**, FIG. **80**B). In addition, RRSP treatment increased phosphorylation of several Signal Transducer and Activator of Transcription (STAT) transcription factors. By contrast, the less responsive SW620 cells showed decreased phosphorylation of several STAT proteins (FIG. **79**, FIG. **80**B,). We also observed a significant fold increase in With No K(lysine)-1 (WNK1) kinase at Thr-60. This kinase is phosphorylated by AKT in HEK293 cells and is best known for regulating ion transport across membranes [151]. However, phosphorylation of Thr-60 has no effect on its kinase activity or its cellular localization [152]. Because RRSP decreases AKT activation (FIG. **80**B), it is unlikely that WNK1 Thr-60 phosphorylation is involved in the growth differences we observe between cell lines.

(469) Thus, we focused on the large fold-change difference observed in phosphorylation at Thr-198 of the cyclin-dependent kinase inhibitor p27 (also known as Kip1) (FIG. **80**B). While HCT-116 cells showed no significant change in p27 phosphorylation in the screen, SW620 showed a threefold increase in p27 phosphorylation. RAS is known to regulate critical components involved in cell cycle. RAS activation is directly linked to hyperphosphorylation of retinoblastoma protein (RB), thereby relieving its repression of E2F transcription factors, allowing transcription of G1 promoting genes, and promoting the cell cycle to progress from G1 to S phase [153]. Previous studies have established that phosphorylation of p27 at Thr-198 is critical for stabilizing p27 expression by preventing ubiquitin-dependent degradation [154]. In fact, aberrant RAS activity in cancer cells causes p27 post-translational downregulation through both ERK and AKT [5, 51, 52]. These data support that inhibition of RAS by RRSP could lead to downstream rescue expression of

p27 expression in the SW620 cells, thereby slowing reversing the hyper-phosphorylation of RB.
(470) To test this possibility, all four cancer cell lines were treated with a sublethal dose of RRSP-DT.sub.B. The treatment increased p27 protein levels in HCT-116, SW620, and SW1463 cells, while in GP5d cells levels remained unchanged (FIG. **80**C,D). Concomitant with increased abundance of p27, all cell lines showed a significant decrease in RB phosphorylation at Ser-807/Ser-811 (FIG. **80**C,E). Unfortunately, total RB was undetectable using commercially available antibodies. To be confident that RB hypo-phosphorylation was not due to low RB expression, we transiently expressed green-fluorescent protein (GFP)-tagged RB in HCT-116 cells (FIG. **81**A). In GFP-RB expressing cells treated with RRSP-DT.sub.B, hypophosphorylation of RB protein compared to PBS and RRSP*-DT.sub.B controls was observed (FIG. **81**B). Protein levels of total GFP-RB were decreased upon treatment with RRSP-DT.sub.B. This result was expected and is consistent with a role of p27 in degradation of RB protein to promote growth arrest [155].
(471) Unexpectedly, hypo-phosphorylation of RB was also observed in GP5d cells despite showing no change in the expression or phosphorylation of p27 (FIG. **80**D,E). The cyclin-dependent kinase inhibitor, p21, also plays a critical role in RB regulation. However, there was also no change in p21 protein levels in RRSP-treated GP5d cells (FIG. **81**C).
(472) RRSP Induces G1 Phase Cell Clyde Arrest
(473) Elevated p27 protein expression in combination with hypo-phosphorylation of RB suggested that RRSP treatment induces cell cycle arrest. Under normal conditions, p27 regulates G1 checkpoint during the cell cycle by preventing entry into S phase through inhibition of CDK2 [27, 156]. To test if RRSP-DT.sub.B treatment induces cell cycle arrest, cell lines were treated for 24 h and the percentage of cells in G1, S, or G2/M phase was monitored. All cell lines that showed reduced RB phosphorylation had significant population of cells locked in the G1 state compared to PBS and RRSP*-DT.sub.B treated samples (FIG. **82**A-D, FIG. **83**A-D, FIG. **84**A-D). The most dramatic increase in G1 arrest was seen in SW620 cells, where nearly 100% of cells remained in the G0/G1 phase following RRSP-DT.sub.B treatment (FIG. **82**C). This G1 cell arrest was dependent on the RAS processing activity of RRSP as the catalytically inactive mutant RRSP*-DT.sub.B did not induce the cell cycle arrest (FIG. **82**A-D). Together, these data illustrate that RRSP cleavage of RAS can induce growth inhibition through inducing cell cycle arrest, after which some cell lines progress to cytotoxic death.
(474) BRAF V600E Cells Treated with RRSP Exhibit Growth Inhibition Independent of RAS
(475) Previously we have demonstrated the broad antitumor properties of RRSP across several different cancer cell lines from multiple tumor types. Through genetic analysis, we observed a correlation between RRSP susceptibility and mutations in the downstream RAS effector BRAF. From this panel 6/8 of the cell lines classified as “low susceptible” to RRSP-DT.sub.B contained the BRAF V600E activating mutation, suggesting that RRSP is inefficient in KRAS-independent tumors [136]. To test this hypothesis in further detail we tested RRSP against two different cell lines: BRAF-dependent BRAF MEFs and HT-29 cells. Similar to RAS-dependent MEFs, BRAF-dependent MEFs lack endogenous RAS but instead rely on BRAF V600E for proliferation. The CRC cell line, HT-29, lacks mutations in RAS but harbors a point mutation in BRAF V600E (FIG. **85**A). For treatment of RAS-dependent BRAF V600E MEFs and HT-29 cells, LF.sub.NRRSP and RRSP-DT.sub.B systems were used respectively. MEFs were treated with LF.sub.NRRSP in combination with PA while HT-29 cells were treated with RRSP-DT.sub.B for 24 hours before lysates were collected for western blot analysis. In RAS-dependent BRAF V600E MEFs these cells do not express RAS therefore RAS cleavage cannot be examined. Instead, RAP1 expression was used as an indicator of RRSP activity and was found to be decreased in only cells treated with LF.sub.NRRSP in combination with PA (data not shown). In HT-29 cells treated with RRSP-DT.sub.B, RAS was found to be significantly cleaved at low and high concentrations compared to PBS and RRSP*-DT.sub.B (FIG. **85**C). We observed no significant differences in HB-EGF expression in HT-29 cells compared to other KRAS mutant cell lines (FIG. **77**A) As expected,

when pERK levels were examined we observed no changes in response to RRSP treatment supporting the activity of the BRAF mutation in both cell lines (FIG. **85**B,C).

(476) We next investigated the proliferative changes RRSP has on BRAF mutant cells. To examine growth sensitivities BRAF dependent BRAF V600E MEFs were treated with either PA alone or in combination with LF.sub.NRRSP/LF.sub.NRRSP* and monitored over several time points through time-lapse imaging. For the HT-29 cell line, cells were treated with increasing concentration of RRSP-DT.sub.B or with RRSP*-DT.sub.B. Contrary to the predicted outcome, in BRAF dependent MEFs, growth inhibition was observed at early timepoints despite no changes in pERK activity (FIG. **85** B-D). However, inhibition observed was temporary and slowly rebounded at extended timepoints. Interestingly, a similar phenotype was observed in HT-29 cells wherein growth cells were highly susceptible to RRSP treatment (FIG. **85**E-G). These data would suggest RRSP can inhibit growth in BRAF mutant cells, albeit to varying degrees with different long-term outcomes on proliferation depending on cell type.

(477) We next investigated the cytotoxic effects RRSP has on BRAF mutant cell lines. Based on the data described above RRSP has no effect on survival in RAS-dependent MEFs. We would expect this phenotype to apply to BRAF dependent MEFS and therefore only HT-29 cells were tested for RRSP cytotoxic effects. We again utilized the CellTiterGlo Assay to quantitatively measure cell survival through the presence of ATP. Contrary to data observed in highly susceptible KRAS mutant CRC cell lines, HT-29 cells showed no adverse survival effects in response to RRSP despite the dramatic growth effects (FIG. **86**A). Data from time-lapse imaging can support these findings, where HT-29 cells treated with RRSP-DT.sub.B formed massively rounded colonies that exhibited no obvious signs of cytotoxicity (FIG. **77**J). To further understand long-term growth effects following RRSP treatment a colony formation was performed. HT-29 cells were treated with either PBS or RRSP-DT.sub.B for 48 hours followed by reseeding cells at a low cell density for 14 days to examine colony formation. We observed a slight decrease in colony formation compared to PBS control, however, this was deemed statistically insignificant, suggesting that RRSP treatment in these cells may be reversible over time (FIG. **86**B). We performed additional analysis for signs of senescence following 48-hour treatment with RRSP-DT.sub.B and observed no changes in β-galactosidase activity (FIG. **86**C). Lastly, we examined cell cycle checkpoint molecule p27 and phosphorylation of RB to determine if short-term growth inhibition was due to cell cycle arrest. In HT-29 cells treated with RRSP-DT.sub.B for 24 hours, no changes in p27 or pRB were observed (FIG. **86**D-F). In addition, cell cycle analysis using flow cytometry demonstrated no induction of cell cycle arrest ruling out this phenotype for RRSP mediated growth inhibition in HT-29 cells (FIG. **86**G-H). Altogether, these results demonstrate that RRSP inhibits growth albeit temporarily in BRAF mutant cells. The mechanism of RRSP-mediated growth inhibition in BRAF mutant cells may be mediated through a cell cycle independent process.

(478) BRAF Dependent ERK Signaling Regulates RAP1a Expression

(479) A critical attribute of BRAF mutant cells is their ability to activate ERK signaling independently of RAS. Therefore, RRSP growth effects in BRAF mutant cells must be independent of RAS and is likely due to inactivation of RAP1. The role of RAP1 is better understood in regulating cell adhesion and migration with some studies highlighting proliferative functions [157]. Our data suggests that temporary growth effects in BRAF dependent MEFs are a result of RAP1 inactivation and that growth rebound is done through ERK driving RAP1 expression. To test this hypothesis BRAF dependent MEFs were treated with either PBS, dimethyl sulfoxide (DMSO), or the MEK inhibitor U0126 for 2 hours, and gene expression of RAP1a was examined. After 2 hours we observed a complete loss of pERK activation in cells treated with UO126 at each concentration (FIG. **87**A). When we examined RAP1a expression following UO126 treatment we observed a decrease in RAP1a transcript levels compared to DMSO control (FIG. **87**B). Notably, DMSO control increased RAP1a transcript levels after short incubation compared to PBS treated sample (FIG. **87**B). These data would suggest that ERK is necessary for RAP1a expression in BRAF

dependent MEFs. These data describe for the first time a mechanism by which RAP1a expression is driven through RAS-regulated ERK signaling and could potentially play critical roles in cell growth in MEFs.

Discussion

(480) It has been over 30 years since the discovery of the importance of RAS for driving tumorigenesis in cancer. Lung, pancreatic, and colorectal cancers remain being the most lethal cancers in the United States with high mutation rates in KRAS, the most commonly mutated isoform. Despite the significant amount of research being conducted on RAS, it still remains a challenging target in the field. Small molecules directed to specific RAS mutants, specially KRAS G12C, have shown promising results in clinical trials, but will only benefit a small subset of patients [158]. Our lab has discovered RRSP as a potent, site specific inhibitor of RAS capable of inhibiting all RAS isoforms simultaneously along with downstream activation. RRSP antitumorigenic effects are well demonstrated in vivo with xenograft models for both breast and colon cancers, wherein tumor growth was stunted and, in some cases, showed regression [136]. Evidence for RRSP as a therapeutic inhibitor of RAS is sufficient, however the mechanism by which RRSP mediates growth inhibition has been an outstanding question. In this study, we examined the signaling consequences of cleavage of all RAS in several CRC cell lines and its downstream implications on cell proliferation and survival. The goal of the study was to understand the mechanism of growth inhibition in response to RRSP, in the absence of cytotoxicity.

(481) First, we examined whether RRSP was a suitable inhibitor across RAS variants. Using the isogenic RAS-dependent MEF model, we demonstrated that all three major RAS isoforms and frequently observed KRAS mutants were equivalent substrates for RRSP. Loss of RAS resulted in reduced ERK activation, which as expected, negatively affected proliferation. Most importantly, in this isogenic study, we observed no significant differences in RAS cleavage between wildtype isoforms and KRAS mutants. RAS-dependent MEFs are useful for studying isolated biochemical properties and signaling of specific oncogenic RAS in a cellular context making it an excellent model to study RRSP targeting of mutant RAS. One of the unavoidable disadvantages of this model is the number of integration sites that vary between RAS-dependent MEF cell lines. For some RAS cell lines a single gene integration could not rescue proliferation and required multiple integrations for stable clonal populations. As a result, in the context of RRSP, expression of multiple genes of a single RAS allele may lead to an underestimation of RRSP effectiveness in cleaving RAS.

(482) We next examined RRSP effectiveness in four CRC cell lines, which displayed variations in susceptibility to RRSP. Two of the cell lines with the greatest RRSP growth sensitivity, HCT-116 and SW1463, had dramatically lower metabolic activity and induction of apoptosis compared to controls. Interestingly, GP5d and SW620 retained normal metabolic activity, yet showed significant inability to form colonies following RRSP treatment, mimicking a senescent-like phenotype. This observation is consistent with prior data for SW620 showing a reduction in cellular material stained with sulforhodamine B after 48 h of treatment at 13.5 nM RRSP, but RRSP was not cytotoxic [136]. A logical hypothesis is that GP5d and SW620 cells have elevated pERK activity and therefore are less susceptible to RRSP. However, our data show the opposite in that moderately susceptible cell lines have significant decreases in pERK and RAS following RRSP treatment. This result would suggest that inhibition of pERK activity is not a predictor of susceptibility to RRSP in many cell lines. Similar results have previously demonstrated that varying growth sensitivities to pharmacological inhibition of mitogen-activated protein kinase kinases (MEKs) do not correlate with pERK activity in KRAS and BRAF mutant CRC cell lines [159, 160]. Further investigation of RRSP in the context of KRAS mutants expressing low levels of pERK would elucidate mechanisms regulating cell fate signals between cell lines.

(483) Mechanisms that link RAS and the cell cycle have been well examined. In quiescent cells, p27 is highly expressed in order to inhibit CDKs activity and to suppress RB phosphorylation [27,

156]. Upon mitogen stimulation, RAS activation suppresses p27 protein expression through post-translational modifications that signal for its ubiquitin-mediated degradation [7, 51, 52]. In RAS-driven human cancers, low levels of p27 are frequently observed. We demonstrated that the growth inhibition in HCT-116, SW1463, and SW620 is the result of G1 cell cycle arrest through the upregulation of p27. These data suggest that RAS cleavage in certain CRC cell lines induces p27 upregulation, leading to a cell cycle arrest state that can induce apoptosis at later timepoints. Transient overexpression of p27 is known to then induce cell cycle arrest and later apoptosis [161, 162]. Although in our studies, only the highly susceptible cell lines showed decreases in viability and induction of apoptosis, whereas SW620 retained metabolic activity and undergoes cell cycle arrest. It is important to note that a prolonged cell cycle arrest through p27 induces a persistent cell cycle arrest through induce senescence [163]. We observe this phenotype in SW620 cells in which RRSP treatment induced a senescent-like phenotype.

(484) Unexpectedly, GP5d cells did not show upregulation of p27 or p21, although RB was hypo-phosphorylated, and cells initiated G1 cell cycle arrest. These data reveal there must be other pathways that drive growth inhibition following RAS cleavage. In fact, RRSP from the insect pathogen *Photorhabdus asymbiotica*, which is identical to RRSP, also cleaves RAS and was recently reported to induce G1 cell cycle arrest [134, 162]. The proposed mechanism did not depend on RAS processing, and instead involved RRSP directly binding to CDK1 when it is overexpressed, essentially functioning as a protein trap. Thus, the multi-domain RRSP may possess at least two mechanisms to inhibit the cell cycle, one by restoring p27 downstream of RAS processing and another by directly binding to CDK1. Notably, because low p27 expression levels have been correlated with poor survival in patients with different types of cancer including colon, the ability of RRSP to restore p27 expression and to initiate cell cycle arrest could have important implications for the treatment of tumors with aberrant RAS signaling.

(485) The most unique phenotype from our data was shown in cells with BRAF mutations, which had differing responses on growth inhibition compared to KRAS mutant cell lines. In both BRAF dependent MEFs and HT-29 CRC cell lines RRSP was capable of inhibiting growth, however, growth inhibition seemed to be temporary. Further investigation determined that growth inhibition did not affect survival or cell cycle progression. Thus, these cells still undergo DNA replication but may continue to expand without undergoing cell division. Due to the activating mutation in BRAF, we hypothesized that RRSP-dependent growth inhibition, in this case, may be RAS-independent. In our preliminary data, we demonstrate for the first time RAP1a mRNA expression is regulated by ERK signaling in MEFs. Because RRSP also targets RAP1, a critical regulator of cytokinesis, cleavage of RAP1 could play a role in growth inhibition through disrupting the final stages of cell division. Previous studies have suggested that RAP1 inactivation could lead to growth abnormalities independently of the cell cycle [164]. The role RAP1 plays in proliferation through regulating mitosis-related activity remains to be investigated but could provide an alternative target for therapeutic strategies.

(486) In summary, a critical finding of this study was that processing of RAS does not result in a single easily tractable cell fate in all cancer cells but drives a suite of alternative overlapping outcomes that can include cytotoxicity, inhibition of cell cycle and senescence. The differences are not driven by the nature of KRAS oncogenic mutation as all mutant forms of KRAS were susceptible to RRSP-driven RAS processing in an isogenic system and RAS was processed in all the CRC cell lines. The impact of RAS processing is thus linked to cancer cell differences in signal pathways downstream of RAS. RRSP anti-tumor effects has some translational applicability to RAS-independent BRAF mutant cells with greater implications for understanding the role of RAP1 during proliferation independently of ERK. A limitation of this study is that only five cell lines were assessed, suggesting that the array of variable outcomes could increase as additional cell lines are considered in the future. However, the key finding here is that, despite differences in the mechanisms underlying RRSP susceptibility, all cells tested were ultimately susceptible and all led

to reduced cell viability, growth, and/or proliferation. Thus, RAS processing or degradation has great promise as a potential broadly applicable therapy against colon cancer.
(487) Methods
(488) Purification and Intoxication of Cells with LF.sub.N-RRSP or RRSP-DT.sub.B in RAS-Less MEFs and Colorectal Cancer Cells
(489) 'RAS-dependent' mouse embryonic fibroblast (MEF) cells were provided by the National Cancer Institute RAS Initiative at Frederick National Laboratory for Cancer Research (FNLCR). HCT-116 cells were purchased from the American Type Culture Collection. HT-29 cells were provided from the Marcus Peter lab at Northwestern University. Other colorectal cancer (CRC) cells lines SW1463, GP5d, and SW620 were provided by the FNLCR. Each CRC cell line was validated by the Northwestern University Sequencing Core by Short Tandem Repeat profiling. All cells were cultured at 37° C. and 5% CO.sub.2 atmosphere. HCT-116, SW1463, GP5d, SW620 cells were cultured in Dulbecco's Modified Eagle Medium (DMEM)-F12 with Glutamax (Gibco) containing 10% fetal bovine serum (FBS; Gemini Bio) and 1% penicillin/streptomycin (P/S; Invitrogen). HT-29 cells were cultured in RPMI-1640 (Gibco) containing 10% FBS and 1% P/S. All MEF cells, except for HRAS RAS-dependent MEFs, were cultured in DMEM (Gibco) with 10% FBS, 1% P/S, and 4 μg/ml of blasticidin (ThermoFisher Scientific).
(490) HRAS RAS-dependent MEFs was cultured in 2.5 μg/ml of puromycin (ThermoFisher Scientific). Recombinant RRSP-DT.sub.B and RRSP*-DT.sub.B were expressed in *E. coli* BL21(DE3) and purified over a HisTrap FF nickel affinity column as described above. For intoxication in CRC cell lines were seeded in 6-well plates (˜70% confluency) overnight, after which medium was replaced with fresh medium containing either RRSP-DT.sub.B or RRSP*-DT.sub.B and incubated at indicated timepoints at 37° C. in the presence of 5% CO.sub.2. Recombinant LF.sub.NRRSP and LF.sub.NRRSP* (previously known as LF.sub.NDUF5 and LF.sub.NDUF5*) were expressed in *Escherichia coli* BL21(DE3) and purified over a HisTrap FF nickel affinity column followed by Superdex 75 size exclusion chromatography using the ÅKTA protein purifier purification system (GE Healthcare), as described above. For intoxication, MEFs were seeded in 6-well plates at 3×10.sup.5 cells per well for 1 h, after which medium was replaced with fresh medium containing with 7 nM Protective antigen (PA) alone (List Labs, #171E) or in the presence of 3 nM LF.sub.NRRSP/LF.sub.NRRSP.sup.H4030A (Also known as LF.sub.NRRSP*) and incubated at indicated timepoints at 37° C. in the presence of 5% CO.sub.2.
(491) Time-Lapse Video Microscopy
(492) For RAS-dependent MEFs and KPCs (6×10.sup.3 cells per well) were cultured in 96-well clear bottom white plates in corresponding complete growth medium and treated after 4 h with LF.sub.NRRSP or RRSP-DT.sub.B respectively. Colorectal cancer cell lines were plated at 80% confluency and cultured in 96-well clear bottom white plates. Complete growth medium with RRSP-DT.sub.B or RRSP*-DT.sub.B was added after overnight cell attachment. All cells were cultured in Nikon Biostation CT and images were taken at indicated timepoints. Cell confluency was quantified using Nikon Elements software. IC50 concentrations were calculated using log(inhibitor) vs. response variable slope (four parameters) function in Graphpad Prism 8.
(493) Cell Viability, Apoptosis, Cell Survival, and Cell Counting Assays
(494) For cell viability assay CRC cell lines were seeded in 96-well clear bottom white plates at ˜80% confluency. Complete growth medium with RRSP-DT.sub.B or RRSP*-DT.sub.B was added after overnight cell attachment. After 72 h, CellTiter-Glo (Promega) reagent was added to each well and luminescence was detected using Tecan Safire2 plate reader. For apoptosis assay, CRC cell lines were seeded at 10,000 cells per well in a 96-well clear bottom white plate. Complete growth medium with RRSP-DT.sub.B or RRSP*-DT.sub.B was added overnight after cell attachment. After 48 h, Caspase-Glo 3/7 Assay (Promega) regent was added to each well and luminescence was detected using Tecan Safire2 plate reader. For crystal violet assays, cells were treated as described above and were incubated for 48 h. Following incubation cells were harvested and reseeded at low

seeding densities in 6-well plates. Colony formation was monitored over 14 days, during which media was replaced every three days. On day 14 colonies were fixed in crystal violet fixing/staining solution (0.05% (g/vol) crystal violet, 1% formaldehyde, 1% (v/v) methanol in PBS. Open source ColonyArea ImageJ plug-in was used for quantitative analysis of the area % covered by the stained colonies [204]. Due to high background from crystal violet staining in SW620 cells, stained wells were dissolved in 10% acetic acid and destained on rocker for 30 min. Absorbance was measured at 590 nm using Tecan Safire2 plate reader.

(495) Proteome Human Phospho-Kinase Array

(496) CRC cell lines were treated as described and washed in 1×PBS. Cells were solubilized using lysis buffer provided by the vendor (R&D Systems) and rocked for 30 min at 4° C. Suspension was spun for 5 min at 14,000×g and supernatant was collected. Concentration of protein in the collected supernatant fluid determined using the BCA assay (ThermoFisher Scientific, no. 23227). 200 μg of sample lysate was applied to nitrocellulose membranes kinase arrays and incubated overnight at 4° C. Provided detection antibodies were incubated with specified concentrations as suggested by the supplier. Membrane arrays were acquired using Odyssey Infrared Imaging System (LI-COR Biosciences) and quantified by densitometry using NIH ImageJ software. Values from densitometry analysis were normalized to HSP60 control. Normalized value was then converted to Log.sub.2 fold change and plotted on heatmap using Graphpad Prism 8.

(497) Flow Cytometry

(498) CRC cell lines were treated as described above. After 24 h of treatment, cells were collected from medium, washed with 1×PBS, and released from well with Trypsin-EDTA (0.25%), phenol red (Invitrogen). Harvested cells were centrifuged at 700×g for 5 min. Cells were washed twice in PBS and spun down at 700 Å˜g for 5 min. PBS was removed and cells were resuspended in 600 μL of ice-cold PBS. Cell were permeabilized with addition of 1.4 mL of ice-cold ethanol slowly and incubated overnight at −20° C. Following two washes with PBS (centrifuged at 700 Å˜g for 5 min), cells were stained in 200 μL PI staining solution (0.1% Triton X-100, 50 μg propidium iodide (BioLegend), 0.2 mg/mL RNase) for 30 min. Samples were analyzed on BD LSR Fortessa 1 Analyzer. At least 10,000 events were collected for each sample. Single cell populations were viewed and gated on cyanine-3 area (Cy3-A) versus cyanine-3 width (Cy3-W) channels, to eliminate doublet events. ModFit LT Software (Version 5) was used for cell cycle analysis.

(499) Quantitative Reverse Transcription PCR

(500) BRAF V600E dependent MEFs were treated with U0126 MEK inhibitor for 24 hours. After incubation mRNA was harvested using the QIAGEN RNeasy kit (QIAGEN, no. 74014) following the manufacturer's instructions. RNA isolated was measured using a NanoDrop 100 spectrophotometer. Reverse transcription was performed using random hexamers (Roche) and SuperScript III Reverse Transcriptase (Invitrogen, no. 18080044) in the presence of RNaseOUT (Invitrogen, no. 10777019) or RNasin (Promega, no. N2611) under the following conditions: 25° C. for 5 min, 55° C. for 6 min, and 95° C. for 5 min. Remaining RNA was hydrolyzed using 1M NaOH. qPCR was performed using iQ DYBR Green Supermix (Bio-Rad, no. 1708880) on the iQ5 Multicolor Real-Time PCR detection system using RAPla and GAPDH gene specific primers. Relative change in gene expression compared to PBS control was determined using the delta delta CT method [205].

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Example 8—Proteolytic RAS Cleavage Leads to Tumor Regression in Patient-Derived Pancreatic Cancer Xenografts

(502) Reference is, Vidimar V., et al. Proteolytic RAS cleavage leads to tumor regression in patient-derived pancreatic cancer xenografts. Abstract Submission.

(503) Effective RAS inhibition and treatment of pancreatic ductal adenocarcinoma (PDAC) represent major unmet medical needs in oncology. RRSP-DT.sub.B is a novel engineered anticancer biologic that enters cells and cleaves the Switch I of all RAS isoforms. RRSP-DT.sub.B was recently shown to reduce breast and colon tumors in xenograft studies. Here, we investigate the anticancer activity of RRSPDT.sub.B in KRAS-mutant PDAC cell lines and patient-derived xenografts (PDXs). We first demonstrate RRSP-DT.sub.B effectively engages RAS and impacts downstream ERK signaling in multiple KRAS-mutant PDAC cell lines. Ras was cleaved in cells with IC.sub.50 ranging from 10-280 picomolar and cell proliferation was inhibited at 63-181 picomolar. A modified RRSP-DT.sub.B that lacks catalytic activity was ineffective, demonstrating loss of proliferation is due to protease activity. We next tested RRSP-DT.sub.B by administration to NSG mice bearing KRAS-mutant PDAC PDXs. Treatment led to ≥95% tumor regression after 29 days. Residual tumors exhibited disrupted tissue architecture, increased fibrosis and fewer proliferating cancer cells compared to controls. Levels of phospho-ERK were also significantly lower, indicating in vivo target engagement. Importantly, tumors that started to regrow in the absence of RRSP-DT.sub.B shrank when treatment resumed, indicating resistance to RRSP-DT.sub.B had not developed. A pharmacokinetic (PK) study showed RRSP-DT.sub.B is active in sera of immunocompetent mice for at least one hour, but absent after 16 hours, justifying use of

daily dosing. Overall, we report that RRSP-DT.sub.B strongly regresses hard-to-treat KRAS-mutant PDX models of pancreatic cancer, warranting further development of this pan-RAS biologic for the management of RAS-addicted tumors.

(504) It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably may be practiced in the absence of any element or elements, limitation or limitations which is not specifically disclosed herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention. Thus, it should be understood that although the present invention has been illustrated by specific embodiments and optional features, modification and/or variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modifications and variations are considered to be within the scope of this invention.

(505) Citations to a number of patent and non-patent references may be made herein. Any cited references are incorporated by reference herein in their entireties. In the event that there is an inconsistency between a definition of a term in the specification as compared to a definition of the term in a cited reference, the term should be interpreted based on the definition in the specification.

Claims

1. A pharmaceutical composition comprising: (a) a fusion protein, the fusion protein consisting of a *Vibrio vulnificus* Ras/Rap1-specific endopeptidase (RRSP) fragment fused to a carrier protein, wherein the RRSP fragment is selected from: SEQ ID NO: 1, or a fragment of SEQ ID NO: 1 comprising SEQ ID NO: 2; and (b) a pharmaceutically acceptable vehicle for delivery to a eukaryotic cell.
2. The pharmaceutical composition of claim 1, wherein the carrier protein facilitates transport of the RRSP fragment into proliferating cells.
3. The pharmaceutical composition of claim 1, wherein the carrier protein is a bacterial toxin.
4. The pharmaceutical composition of claim 3, wherein RRSP fragment is fused at its N-terminus to the carrier protein.
5. The pharmaceutical composition of claim 3, wherein the bacterial toxin is anthrax toxin lethal factor N-terminus (LF.sub.N).
6. The pharmaceutical composition of claim 5, further comprising anthrax toxin protective antigen (PA) which forms a complex with the fusion protein and the complex is delivered to the cytosol of proliferating cells.